COURSE

Adaptive and Autonomous Aerospace Systems

Adaptive Control of a Hexacopter for Packages Delivery

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TASK 1

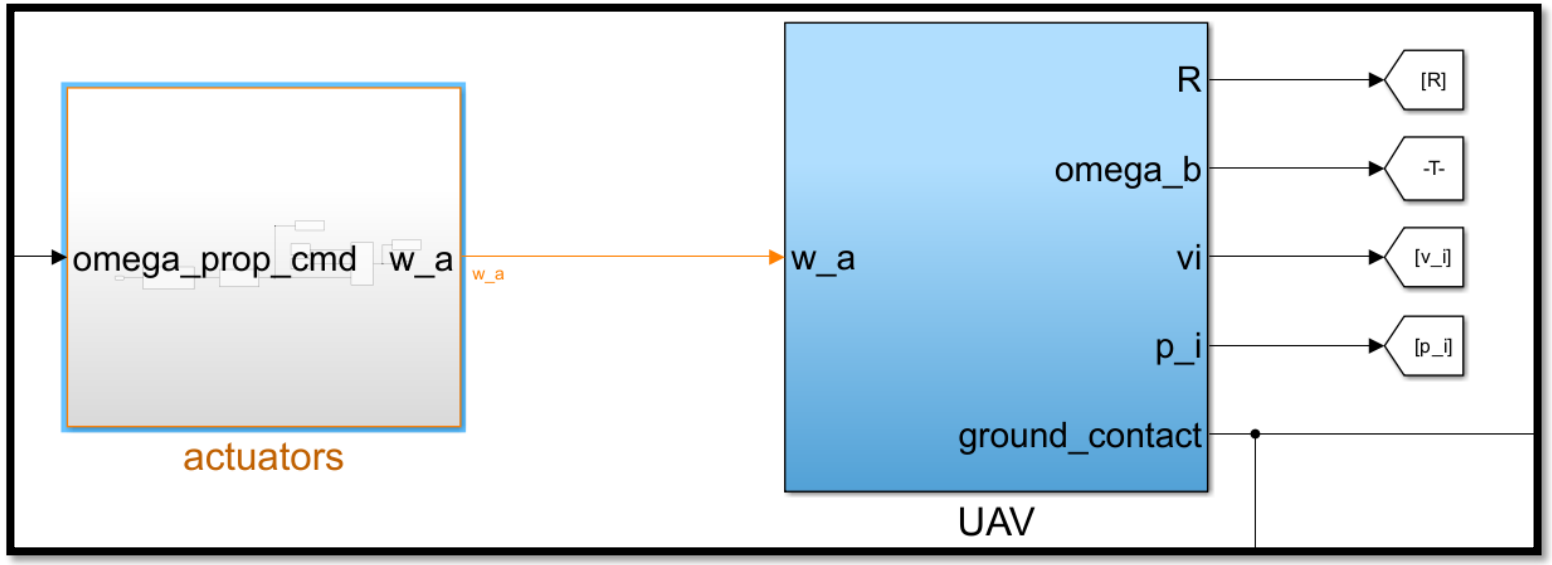
Model Implementation

Hexacopter dynamics · Position trim · Velocity trim

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UAV Simulink model — Overview

SIMULINK MODEL — ACTUATOR DYNAMICS AND UAV

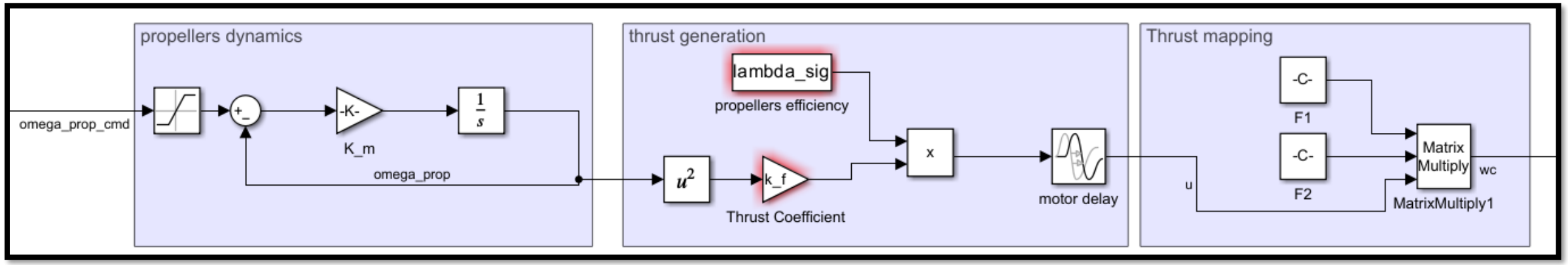


MODEL STRUCTURE

- Actuator dynamics: commanded propeller speed $\Omega_{r,cmd} \rightarrow$ control wrench $\boldsymbol{w}_c$, which drives the UAV block ($\boldsymbol{w}_a$).
- UAV block outputs: R, ω_b, v_i, p_i, v_b and the Euler angles.

UAV Simulink model — Actuators dynamics

ACTUATOR SUBSYSTEM — BLOCK SCHEME



PROPELLER DYNAMICS

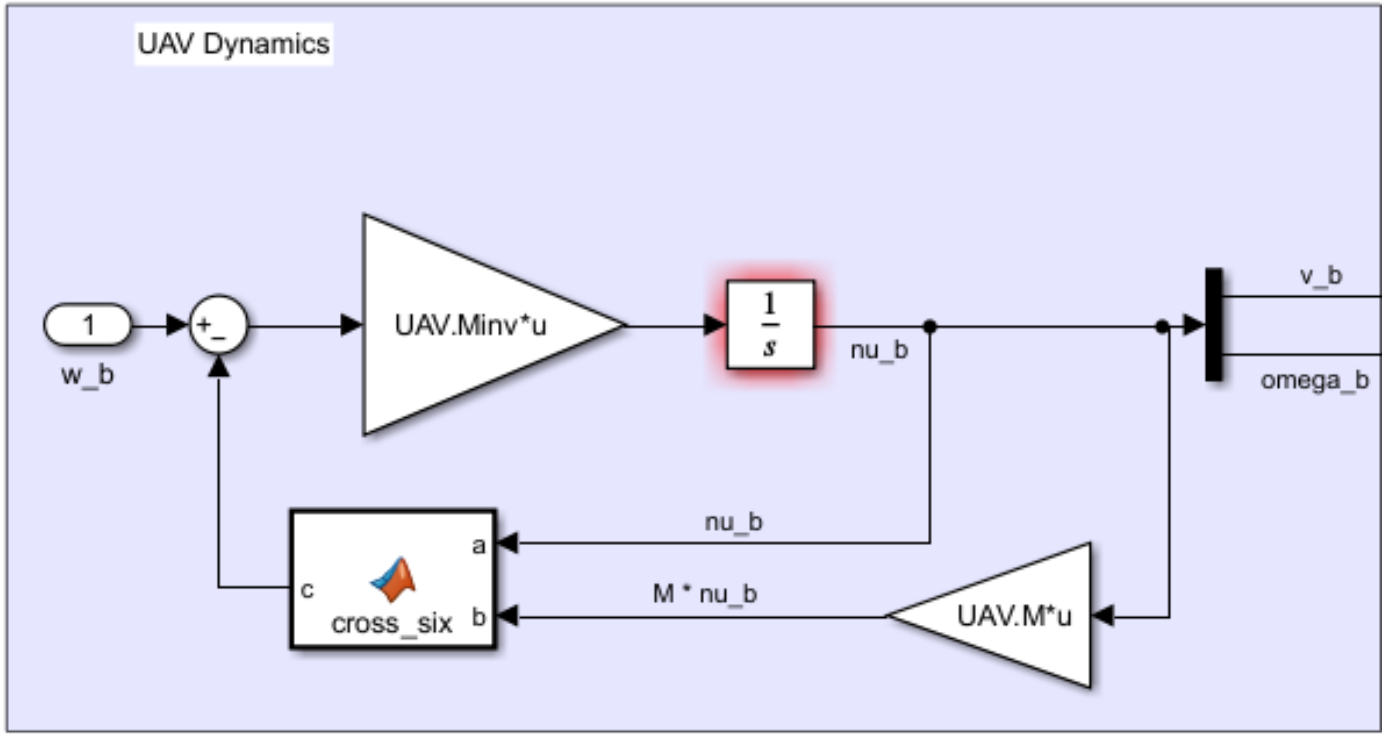
$$\begin{cases} \dot{\Omega}_r = -k_{r,m}(\Omega_r - \Omega_{r,cmd}) \\ T_r = \text{sat}_{T_m}^{T_M}\left(\lambda_r k_f \Omega_r^2\right) \\ u = \begin{bmatrix} T_1 & \cdots & T_N \end{bmatrix}^\top \end{cases}$$

SYMBOLS

- $\Omega_{r,cmd}$: commanded rotor speed
- $k_{r,m} = 1/0.05$ s, inverse motor time constant
- λ_r: propeller effectiveness (fault injection)
- sat$[T_m, T_M]$: envelope, applied to the speed command $[\Omega_{min}, \Omega_{max}]$
- 10 ms transport delay on the thrust

UAV Simulink model — rigid-body dynamics

SIMULINK MODEL — UAV RIGID-BODY DYNAMICS



6-DOF GENERALIZED VELOCITY

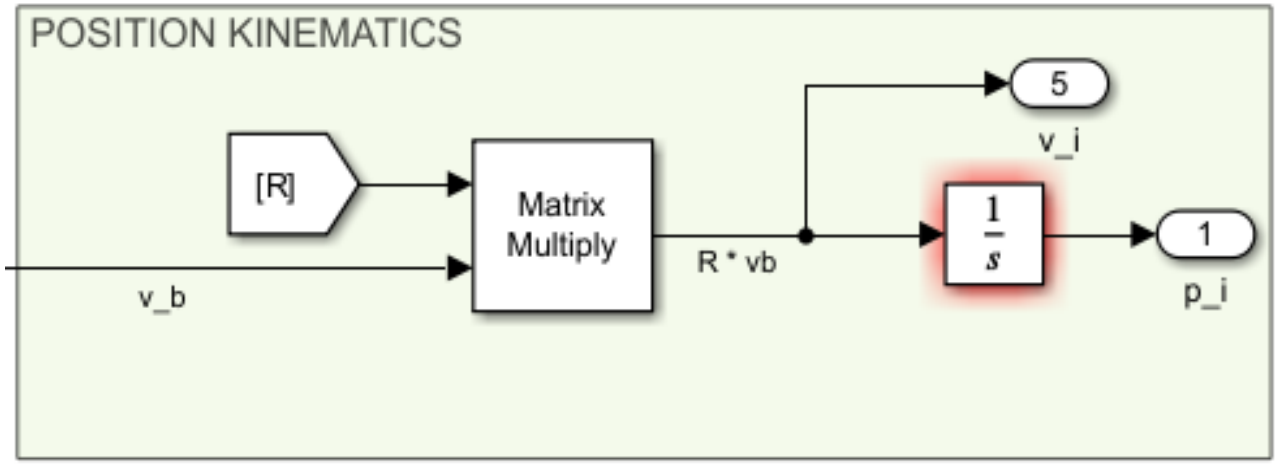
$$M\dot{\boldsymbol{\nu}}_b + \boldsymbol{\nu}_b^{\times} M\boldsymbol{\nu}_b = -D\boldsymbol{\nu}_b + \boldsymbol{w}_g + Fu$$

SYMBOLS

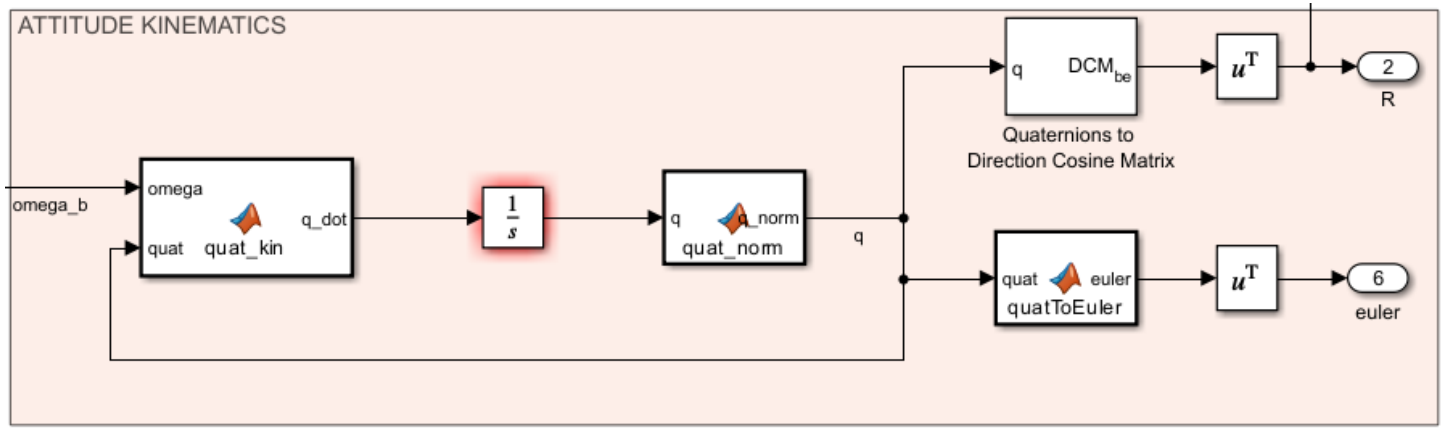
- M: generalized mass matrix $[mI_3,\ S^{\top};\ S,\ J]$
- D: linear + rotational aerodynamic damping
- $\boldsymbol{w}_g$: gravity wrench, resolved in body axes
- Fu: control wrench (rotor thrusts, $F_1 \cdot F_2$ map)

UAV Simulink model — kinematics

POSITION KINEMATICS



ATTITUDE KINEMATICS



POSITION & ATTITUDE KINEMATICS

$$\begin{cases} \dot{p} = R\, v_b \\ \dot{q} = \frac{1}{2} \begin{bmatrix} -\varepsilon^\top \\ \varepsilon^\times + \eta I_3 \end{bmatrix} \omega_b \end{cases}$$

SYMBOLS

- R: rotation matrix (body $\rightarrow$ inertial)
- $q = [\eta,\ \varepsilon^\top]^\top$: unit quaternion
- Normalization: $q \leftarrow q/\|q\|$, applied to the integrator output before use.

Trim commands for hover at p = [0, 0, 0]^T

$$p(t) = p_d, \quad \dot{p}(t) = 0, \quad \omega_b(t) = 0, \quad q(t) = q_0$$

- This requires the total external wrench to be zero.

TRIM DERIVATION

$$\begin{cases} M\dot{\boldsymbol{\nu}}_b^0 + \boldsymbol{\nu}_b^{0\times} M\boldsymbol{\nu}_b^0 = \boldsymbol{w}_g + \boldsymbol{w}_{r,a}^0 + \boldsymbol{w}_c \;\Rightarrow\; \boldsymbol{w}_c = -\boldsymbol{w}_g \\ \boldsymbol{w}_g = -\begin{bmatrix} 0 & 0 & mg & 0 & 0 & 0 \end{bmatrix}^\top \;\Rightarrow\; \boldsymbol{w}_c = \begin{bmatrix} 0 & 0 & mg & 0 & 0 & 0 \end{bmatrix}^\top \\ \dot{\Omega}_r^0 = -k_{r,m}\left(\Omega_r - \Omega_{r,cmd}\right) = 0 \;\Rightarrow\; \Omega_r = \Omega_{r,cmd} \end{cases}$$

WRENCH COMPONENTS

$$\begin{cases} F_2\, u = \begin{bmatrix} T & \tau_x & \tau_y & \tau_z \end{bmatrix}^\top \;\Rightarrow\; \boldsymbol{w}_c = F_1 F_2\, u = \begin{bmatrix} 0 & 0 & T & \tau_x & \tau_y & \tau_z \end{bmatrix}^\top \\ \begin{bmatrix} 0 & 0 & T & \tau_x & \tau_y & \tau_z \end{bmatrix}^\top = \begin{bmatrix} 0 & 0 & mg & 0 & 0 & 0 \end{bmatrix}^\top \;\Rightarrow\; T = mg, \quad \tau = 0 \end{cases}$$

- Coplanar rotors give no force along x_b, y_b, so the first two components of $\boldsymbol{w}_c$ are zero.

– Convention: z_b up, thrust positive along $+z_b$. run_task1_trim.m uses thrust along $-z_b$, so it prints the same trim with opposite sign: $T = -mg = -11.772$ N.

ALLOCATION AND COMMAND

- Allocation: 6 identical, evenly-spaced coplanar rotors ⇒ the min-norm pseudo-inverse F_2^+ splits the collective equally.

$$\begin{cases} T_r = \dfrac{T}{N_r} = \dfrac{mg}{6} \\ T_r = k_f\, \Omega_{r,cmd}^2 \end{cases} \Longrightarrow\; \Omega_{r,cmd} = \sqrt{\frac{mg}{6\, k_f}}$$

$$\Omega_{r,cmd} = \sqrt{\frac{1.2 \cdot 9.81}{6 \cdot 3.65 \cdot 10^{-6}}} = 733.2 \text{ rad/s} \approx 7001 \text{ rpm}$$

Hover trim — results

KNOWN RESULTS

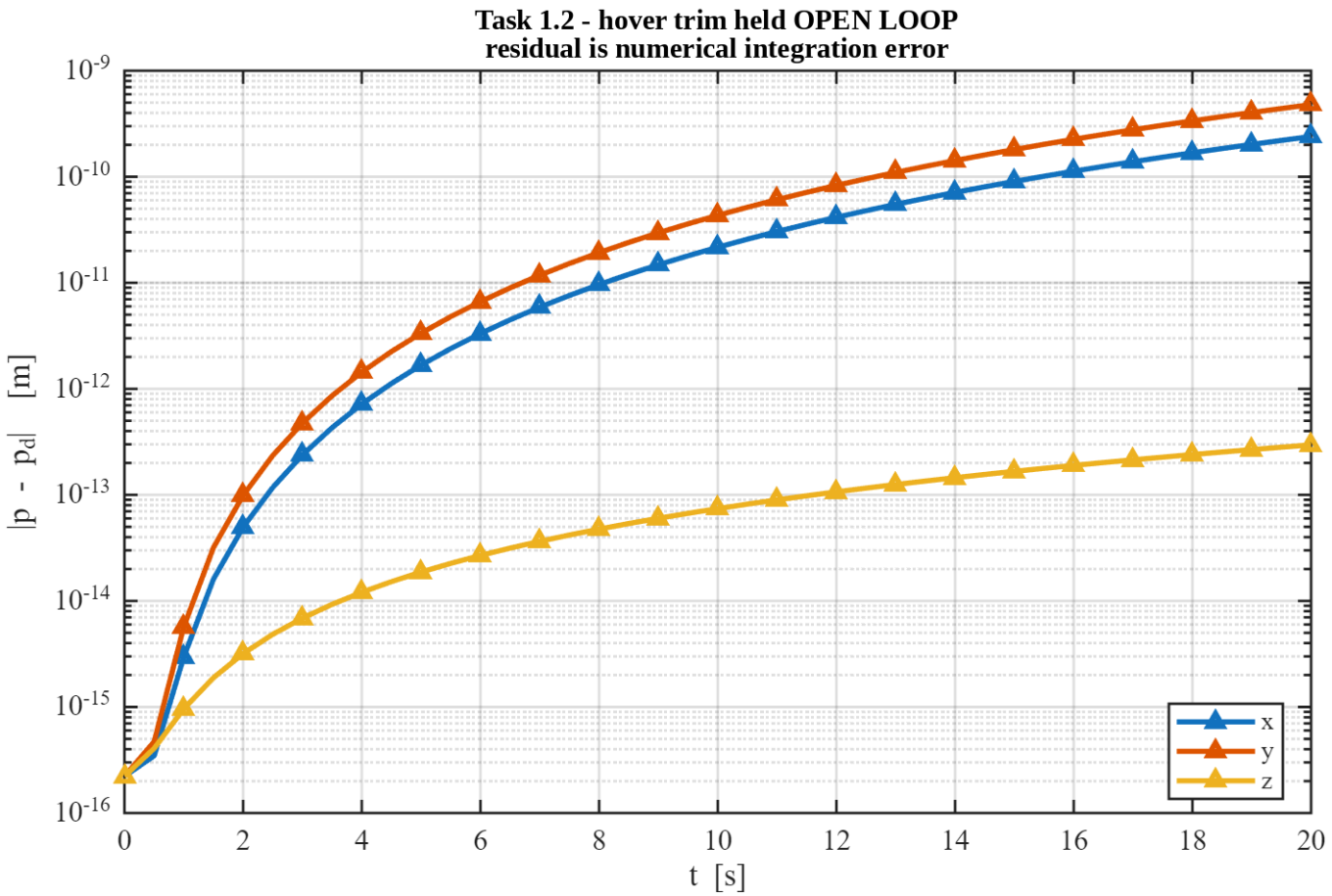
$$\begin{cases} T = mg = 11.772 \text{ N} \\ T_r = \dfrac{T}{6} = 1.962 \text{ N} \\ \Omega_r = 733.2 \text{ rad/s} \approx 7001 \text{ rpm} \\ \tau = 0 \quad \text{(level, symmetric)} \end{cases}$$

CHECKS

– Position error stays below 10^{-9} m over 20 s (numerical integration residual).

- Ω_r lies inside the actuator range $[131.9,\ 1078.6]$ rad/s.

POSITION ERROR — OPEN-LOOP, SEMILOG (x, y, z)



TASK 2

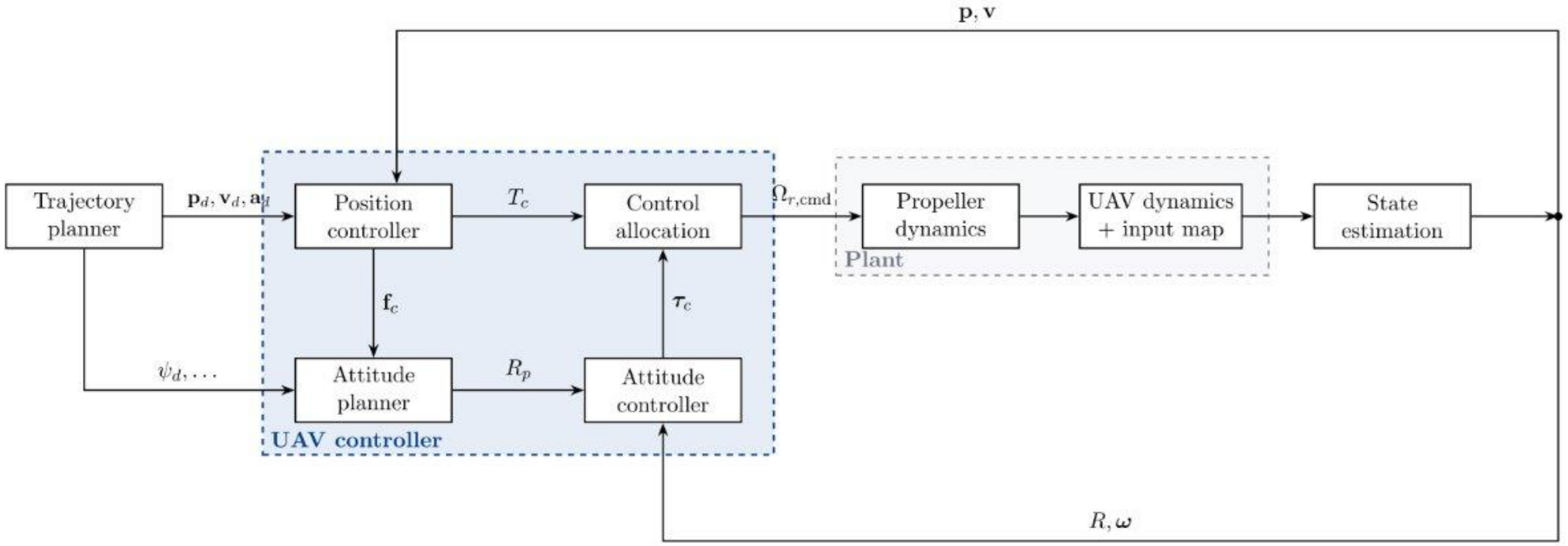
Baseline Controller

Cascade architecture · Loop-shaping tuning · Time-domain validation

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Control architecture

CASCADED CONTROL ARCHITECTURE



CASCADE PRINCIPLE

– Cascaded (inner–outer) design: the position loop treats the commanded thrust direction as achieved instantly by the much faster attitude loop.

– This time-scale separation is checked explicitly in T2.2 (successive loop closure).

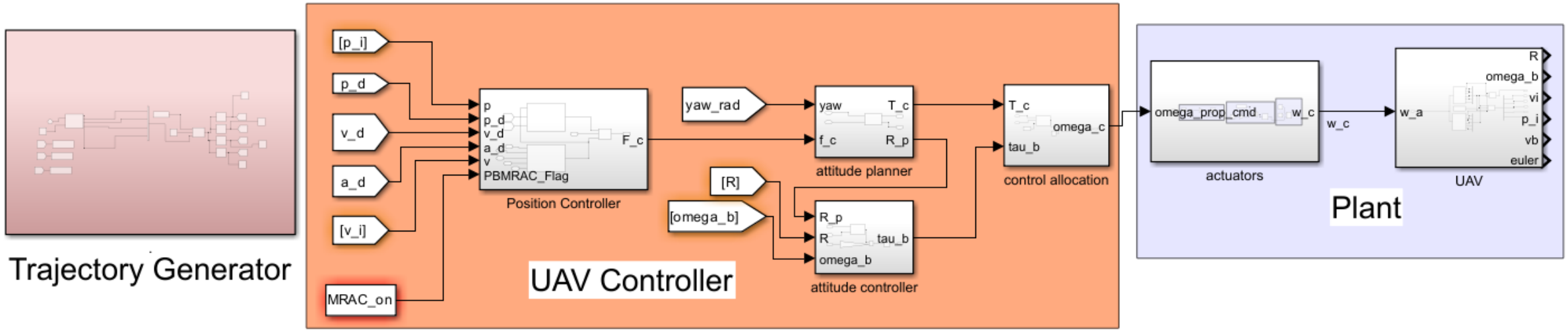
OUTER AND INNER LOOPS

– Outer loop (position): PID on position error; the integral term rejects a constant force bias (unmodeled package weight in Task 3, or steady headwind).

– Inner loop (attitude): geometric control on $SO(3)$.

Control architecture

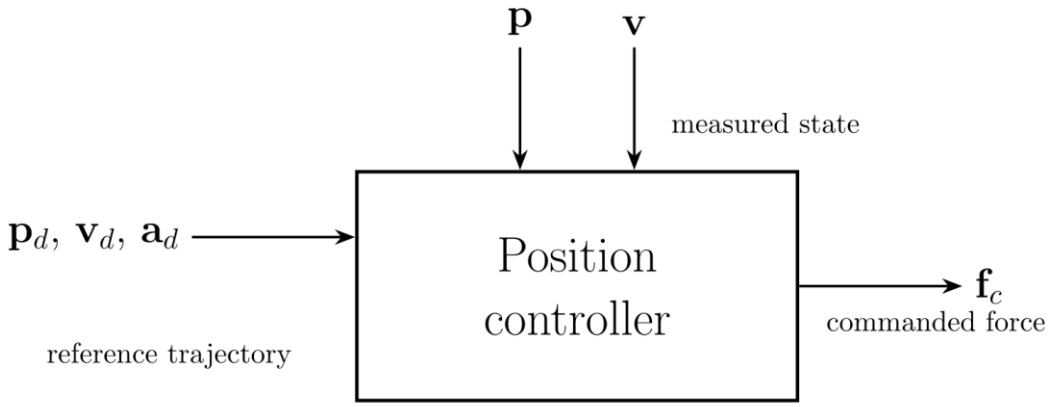
SIMULINK MODEL — UAV CONTROLLER AND PLANT



MODEL STRUCTURE

– Trajectory planner → UAV controller (position controller, attitude planner, attitude controller, control allocation) → plant (actuators, UAV).
– Controller inputs: p_i, p_d, v_i, v_d, a_d, yaw reference, measured R and ω_b, and the MRAC flag; outputs: Ω_c to the actuators.
– UAV outputs: R, ω_b, v_i, p_i, v_b and Euler angles.

Position controller



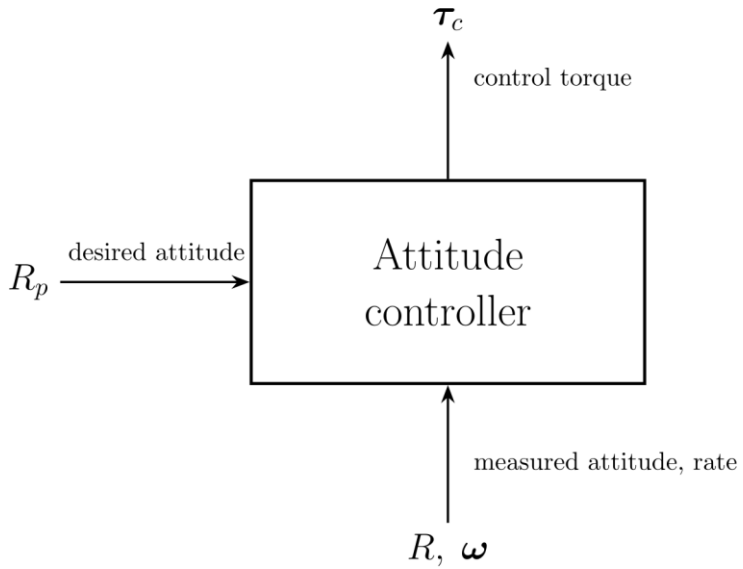
POSITION CONTROL LAW

$$\begin{cases} \mathbf{f}_c = m\,(g\,\mathbf{e}_3 + \ddot{\mathbf{p}}_d) - m\,(K_p\,\mathbf{e}_p + K_v\,\mathbf{e}_v + K_i\,\boldsymbol{\xi}_p) \\ \dot{\boldsymbol{\xi}}_p = \mathbf{e}_p \end{cases}$$

- $e_p = p - p_d\,, \quad e_v = v - v_d$ (inertial-frame errors)
- $K_p,\ K_v,\ K_i$: diagonal PID gain matrices

- Feedforward term $m(g\,\mathbf{e}_3 + \ddot{\mathbf{p}}_d)$: gravity cancellation + desired acceleration.
- Feedback term: PID on the position error.
- Integral state $\boldsymbol{\xi}_p$ accumulates position error to reject constant force disturbances (e.g. unmodeled payload weight, steady wind).

Attitude controller



- Geometric control on $SO(3)$:

  no Euler angles, so no gimbal lock; defined on all of SO(3).
- $\mathbf{e}_R = 0$ at $R = R_d$, but also for 180° rotations about the gain axes (unstable): almost-global, not global.

ATTITUDE ERROR

$$\begin{cases} R_e = R_d^\top R \\ \boldsymbol{e}_R = \sum_{i=1}^{3} K_{R,ii} \left((R_e^\top \boldsymbol{e}_i) \times \boldsymbol{e}_i \right) \end{cases}$$

CONTROL TORQUE

$$\boldsymbol{\tau}_c = -\boldsymbol{e}_R - K_\omega \boldsymbol{\omega}$$

- $\boldsymbol{e}_i,\ i=1,2,3$: basis vectors of the body frame
- $K_{R,ii}$: per-axis attitude gain (diagonal)

  K_ω: rate-error gain

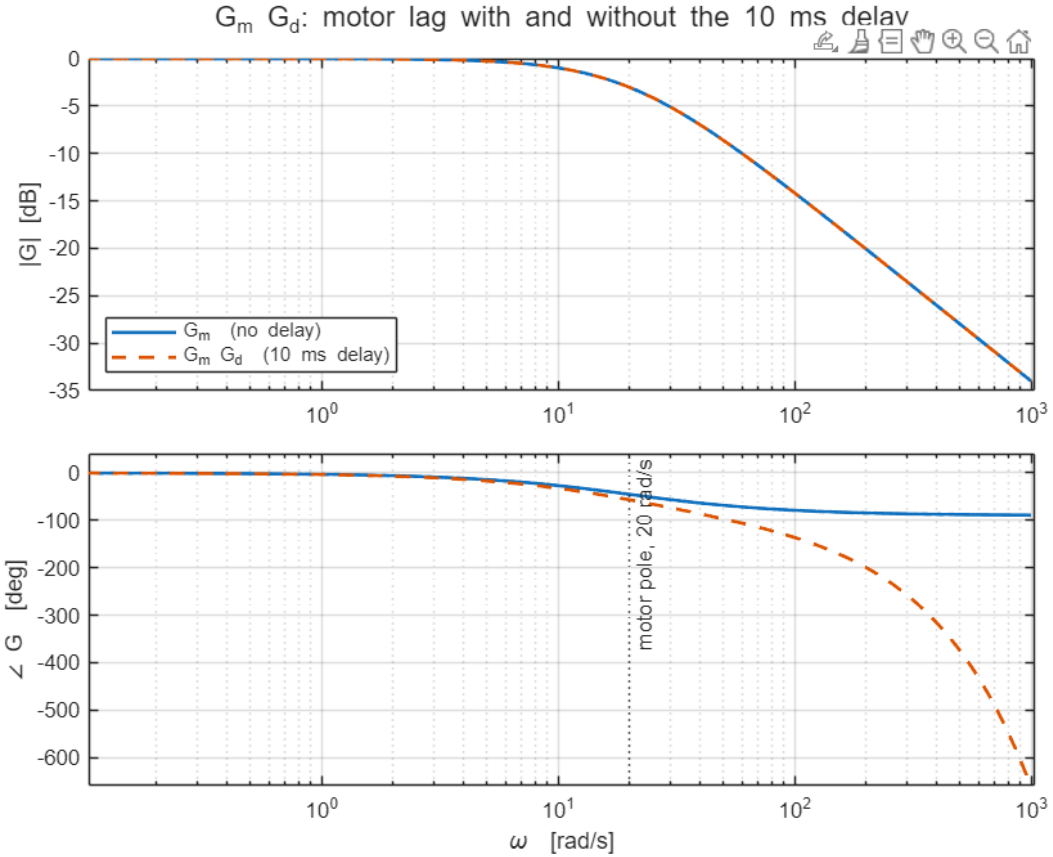
Design plant — actuator + delay + rotational dynamics

DESIGN PLANT

$$\begin{cases} G_m(s) = \dfrac{20}{s+20} \\ G_d(s) = e^{-0.01\,s} \\ G(s) = \dfrac{G_m(s)\,G_d(s)}{J\,s^2 + D\,s} \end{cases}$$

SIMPLIFIED ROTATIONAL DYNAMICS

$$J\,\dot{\omega} + D\,\omega = u$$

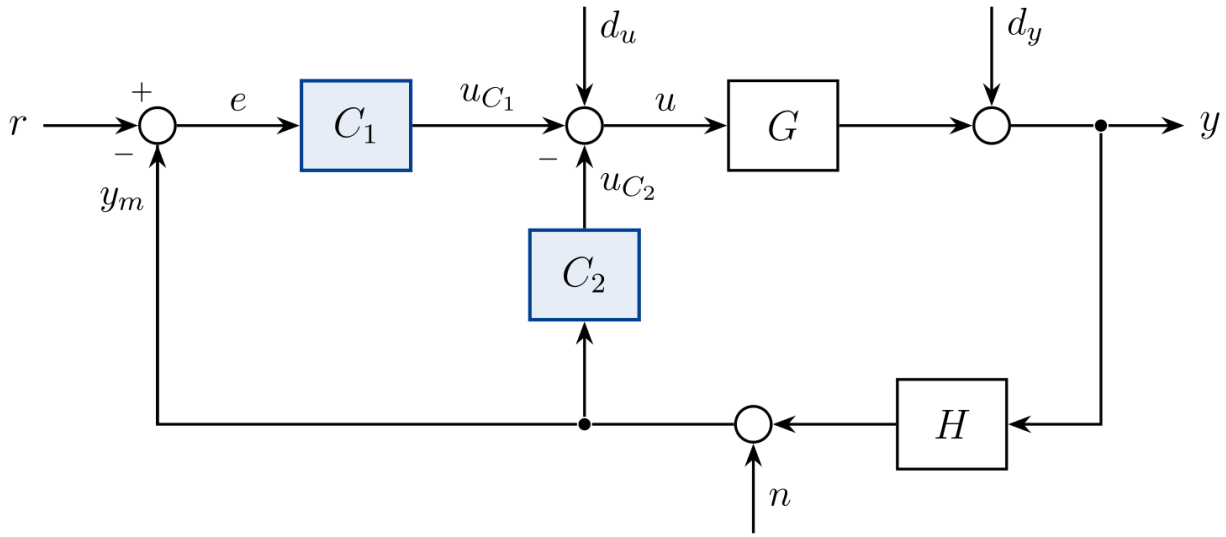


- G_m: first-order motor lag, $\tau_m = 0.05$ s
- G_d: computation / sensor delay, $t_d = 0.01$ s
- $J = 0.01$ kg·m^2, $D = 3.6\times10^{-4}$ (pitch/roll), 1.2×10^{-2} (yaw)

- sisotool / controlSystemDesigner shapes a **LINEAR SISO** transfer function on Bode / root-locus.

Loop-shaping scheme — 2-DOF architecture (sisotool / CSD)

2-DOF LOOP STRUCTURE



- r: reference (r=0 for regulation, tracked via feedforward elsewhere)
- C_1: forward-path controller
  C_2: inner-loop shaping (derivative-type)
- G: design plant from the previous slide

CLOSED-LOOP RELATIONS

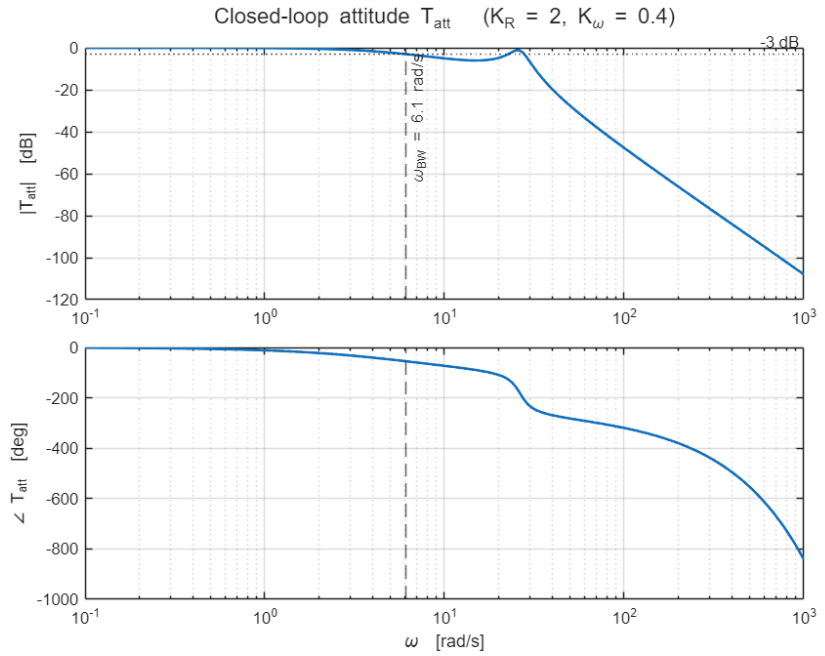
$$\begin{cases} u = C_1(r - y) - C_2\, y \\ H = 1 + \dfrac{C_2}{C_1} \\ T(s) = \dfrac{C_1\, G}{1 + G\,(C_1 + C_2)} \end{cases}$$

DESIGN OBJECTIVES

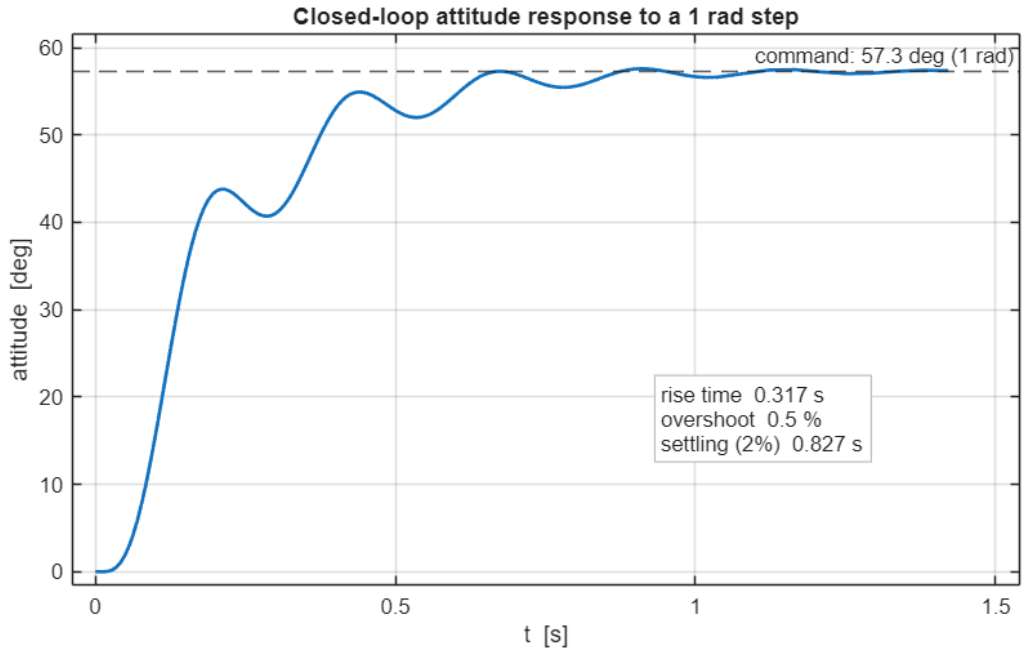
Objective	Target
Overshoot (lateral / vertical)	$\leq 35\,\%$
Settling time (lateral / vertical)	≤ 15 s
Loop margins	adequate
Successive loop closure	$\omega_{att} \geq 5\,\omega_{pos}$
Disturbance rejection	bounded excursion
Actuator envelope	within $[\Omega_{min}, \Omega_{max}]$

Attitude loop (pitch/roll) — tuning results

CLOSED-LOOP FREQUENCY RESPONSE — ATTITUDE (PITCH/ROLL)



CLOSED-LOOP STEP RESPONSE — 1 rad COMMAND



RESULTING CONTROLLERS

$$\begin{cases} C_1 = 2 \\ C_2 = 0.4\,\mathrm{s} \end{cases}$$

TUNED GAINS

$$\begin{cases} K_R = 2 \\ K_\omega = 0.4 \end{cases}$$

STABILITY MARGINS

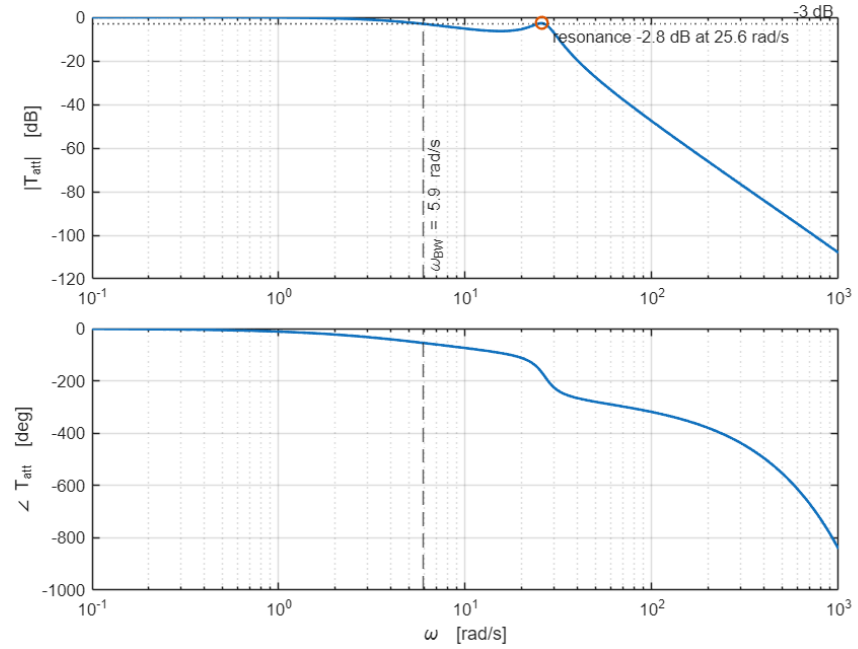
$$\begin{cases} PM = 82.4^\circ \\ \omega_c = 5.16\ \mathrm{rad/s} \end{cases}$$

STEP-RESPONSE METRICS

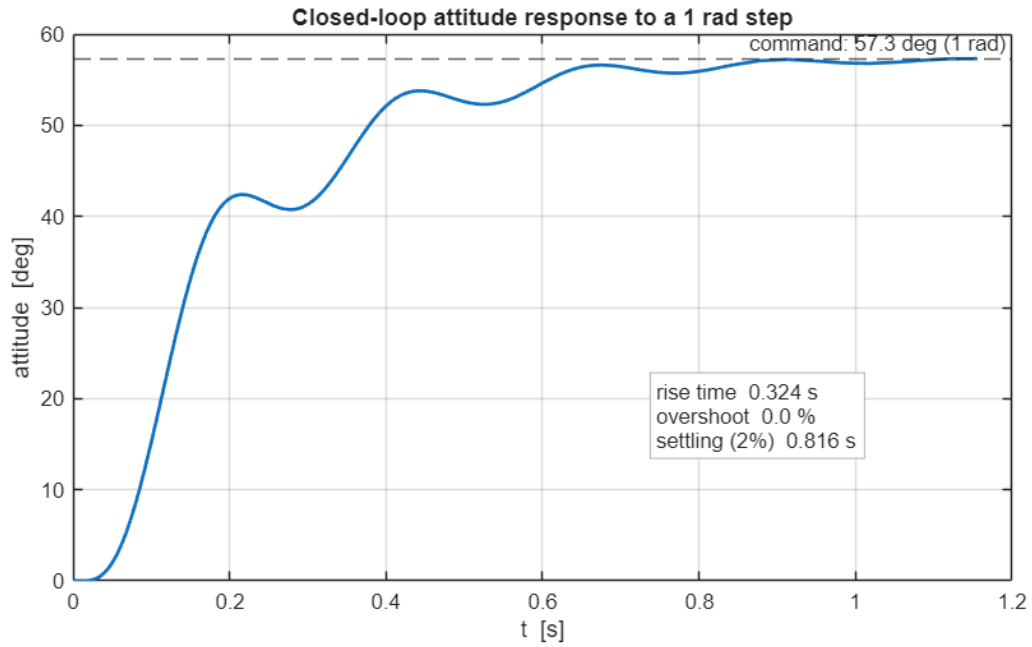
$$\begin{cases} t_r = 0.317\ \mathrm{s} \\ \mathrm{OS} = 0.5\ \% \\ T_s^{2\%} = 0.827\ \mathrm{s} \end{cases}$$

Yaw loop — tuning results

CLOSED-LOOP FREQUENCY RESPONSE — YAW



CLOSED-LOOP STEP RESPONSE — 1 rad COMMAND



STABILITY MARGINS

$$\begin{cases} PM = 82.4^\circ \\ \omega_c = 5 \text{ rad/s} \end{cases}$$

STEP-RESPONSE METRICS

$$\begin{cases} t_r = 0.324 \text{ s} \\ \text{OS} = 0.0 \ \% \\ T_s^{2\%} = 0.816 \text{ s} \end{cases}$$

DESIGN NOTE

– Same 2-DOF sisotool architecture as pitch/roll, shaped independently on the yaw-axis plant (J, D from the yaw row of $D_{\tau\omega}$).

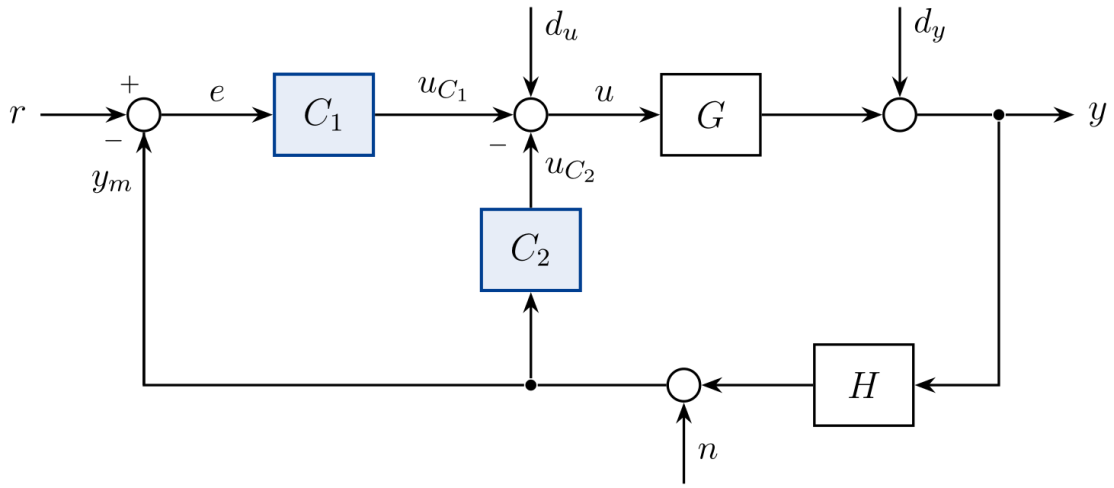
Position loop — cascaded design plant

CASCADED DESIGN PLANT

$$\begin{cases} P_{xy} = T_{att} \cdot P_{trans} \\ P_{trans} = \dfrac{1}{m\,s^2 + D_v\,s} \\ P_z = \dfrac{G_m\,G_d}{m\,s^2} \end{cases}$$

- T_{att}: the position loop is shaped on the cascade (attitude dynamics included), not on the translational plant alone.
- Lateral/longitudinal (x, y) channels use P_{xy}.
- Vertical (z) is driven directly by collective thrust — no attitude cascade in front of it, so P_z is simpler.

TWO-DEGREE-OF-FREEDOM LOOP

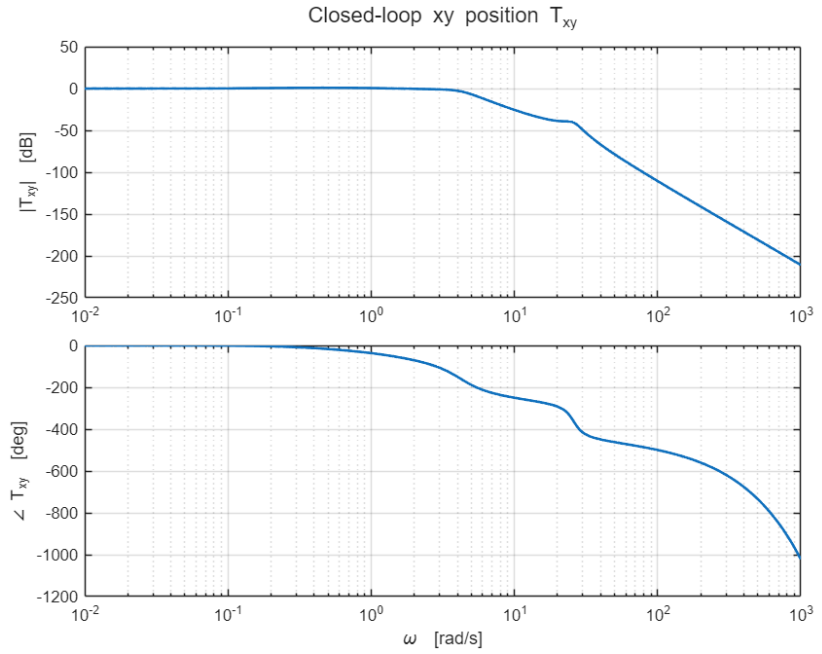


FEEDBACK DESIGN

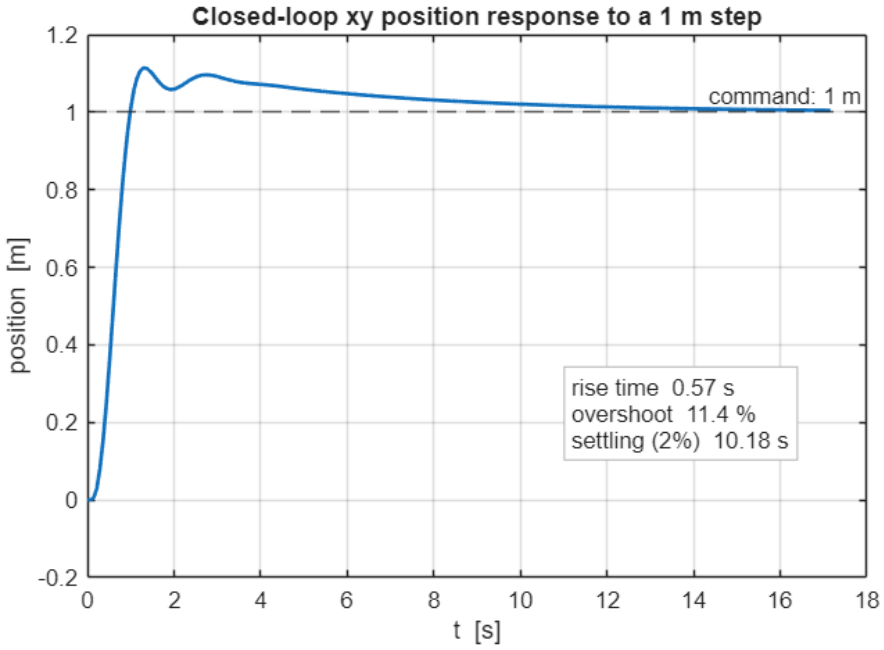
Position feedback gains use proportional–integral feedback on position error plus measured-velocity feedback. For trajectory tracking, the desired velocity enters through a reference feedforward term, so the velocity contribution is proportional to the velocity error.

Position loop — tuning results

CLOSED-LOOP FREQUENCY RESPONSE — XY POSITION



CLOSED-LOOP STEP RESPONSE — 1 m COMMAND



TUNED GAINS

	K_p	K_v	K_i
x, y	8.638	5.2135	1.5747
z (designed)	10.274	4.4458	1.4

RESULTING CONTROLLERS

$$\begin{cases} C_1(s) = K_p + \dfrac{K_i}{s} = 8.638 + \dfrac{1.5747}{s} \\ C_2(s) = K_v\, s = 5.2135\, s \end{cases}$$

STABILITY MARGINS

$$\begin{cases} GM = 9.36 \text{ dB} \ @\ 4.75 \text{ rad/s} \\ PM = 60.5^\circ \ @\ 1.66 \text{ rad/s} \end{cases}$$

STEP-RESPONSE METRICS

$$\begin{cases} t_r = 0.57 \text{ s} \\ \text{OS} = 11.4\ \% \\ T_s^{2\%} = 10.18 \text{ s} \end{cases}$$

Successive loop closure

– The cascade is only valid if the attitude (inner) loop is meaningfully faster than the position (outer) loop — otherwise the position loop's assumption that thrust direction is achieved instantly breaks down.

A Good Practice:

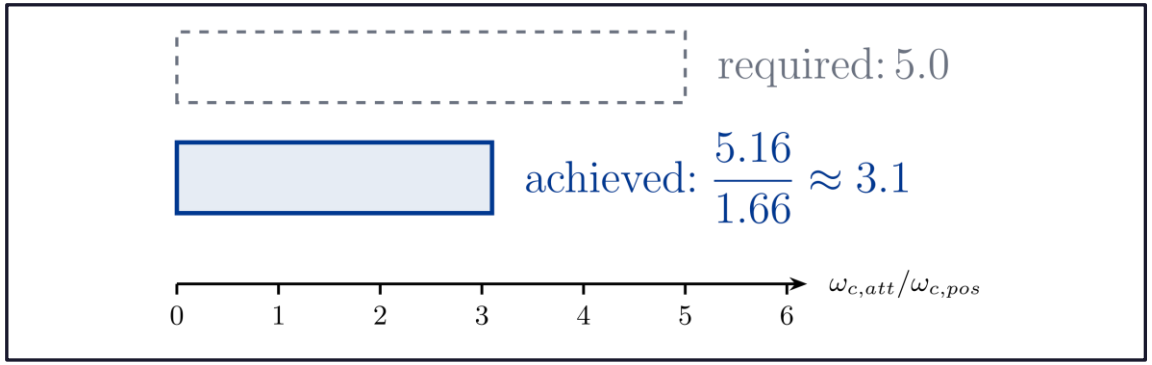
$$\omega_{c,att} \geq 5 \cdot \omega_{c,pos}$$

ACHIEVED CROSSOVER FREQUENCIES

$\omega_{c,att}$ (attitude loop)
$5.16\,\mathrm{rad/s}$

$\omega_{c,pos}$ (position loop)
$1.66\,\mathrm{rad/s}$

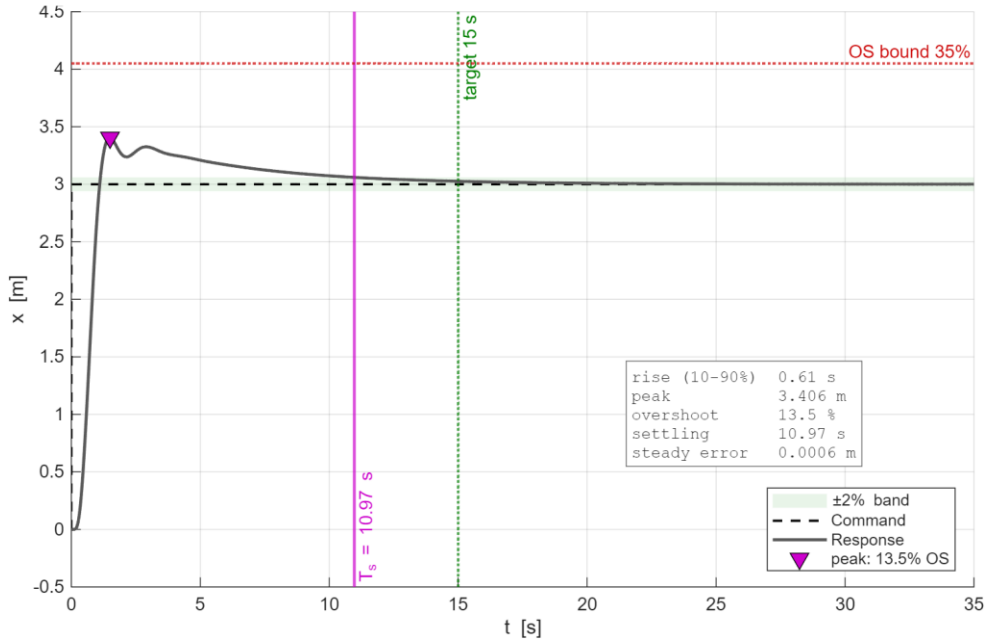
CROSSOVER-FREQUENCY SEPARATION



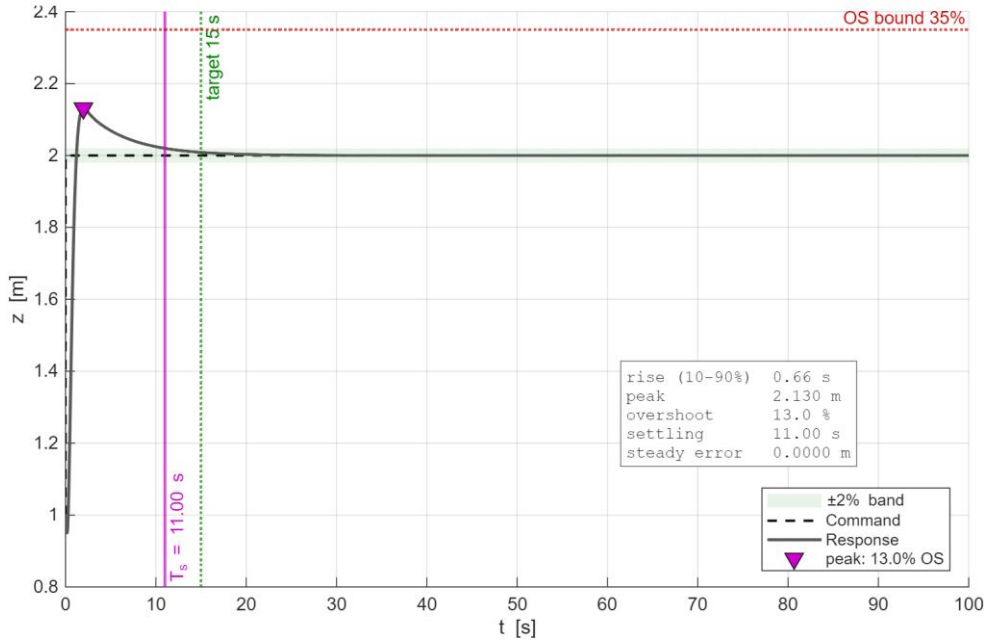
The 5× separation requirement is not met (achieved ≈ 3.1×). However, the attitude loop settles much faster than the position loop, and the controller still delivers the required performance.

Setpoint tracking — lateral & vertical step (nonlinear model)

LATERAL STEP — 3 m IN x (NONLINEAR MODEL)



VERTICAL STEP — 1 m IN z (NONLINEAR MODEL)



RESULTS

$$\begin{cases} T_s = 10.97 \text{ s}, \quad \text{OS} = 13.5\ \% & \text{(lateral)} \\ T_s = 11.00 \text{ s}, \quad \text{OS} = 13.1\ \% & \text{(vertical)} \end{cases}$$

DESIGN TARGETS

$$\begin{cases} \text{OS} \leq 35\ \% \\ T_s \leq 15 \text{ s} \end{cases}$$

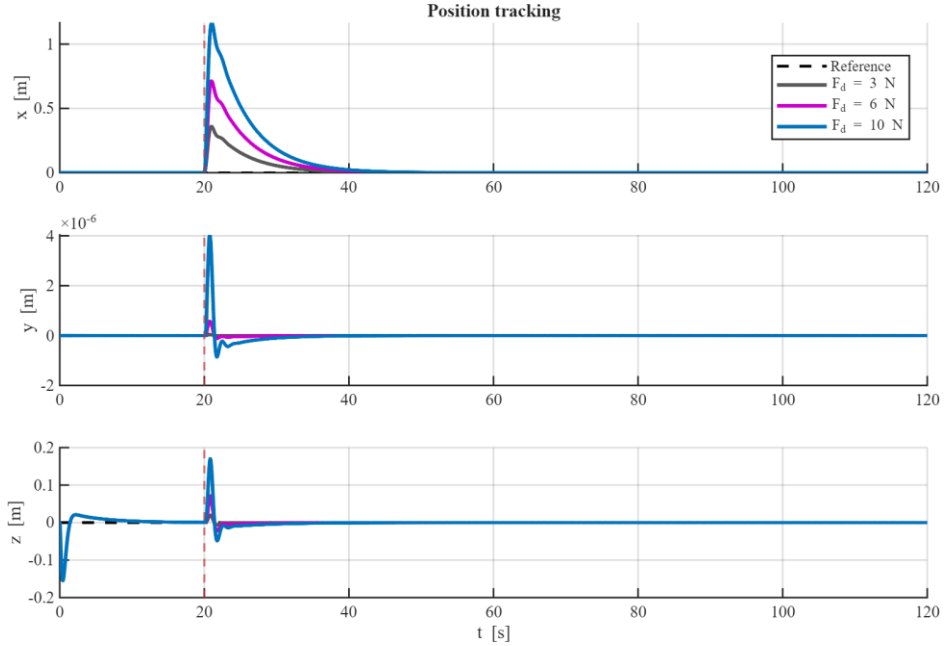
NOTE

– Same objectives the linear design was shaped for, now checked on the full nonlinear model.

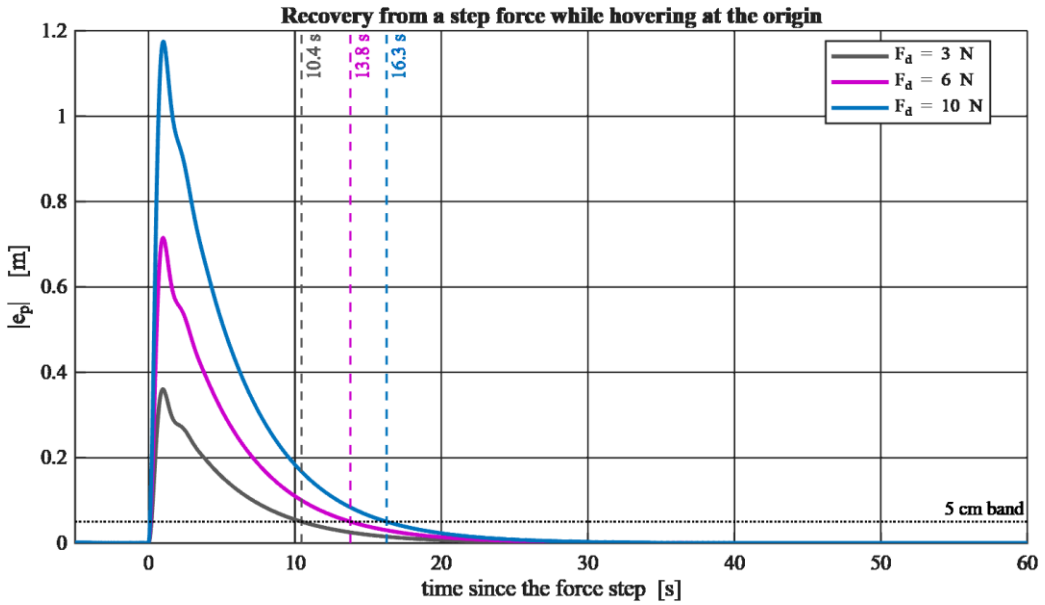
– Lateral (x) goes through the attitude cascade; vertical (z) is driven by the collective thrust directly.

Disturbance rejection

POSITION TRACKING UNDER FORCE STEPS (3, 6, 10 N)



RECOVERY FROM A STEP FORCE — HOVER AT THE ORIGIN



TRACKING ERROR UNDER THE FORCE DISTURBANCE

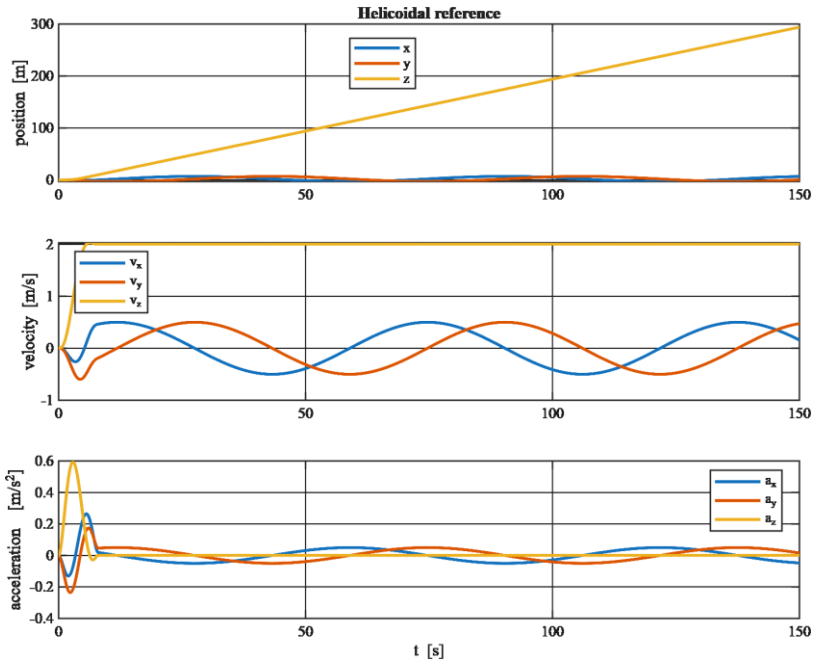
Case	RMS $\|e_p\|$	peak $\|e_p\|$	ss $\|e_p\|$	RMS $\|e_v\|$	peak in window
No disturbance	0.0122	0.1554	0.0071	0.0262	0.0071
With disturbance	0.1576	1.1377	0.0066	0.1165	1.1377
With dist. vs. none	−1194.8 %	−632.1 %	+7.0 %	–	–

DISTURBANCE DETAILS

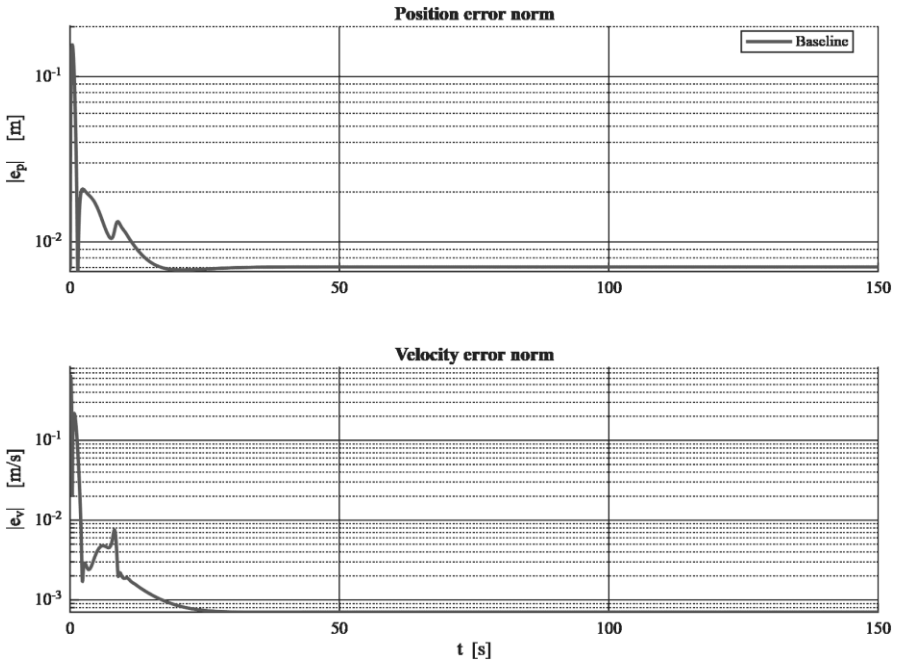
$$\begin{cases} \mathbf{F}_d = [10 \;\; 0 \;\; 0]^\top \text{ N} \;\; \text{(inertial, } x\text{-axis)} \\ \Delta\mathbf{f}_c = [-9.744 \;\; -0.001 \;\; 0.207]^\top \text{ N} \\ \overline{\|e_p\|} = 0.0066 \text{ m} \;\; \text{(undisturbed: } 0.0071 \text{ m)} \end{cases}$$

Trajectory tracking — helicoidal reference

HELICOIDAL REFERENCE — POSITION, VELOCITY, ACCELERATION



TRACKING ERROR NORMS — POSITION AND VELOCITY



HELIX PARAMETERS

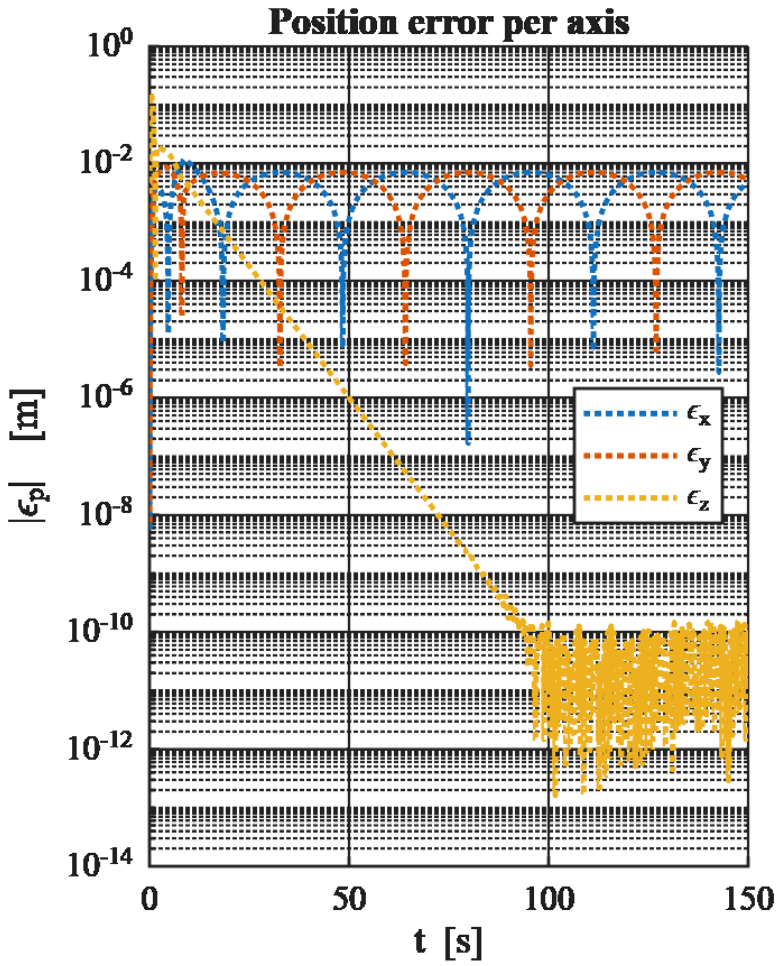
$$\begin{cases} R = 5 \text{ m} \\ \omega = 0.1 \text{ rad/s} \\ v_z = 2 \text{ m/s} \\ \text{approach: } 8 \text{ s} \\ \text{entry: searched} \end{cases}$$

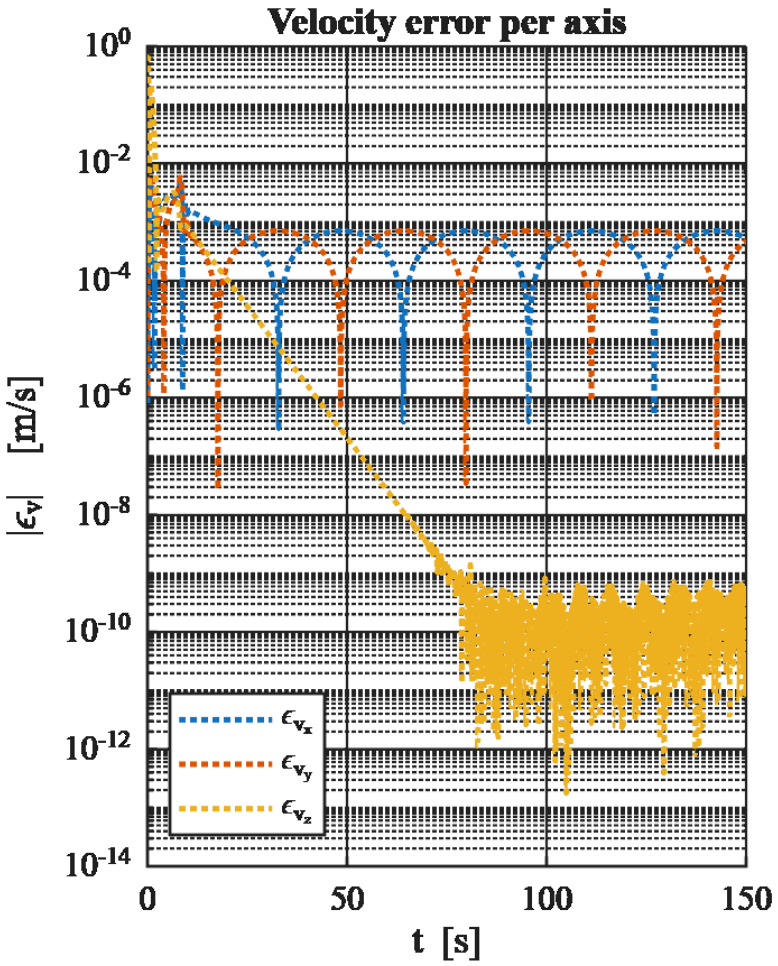
NOMINAL TRACKING ERROR

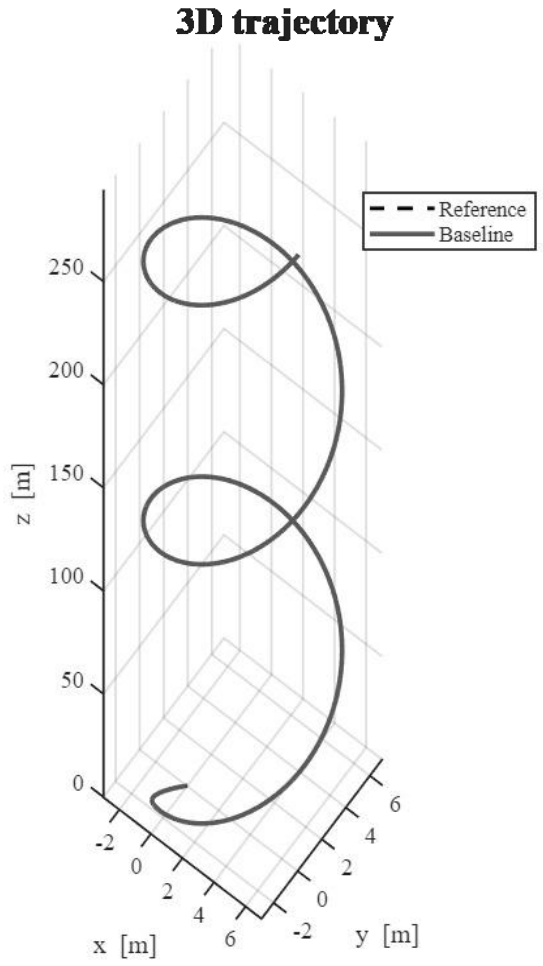
$$\begin{cases} \text{RMS } \|e_p\| = 0.0122 \text{ m} \\ \text{peak } \|e_p\| = 0.1554 \text{ m} \\ \text{ss } \|e_p\| = 0.0071 \text{ m} \\ \text{RMS } \|e_v\| = 0.0262 \text{ m/s} \end{cases}$$

Trajectory tracking — helicoidal reference

PER-AXIS POSITION AND VELOCITY ERRORS AND 3D TRAJECTORY — HELICOIDAL REFERENCE

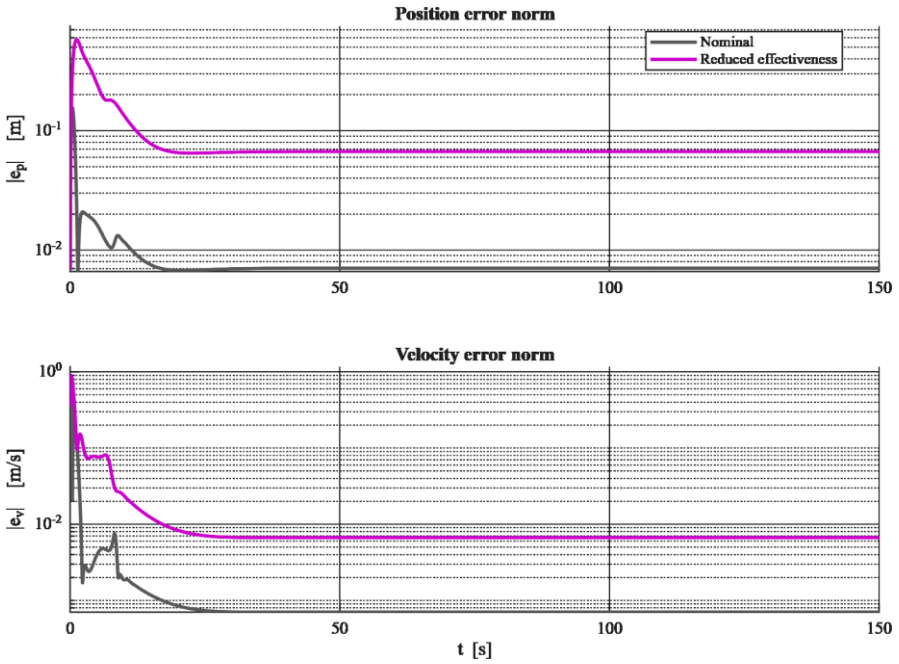






Reduced propeller effectiveness

TRACKING ERROR — NOMINAL VS REDUCED EFFECTIVENESS

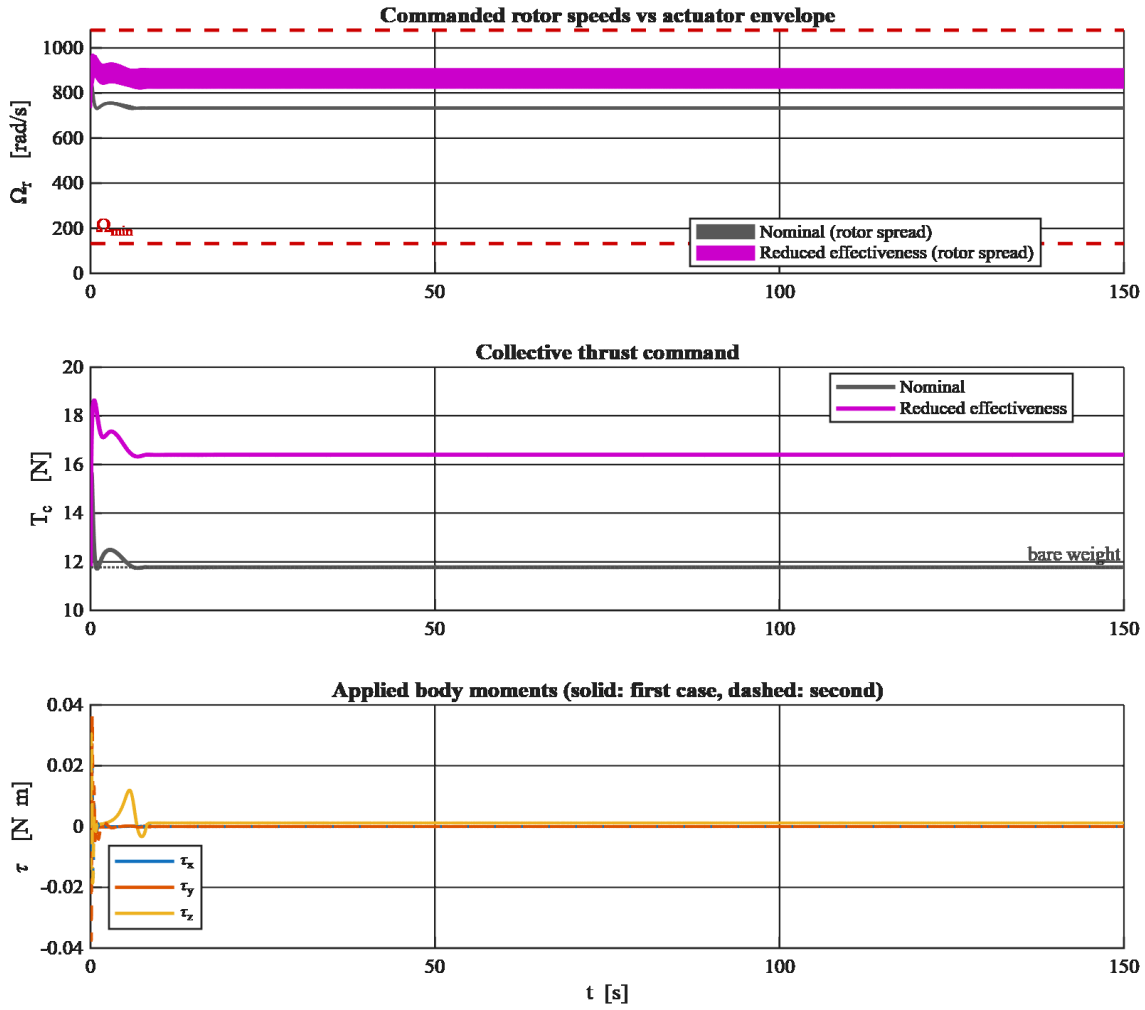


FAULT VECTOR

$$\boldsymbol{\lambda}_r = \begin{bmatrix} 0.8 & 0.6 & 0.75 & 0.78 & 0.5 & 0.9 \end{bmatrix}^\top$$

– Integral action absorbs the constant part but not the reduced authority — this motivates the adaptive augmentation of Task 3.

ACTUATOR RESPONSE — ROTOR SPEEDS, THRUST, BODY MOMENTS



Reduced propeller effectiveness — numerical impact

TRACKING ERROR — HELICOIDAL REFERENCE

Case	RMS $\|e_p\|$	peak $\|e_p\|$	ss $\|e_p\|$	RMS $\|e_v\|$
Nominal	0.0122	0.1554	0.0071	0.0262
Degraded	0.1048	0.5785	0.0669	0.0553
Degraded vs. nominal	−761.1 %	−272.3 %	−847.6 %	–

Relative change: positive = degraded case tracks better.

ACTUATOR ENVELOPE (LIMITS 132–1079 RAD/S)

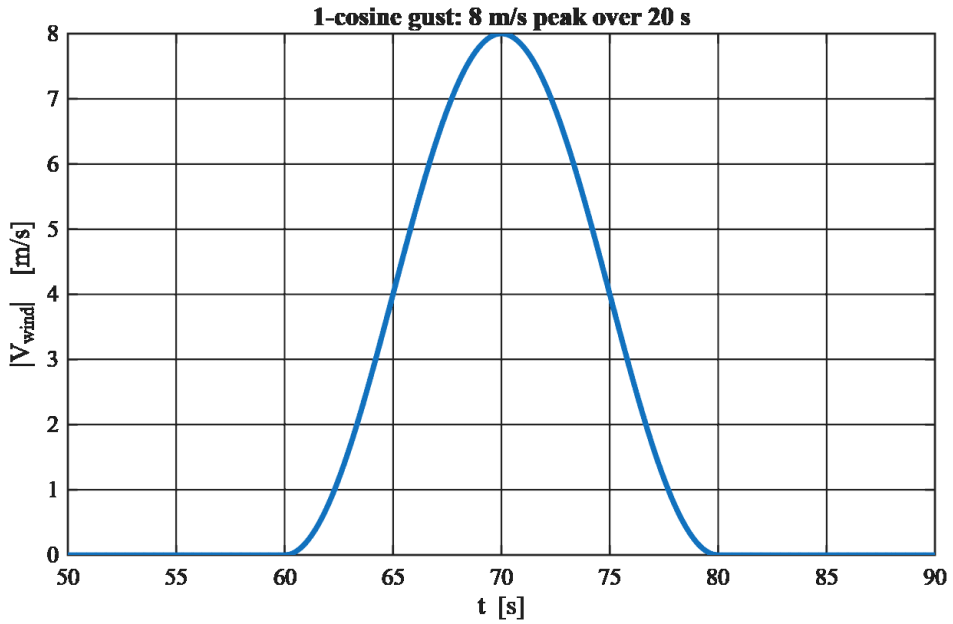
Case	min Ω	max Ω	max spread	saturated
Nominal	730.8	846.4	8.3	0.0 %
Reduced effectiveness	733.2	968.9	120.9	0.0 %

SUMMARY

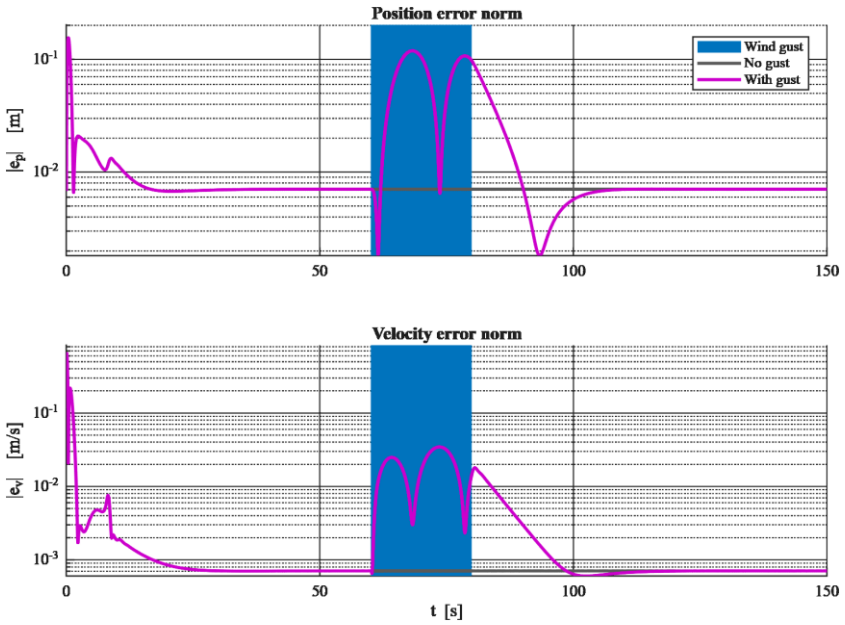
- Fault vector $\lambda_r = [0.8 \;\; 0.6 \;\; 0.75 \;\; 0.78 \;\; 0.5 \;\; 0.9]^\top$.
- RMS position error grows from 0.0122 m to 0.1048 m; steady-state error from 0.0071 m to 0.0669 m.
- No rotor saturates (0.0 %): max Ω rises from 846.4 to 968.9 rad/s, inside the 132−1079 rad/s envelope.
- Rotor-speed spread grows from 8.3 to 120.9 rad/s as the controller compensates the asymmetry.

Wind gust disturbance

GUST VELOCITY PROFILE — 1-COSINE, 8 m/s PEAK OVER 20 s



TRACKING ERROR NORMS — WITH AND WITHOUT GUST



GUST MODEL

$$\begin{cases} \text{shape: 1-cosine} \\ V_{\text{peak}} = 8.0 \text{ m/s} \\ \text{direction} = [1 \;\; 0 \;\; 0] \\ \text{duration} = 20.0 \text{ s}, \quad \text{onset} = 60.0 \text{ s} \end{cases}$$

TRACKING ERROR WITH AND WITHOUT GUST

Case	RMS $\|e_p\|$	peak $\|e_p\|$	ss $\|e_p\|$	RMS $\|e_v\|$	peak in window
No gust	0.0122	0.1554	0.0071	0.0262	0.0071
With gust	0.0332	0.1554	0.0065	0.0274	0.1188
With gust vs. no gust	−172.7 %	+0.0 %	+8.4 %	–	–

Disturbances — attitude excursions and actuator demand

ATTITUDE EXCURSIONS

Case	max |roll|	max |pitch|	max $\|\omega\|$	yaw RMS err
No disturbance	1.71 °	1.86 °	0.642	1.49 °
With force disturbance (10 N)	55.07 °	39.49 °	2.673	7.86 °
No gust	1.71 °	1.86 °	0.642	1.49 °
With gust (8 m/s)	5.33 °	8.14 °	0.642	1.51 °

ACTUATOR ENVELOPE (LIMITS 132–1079 RAD/S)

Case	min Ω	max Ω	max spread	saturated
No disturbance	730.8	846.4	8.3	0.0 %
With force disturbance (10 N)	645.0	930.2	185.7	0.0 %
No gust	730.8	846.4	8.3	0.0 %
With gust (8 m/s)	728.0	846.4	8.3	0.0 %

SUMMARY

- Gust: attitude excursions stay below 9 ° and no rotor saturates.
- Force disturbance: excursions reach 55 ° roll / 39 ° pitch, yet rotor speeds stay inside the envelope (0.0 % saturated).
- Actuator limits for all tests: 132–1079 rad/s.

Verdict — design goals vs. achieved

Objective	**Target**	**Achieved**	
Overshoot (lateral / vertical step)	≤ 35 %	13.5 % / 13.1 %	✓
Settling time (lateral / vertical)	≤ 15 s	10.97 s / 11.01 s	✓
Attitude loop margins	adequate margin	$PM = 82.4^\circ$ at 5.16 rad/s	✓
Position loop margins	adequate margin	$GM = 9.36$ dB, $PM = 60.5^\circ$	✓
Successive loop closure	$\omega_{att} \geq 5\,\omega_{pos}$	$5.16/1.66 \approx 3.1$	×
Disturbance rejection	bounded excursion	peak $\|e_p\| = 1.14$ m (10 N force)	✓
Actuator envelope (helix, disturbances)	within $[\Omega_{min}, \Omega_{max}]$	0 % saturated, 635–969 rad/s	✓

All design goals are met except the 5× bandwidth separation between the attitude and position loops (3.1× achieved). Even so, the attitude loop settles much faster than the position loop, and the cascade delivers the required tracking and disturbance-rejection performance.

TASK 3

Adaptive Augmentation

Package addition · Predictor-based MRAC · Statistical validation

POLITECNICO
MILANO 1863

Why add adaptation

HEXACOPTER CARRYING THE PACKAGE



CONTEXT

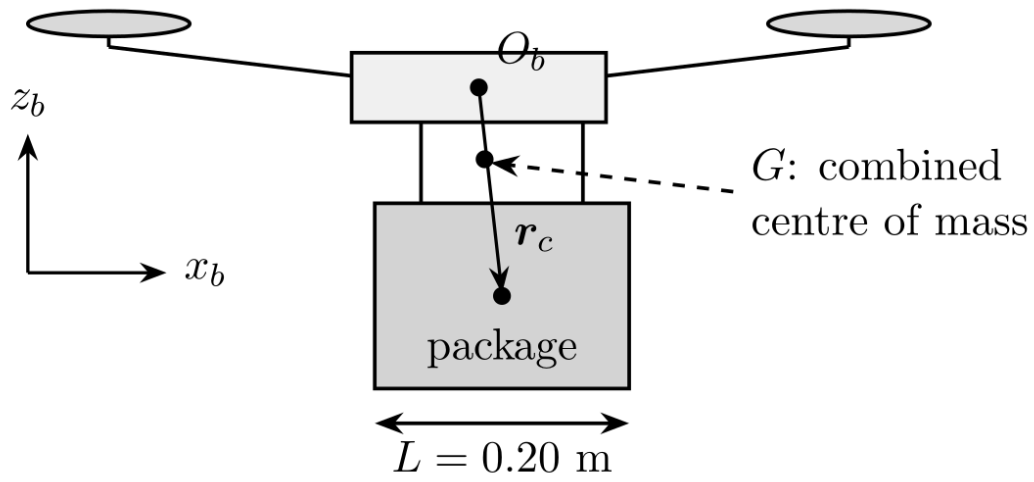
- Nominal vehicle mass: $m_0 = 1.2$ kg.
- Unknown payload changes mass, centre of mass and inertia.
- Rotor degradation changes available thrust and moments.

OBJECTIVE

Objective: recover tracking performance under unknown payload-induced dynamics without retuning the baseline controller.

Model the package as a rigid body

PACKAGE GEOMETRY



PACKAGE RIGID-BODY MODEL

$$\begin{cases} m_{total} = m_0 + m_p \\ \boldsymbol{r}_G = \dfrac{m_0\,\boldsymbol{r}_{G0} + m_p\,\boldsymbol{r}_c}{m_{total}} \\ J_{package,centre} = \dfrac{m_p L^2}{6}\, I_3 \\ J_{total} = J_0 + J_{package,centre} + m_p\left[(\boldsymbol{r}_c^\top \boldsymbol{r}_c) I_3 - \boldsymbol{r}_c \boldsymbol{r}_c^\top\right] \end{cases}$$

PACKAGE PARAMETERS

- Uniform cube, side $L = 0.20$ m.
- Attachment centre $\mathbf{r}_c = [0.02,\ 0.02,\ -0.15]^\top$ m in body coordinates.
- Main deterministic comparison: $m_p = 0.40$ kg.
- Payload sweep: $m_p = 0.5,\ 0.6,\ 0.7$ kg.

NOTES

- Parallel-axis term accounts for the distance between the package centre and the body origin.
- Small x, y offsets also shift the centre of mass — the package changes more than vertical acceleration.

The payload changes the coupled dynamics

INERTIA-COUPLED MASS MATRICES — TRUE PLANT vs CONTROLLER MODEL

True plant $(m_0 + m_p)$

1.60	0	0	0	-0.0600	-0.0080
0	1.60	0	0.0600	0	0.0080
0	0	1.60	0.0080	-0.0080	0
0	0.0600	0.0080	0.0218	-0.0002	0.0012
-0.0600	0	-0.0080	-0.0002	0.0218	0.0012
-0.0080	0.0080	0	0.0012	0.0012	0.0210

Controller model (m_0)

1.20	0	0	0	0	0
0	1.20	0	0	0	0
0	0	1.20	0	0	0
0	0	0	0.0100	0	0
0	0	0	0	0.0100	0
0	0	0	0	0	0.0180

EFFECT OF THE PACKAGE

- The nominal vehicle has $S = 0$.
- The attached package generally makes S nonzero.
- Translation and rotation become coupled about the original body origin.

COUPLED GENERALIZED-VELOCITY EQUATION

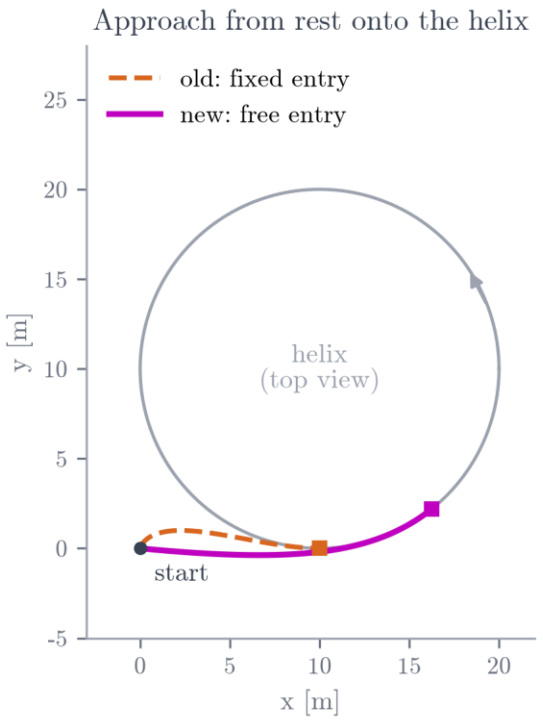
$$\begin{cases} S = m_{total}\,[\boldsymbol{r}_G]^{\times} \\ M = \begin{bmatrix} m_{total} I_3 & S^{\top} \\ S & J_{total} \end{bmatrix} \\ M\dot{\boldsymbol{\nu}} + \boldsymbol{\nu}^{\times} M\boldsymbol{\nu} = \boldsymbol{w}_{gravity} + \boldsymbol{w}_{aero} + \boldsymbol{w}_{control} \end{cases}$$

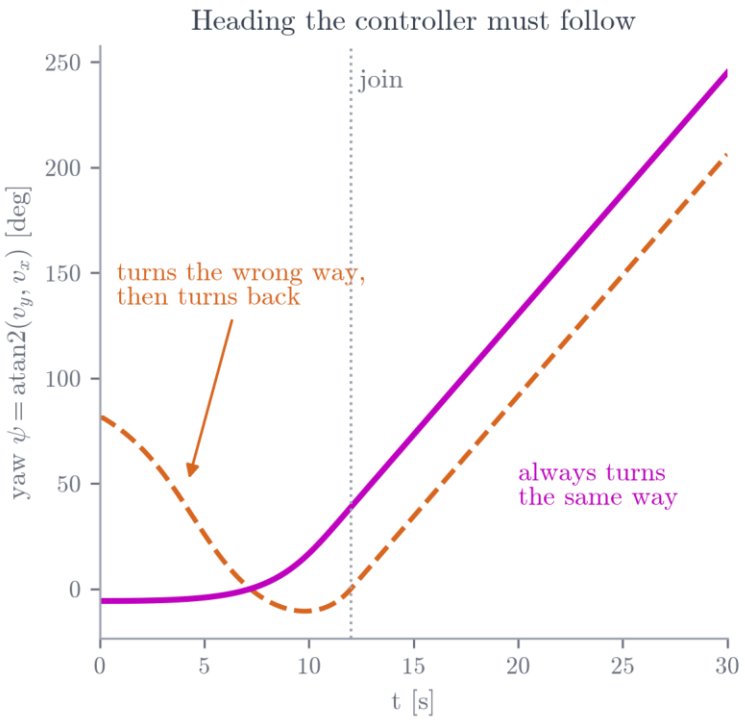
KEY POINT

The plant uses the package parameters; the controller retains nominal parameters.

A fair reference: choosing the helix entry

OLD vs NEW APPROACH — TASK 3 HELIX (R = 10 m, 0.2 rad/s, 12 s APPROACH)





THE PROBLEM

- Yaw follows the velocity: $\psi = \text{atan2}(v_y, v_x)$.
- With a fixed entry phase the approach can swing the heading the wrong way, then snap back at the join, although $\mathbf{p}, \mathbf{v}, \mathbf{a}$ are continuous.
- That yaw transient tests the reference, not the controller under package uncertainty.

THE FIX

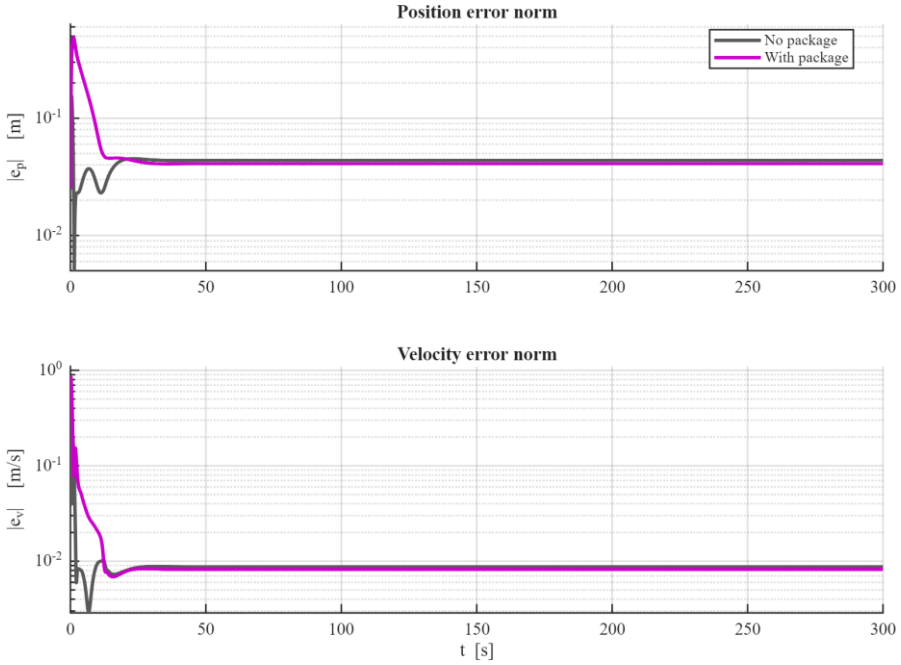
- One 7th-order approach from rest; $\mathbf{p}, \mathbf{v}, \mathbf{a}$ AND jerk matched at the join.
- Entry phase searched (1440 candidates); any phase that reverses the heading is rejected.
- The feasible entry with the least snap is kept (here $\phi_E = -51^\circ$; old fixed entry -90°).

HOW THE ENTRY IS CHOSEN (traj.helix)

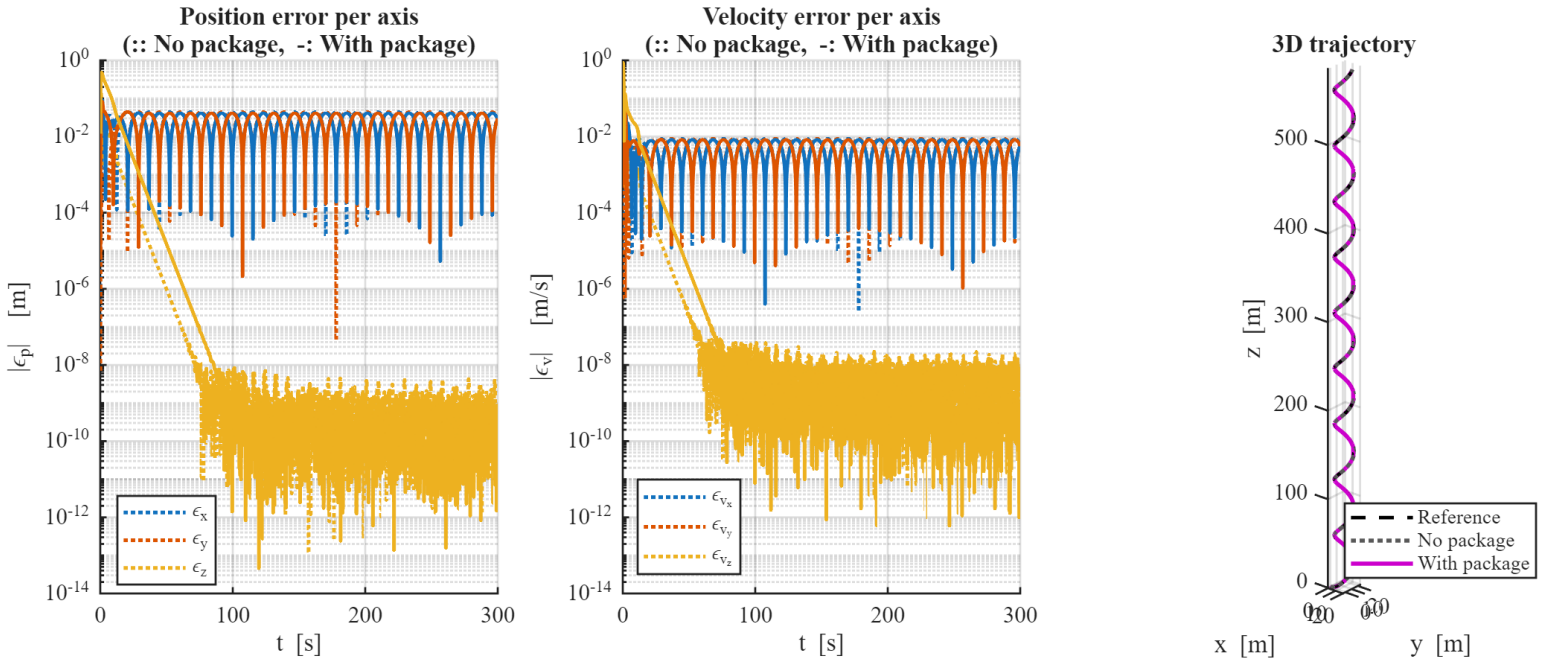
$$\mathbf{p}(t) = \sum_{k=0}^{7} \mathbf{c}_k t^k, \; t \in [0, T_a]; \quad J(\phi) = \int_0^{T_a} \|\mathbf{p}^{(4)}\|^2 dt; \quad \phi_E = \arg\min_{\phi} J(\phi) \;\; \text{s.t.} \;\; \dot{\psi}\,\omega \geq 0 \text{ on } [0, T_a]$$

Establish the baseline payload penalty

ERROR NORMS — NO PACKAGE vs 0.40 kg PACKAGE



PER-AXIS ERRORS AND 3D TRAJECTORY



REFERENCE TRAJECTORY

$0 - 12$ s: minimum-snap approach from rest (slide 5)

$$\mathbf{p}_d = [\,10 + 10\cos(\phi_E + 0.2\tau),\ 10 + 10\sin(\phi_E + 0.2\tau),\ 10 + 2\tau\,]^\top \text{ m}$$

$$\tau = t - 12 \text{ s}, \quad \phi_E = -51^\circ, \quad \|\mathbf{v}_d\| = 2.83 \text{ m/s}$$

CASES

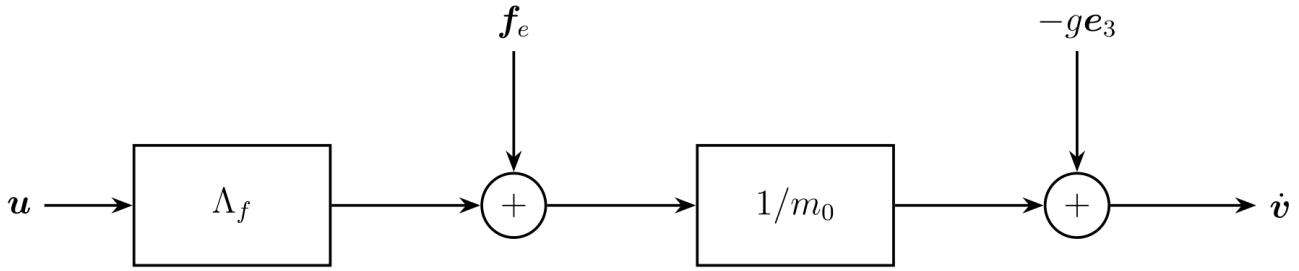
- Case A: no package, PBMRAC off.
- Case B: 0.40 kg package, PBMRAC off.
- Peak error 0.15 $\rightarrow$ 0.49 m; RMS 0.044 $\rightarrow$ 0.062 m.

The uncertain translational model

DESIGN-MODEL APPROXIMATION

$$\begin{cases} \dot{\boldsymbol{p}} = \boldsymbol{v} \\ \dot{\boldsymbol{v}} = -g\boldsymbol{e}_3 + \dfrac{\Lambda_f \boldsymbol{u} + \boldsymbol{f}_e}{m_0} \\ \Lambda_f = \mathrm{diag}(\lambda_x, \lambda_y, \lambda_z) \end{cases}$$

BLOCK DIAGRAM — DESIGN APPROXIMATION



SYMBOLS

Symbol	Meaning
$\boldsymbol{u}$	total requested inertial force (N)
Λ_f	effective translational input gain
$\boldsymbol{f}_e$	matched additive disturbance force (N)
m_0	controller nominal mass (1.2 kg)

WHY ONLY THE POSITION LOOP

- **Time-scale separation:** the attitude loop is faster than the position loop, so the position loop sees the attitude as nearly ideal.
- **Direct actuation:** the attitude is commanded directly by the torque τ_c.
- **Moderate uncertainty:** the package effects on the torque channel are small enough to be absorbed by the robustness of the baseline attitude controller.
- **Consequence:** $\boldsymbol{f}_e$ and Λ_f appear only in the translational model, so the adaptation acts on the position loop only.

The predictor measures model mismatch

PREDICTOR DYNAMICS

$$\begin{cases} \varepsilon_p = \boldsymbol{p} - \hat{\boldsymbol{p}} \qquad \varepsilon_v = \boldsymbol{v} - \hat{\boldsymbol{v}} \\ \dot{\hat{\boldsymbol{p}}} = \hat{\boldsymbol{v}} + L_p\, \varepsilon_p \\ \dot{\hat{\boldsymbol{v}}} = -g\boldsymbol{e}_3 + \dfrac{\hat{\Lambda}_f \boldsymbol{u} + \hat{\boldsymbol{\theta}}}{m_0} + L_v\, \varepsilon_v \end{cases}$$

PREDICTOR DYNAMICS — STATE-SPACE FORM

$$\begin{cases} \dot{\hat{\boldsymbol{x}}}_p = A_p\, \hat{\boldsymbol{x}}_p + B_p\big(\hat{\Lambda}_f\, \boldsymbol{u} + \varphi_p^\top \hat{\boldsymbol{\theta}}_p\big) + \boldsymbol{b}_g + L\, \hat{\boldsymbol{e}}_p \\ \boldsymbol{x}_p = \begin{bmatrix} \boldsymbol{p} \\ \boldsymbol{v} \end{bmatrix}, \quad \hat{\boldsymbol{e}}_p = \boldsymbol{x}_p - \hat{\boldsymbol{x}}_p = \begin{bmatrix} \varepsilon_p \\ \varepsilon_v \end{bmatrix} \\ A_p = \begin{bmatrix} 0 & I_3 \\ 0 & 0 \end{bmatrix}, \quad B_p = \dfrac{1}{m_0} \begin{bmatrix} 0 \\ I_3 \end{bmatrix}, \quad \varphi_p = I_3, \quad \hat{\boldsymbol{\theta}}_p = \hat{\boldsymbol{f}}_e \\ \boldsymbol{b}_g = \begin{bmatrix} 0 \\ -g\, \boldsymbol{e}_3 \end{bmatrix}, \quad L = \begin{bmatrix} L_p I_3 & 0 \\ 0 & L_v I_3 \end{bmatrix} \end{cases}$$

ERROR DEFINITIONS

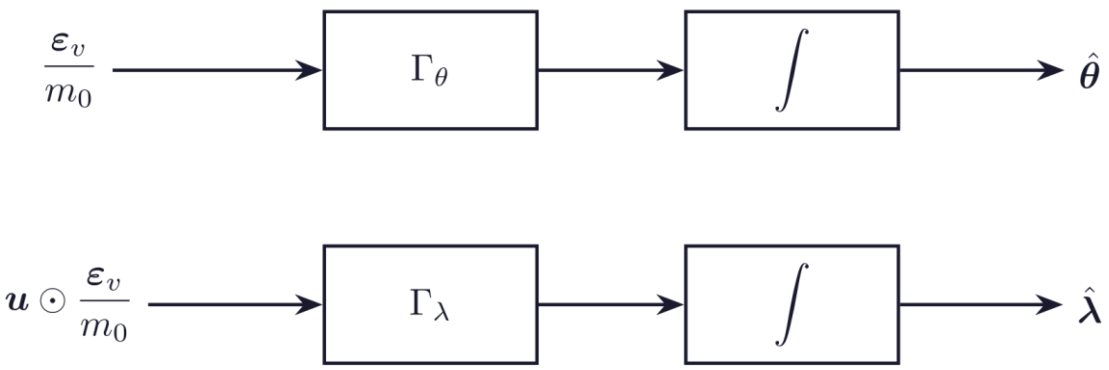
- Tracking error compares actual motion with the desired trajectory.
- Prediction error compares actual motion with the predictor.

PARAMETER UPDATES

- PBMRAC parameter updates use ε_v.
- $\hat{\boldsymbol{\theta}}$ estimates an additive force; $\hat{\Lambda}_f$ estimates effective input gains.

The nominal update laws in Simulink

UPDATE-LAW BLOCKS IN SIMULINK



robust_on $= 1$ adds e-modification to $\hat{\boldsymbol{\theta}}$ and projection to $\hat{\boldsymbol{\lambda}}$

NOMINAL UPDATE LAWS (robust_on = 0)

$$\begin{cases} \dot{\hat{\boldsymbol{\theta}}} = \Gamma_\theta \dfrac{\boldsymbol{\varepsilon}_v}{m_0} \\ \dot{\hat{\lambda}}_i = \gamma_{\lambda,i}\, u_i \dfrac{\varepsilon_{v,i}}{m_0}, \qquad i = x, y, z \end{cases}$$

INITIAL CONDITIONS AND LIMITS

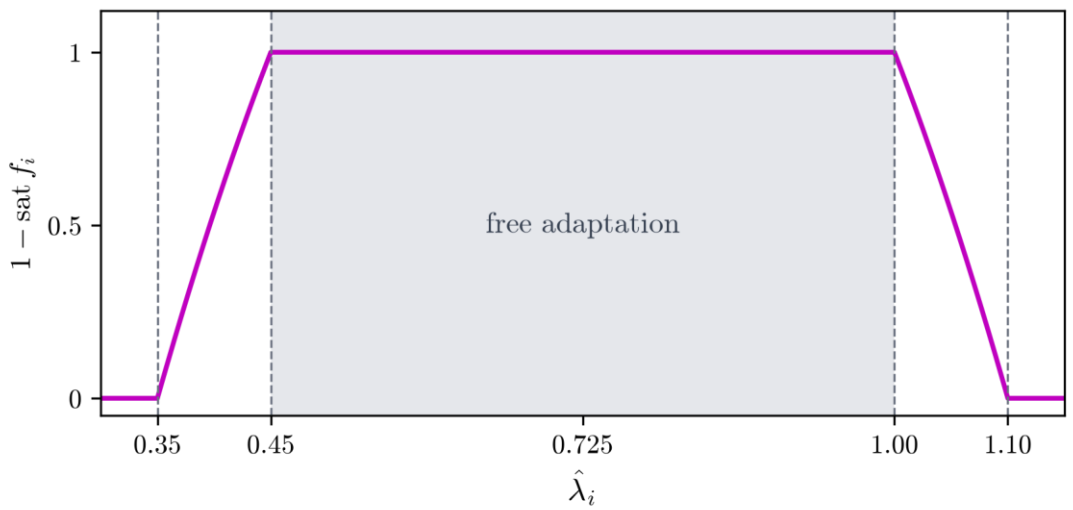
- $\hat{\lambda}_i$ starts at 1, $\hat{\boldsymbol{\theta}}$ at zero.
- The integrators are no longer saturated: the old hard limit $\hat{\lambda}_i \in [0.5, 1]$ is replaced by projection (next slide).

INTERPRETATION

- If measured velocity exceeds the prediction, the force estimate is adjusted in that direction.
- The multiplicative estimate is updated using the input–prediction-error product.

Robust modifications: e-modification and projection

PROJECTION: SCALING OF OUTWARD ADAPTATION



ROBUST UPDATE LAWS (robust_on = 1)

$$\dot{\hat{\boldsymbol{\theta}}} = \Gamma_\theta \left(\varepsilon_v - \sigma_e \, \|\varepsilon_v\| \, \hat{\boldsymbol{\theta}} \right)$$

$$\dot{\hat{\lambda}}_i = \text{Proj}(\hat{\lambda}_i, \; d_i), \qquad d_i = \gamma_{\lambda,\,i} \, u_i \, \varepsilon_{v,\,i}$$

$$f_i = \left((\hat{\lambda}_i - c)^2 - r_0^2 \right) / \left(r_1^2 - r_0^2 \right)$$

$$\text{Proj} = (1 - \text{sat}_{[0,\,1]} \, f_i) \, d_i \quad \text{if } f_i > 0, \;\; d_i(\hat{\lambda}_i - c) > 0$$

$$\text{Proj} = d_i \quad \text{otherwise}$$

PARAMETERS

σ_e (e-modification)	0.02
$c, \; r_0, \; r_1$	0.725, 0.275, 0.375
Free adaptation	$\hat{\lambda}_i \in [0.45, \; 1.00]$
Hard wall	$\hat{\lambda}_i \in [0.35, \; 1.10]$
Γ_θ	diag(144, 144, 57.6)
γ_λ	0.75, 0.75, 1.5

WHY THESE TERMS

- e-modification damps $\hat{\boldsymbol{\theta}}$ in proportion to $\|\varepsilon_v\|$: it limits drift under disturbances and vanishes as the prediction error goes to zero.
- $\hat{\Lambda}_f$ is inverted in the control law: projection keeps $\hat{\lambda}_i \geq 0.35$, so each inverse gain is ≤ 2.9.
- Selected with robust_on = 1; neither term gives parameter convergence without persistent excitation.

Saturation hedging of the adaptive laws

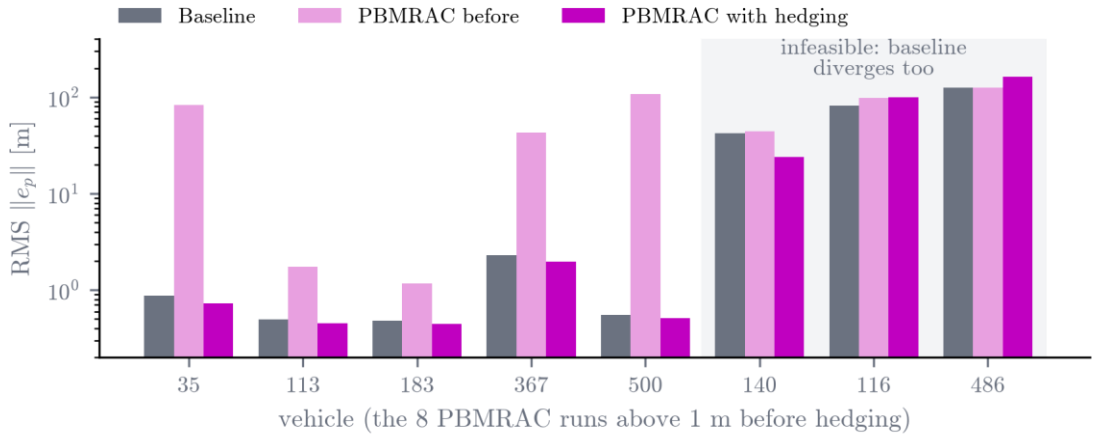
WHY IT IS NEEDED, AND HOW IT WORKS

- Collective capped at $T_{max} = 22.93$ N (attitude planner).
- Using $\mathbf{u}$ beyond the cap, ε_v measures the cap: $\hat{\lambda} \to 0.35$, wind-up.

$$\mathbf{u}_h = H(\mathbf{u}), \qquad H(\mathbf{u}) = \begin{cases} \mathbf{u}, & \|\mathbf{u}\| \le T_{max} \\ T_{max} \dfrac{\mathbf{u}}{\|\mathbf{u}\|}, & \text{otherwise} \end{cases}$$

- $\mathbf{u}_h$ drives the predictor and the $\hat{\theta}$, $\hat{\Lambda}$ laws; the control law still uses $\mathbf{u}$.
- Below the cap $\mathbf{u}_h = \mathbf{u}$: nominal behaviour unchanged.

THE 8 VEHICLES THAT DIVERGED BEFORE



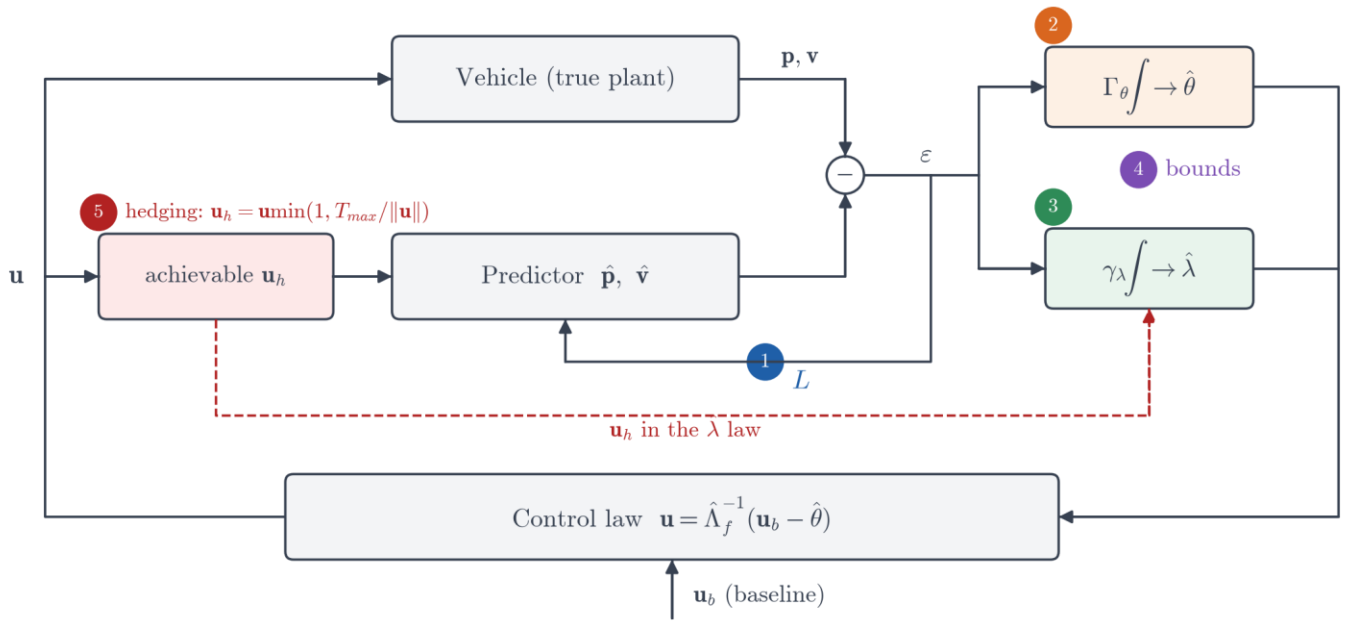
EVIDENCE — MONTE CARLO, SAME 500 VEHICLES

Same 500 vehicles, same gains	Before	After
Median RMS (PBMRAC)	0.069 m	0.069 m
p95 RMS (PBMRAC)	0.257 m	0.203 m
Mean RMS (PBMRAC)	1.091 m	0.660 m
PBMRAC better than baseline	488 / 500	498 / 500
PBMRAC runs $>$ 1 m (baseline: 4)	8	4
Tail vehicles (tilt rule 30°)	13	9
$\hat{\lambda}$ at the 0.35 wall	28	3
Largest PBMRAC tilt	141°	49°

Monte Carlo campaign of Task 3.4 (slides ahead), measured at the previous gain set.

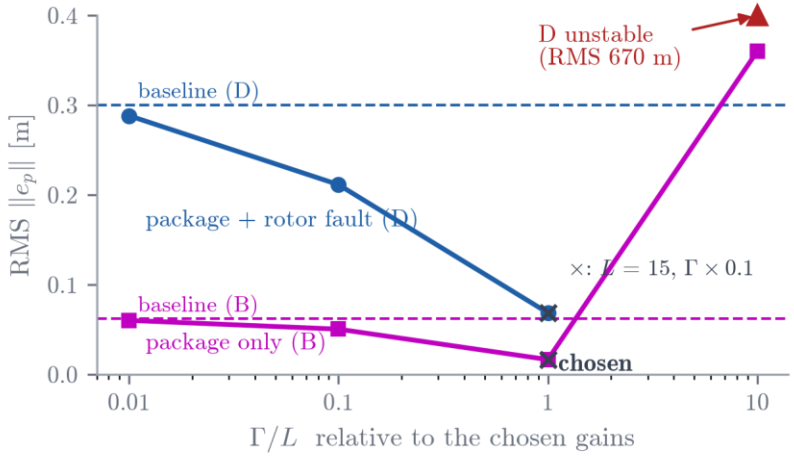
Choosing the adaptive gains

WHERE EACH GAIN ACTS



1. L: how tightly the predictor follows the vehicle
2. Γ_θ: how fast the force estimate $\hat{\theta}$ learns
3. γ_λ: how fast the effectiveness $\hat{\lambda}$ learns
4. σ_e, projection: keep $\hat{\theta}$, $\hat{\lambda}$ bounded

THE TRADE-OFF — MEASURED



- Measured on the Task 3 runs: what matters is Γ/L; $L = 15$ with $\Gamma \times 0.1$ gives the same RMS.
- Ratio $\times 0.1$: the benefit mostly disappears (fault case RMS 0.069 $\to$ 0.211 m). Ratio $\times 10$: B oscillates, D diverges.
- Rule: choose L, then raise Γ until the adaptive force oscillates or the rotors near their limit; back off.

CHOSEN VALUES (TASK 3 RUNS)

$L_p = L_v = 150$ $\quad \Gamma_\theta = \mathrm{diag}(144, 144, 57.6)$ $\quad \gamma_\lambda = [0.75, 0.75, 1.5]$ $\quad \sigma_e = 0.02$ $\quad \hat{\lambda}_i \in [0.35,\ 1.10]$

Derive the adaptive force augmentation

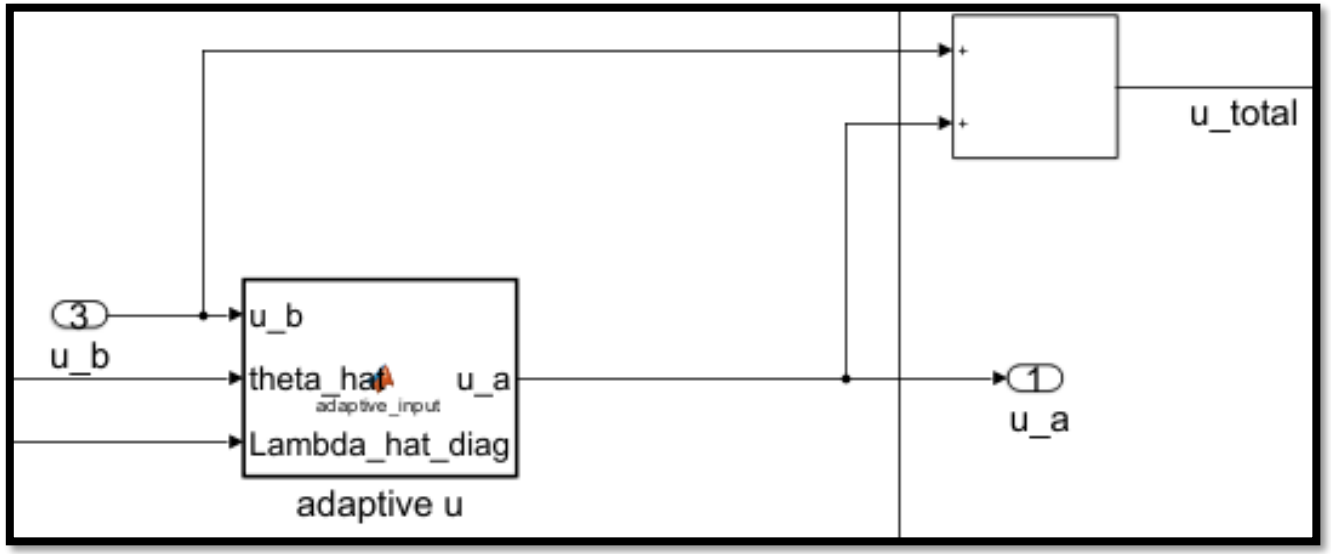
ADAPTIVE FORCE AUGMENTATION

$$\begin{cases} \hat{\Lambda}_f \boldsymbol{u} + \hat{\boldsymbol{\theta}} = \boldsymbol{u}_b \\ \boldsymbol{u} = \hat{\Lambda}_f^{-1}(\boldsymbol{u}_b - \hat{\boldsymbol{\theta}}) \\ \boldsymbol{u}_{ad} = \boldsymbol{u} - \boldsymbol{u}_b = (\hat{\Lambda}_f^{-1} - I_3)\boldsymbol{u}_b - \hat{\Lambda}_f^{-1}\hat{\boldsymbol{\theta}} \end{cases}$$

NOTES

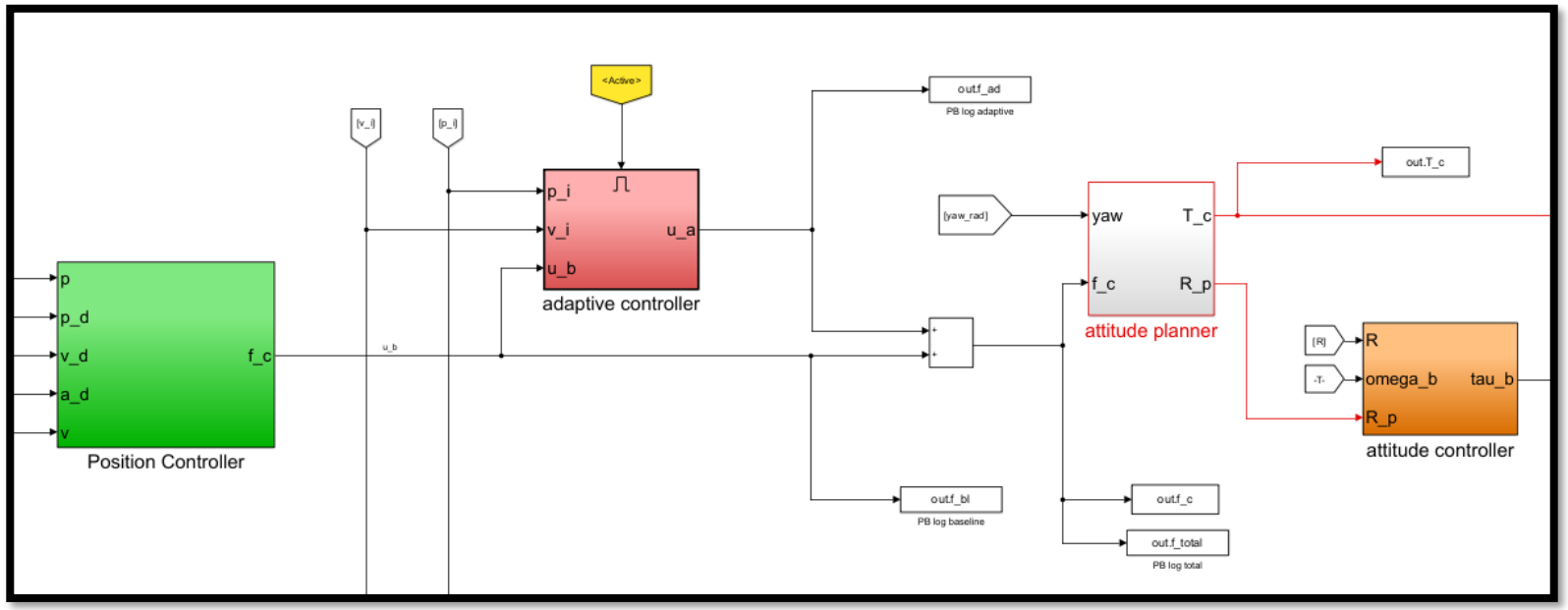
- $\mathbf{u}_b$, $\mathbf{u}_{ad}$, $\mathbf{u}$ are all inertial-force vectors, in N.
- $\hat{\Lambda}_f$ is diagonal, so inversion is componentwise division.
- Projection keeps $\hat{\lambda}_i \geq 0.35$: no division by zero, each inverse gain at most 2.9.

SIMULINK — ADAPTIVE LAWS



Where PBMRAC sits in the control architecture

SIMULINK — CONTROL ARCHITECTURE WITH ADAPTIVE CONTROLLER



SIGNAL FLOW

Position baseline $\rightarrow \boldsymbol{u}_b + \boldsymbol{u}_{ad} \rightarrow$ attitude planner $\rightarrow T_c$ and R_d

R_d and measured attitude $\rightarrow$ attitude controller $\rightarrow \boldsymbol{\tau}_b$

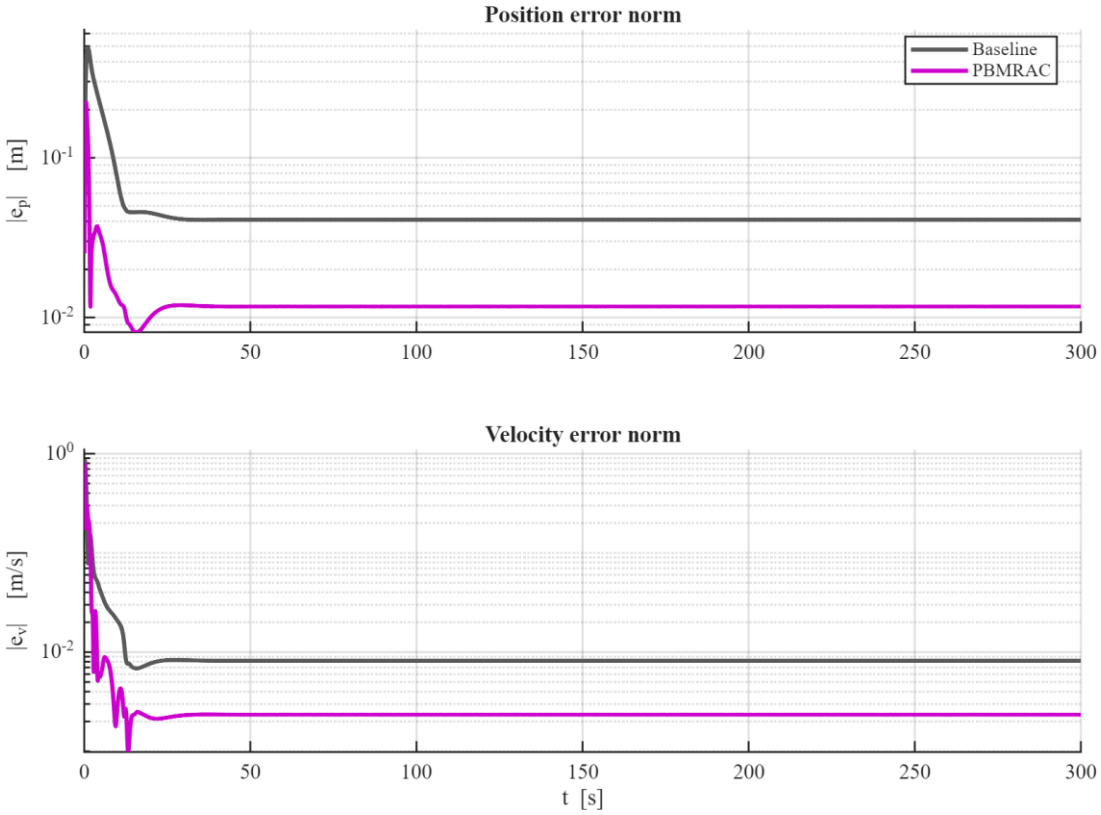
$[T_c; \boldsymbol{\tau}_b] \rightarrow$ rotor allocation $\rightarrow \Omega_{cmd} \rightarrow$ motors $\rightarrow$ nonlinear plant

SIGNALS

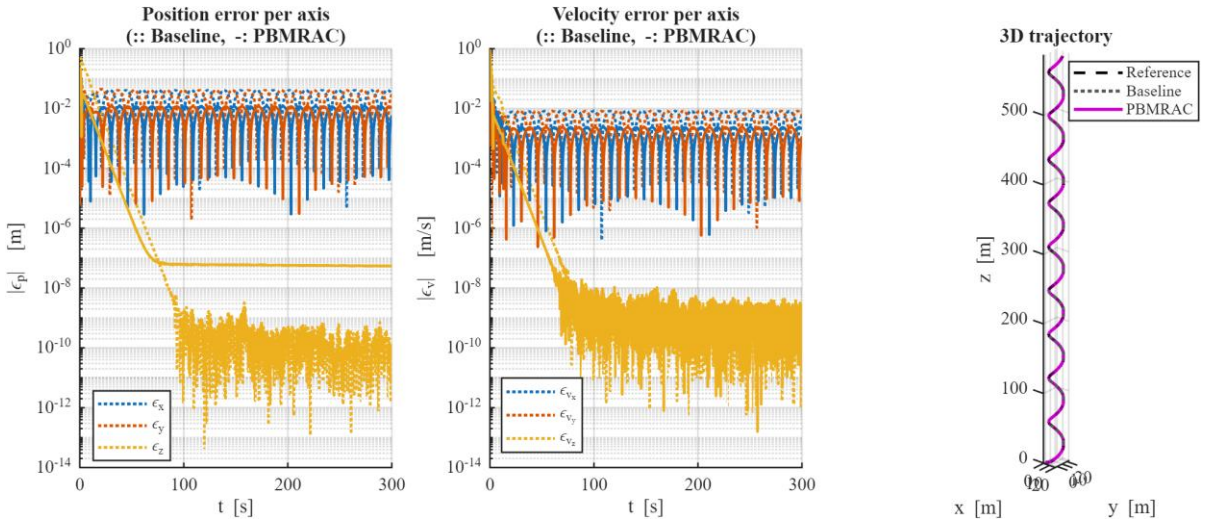
Signal	Meaning
$\boldsymbol{u}$	inertial requested force
R_d	desired body-to-inertial rotation
$\boldsymbol{\tau}_b$	body-axis moments
$[T_c; \boldsymbol{\tau}_b]$	collective thrust and three moments
Ω_{cmd}	rotor angular-speed commands

Tracking improves on the loaded vehicle

POSITION AND VELOCITY ERROR NORMS



PER-AXIS ERRORS AND 3D PATH (:: BASELINE, — PBMRAC)

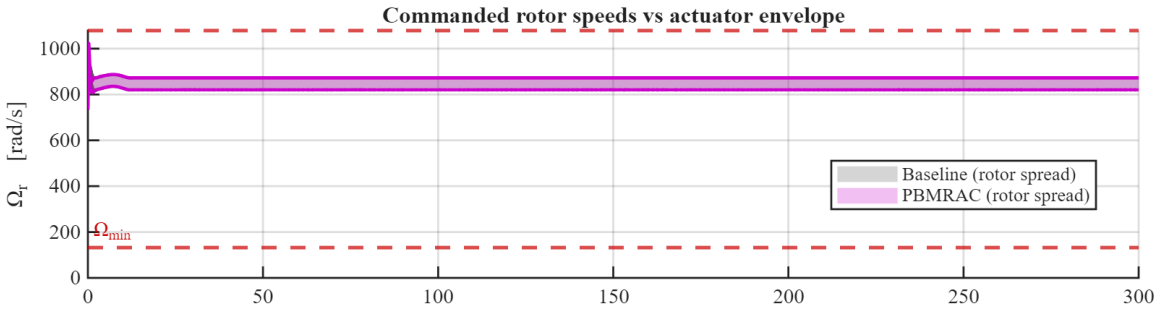


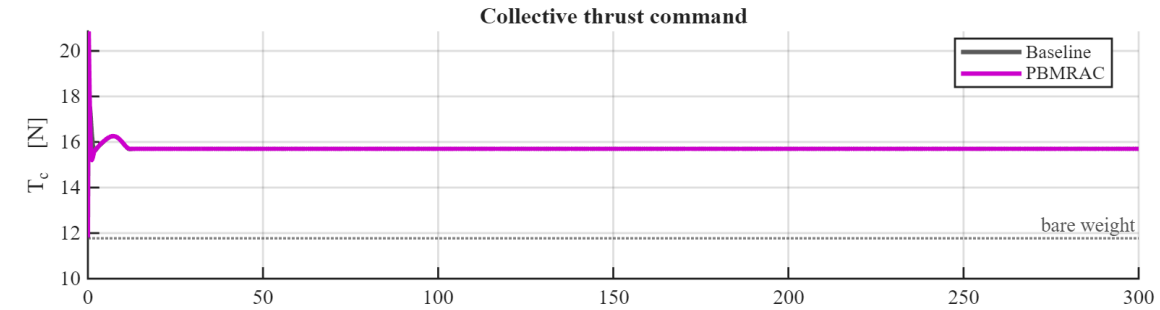
TEST CONDITIONS

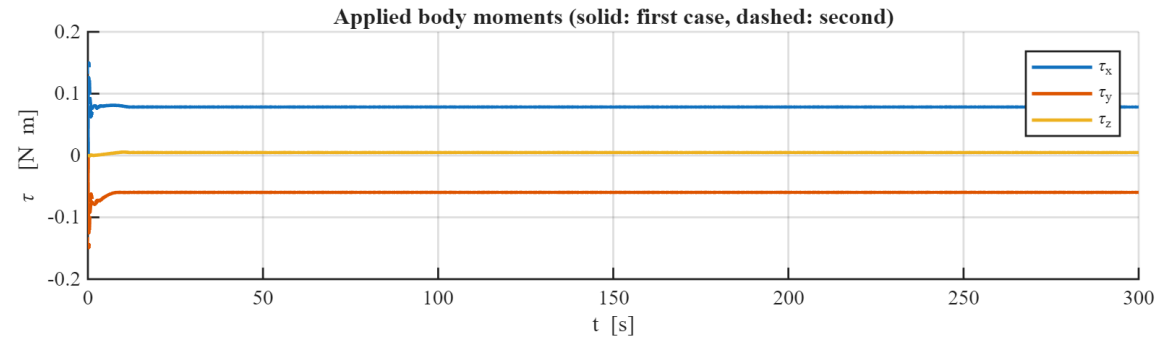
- Same 0.40 kg package, helix and baseline gains; robust PBMRAC.
- Steady error 0.041 $\rightarrow$ 0.012 m (-71 %); RMS 0.062 $\rightarrow$ 0.017 m (-73 %).

Check attitude and actuator demand

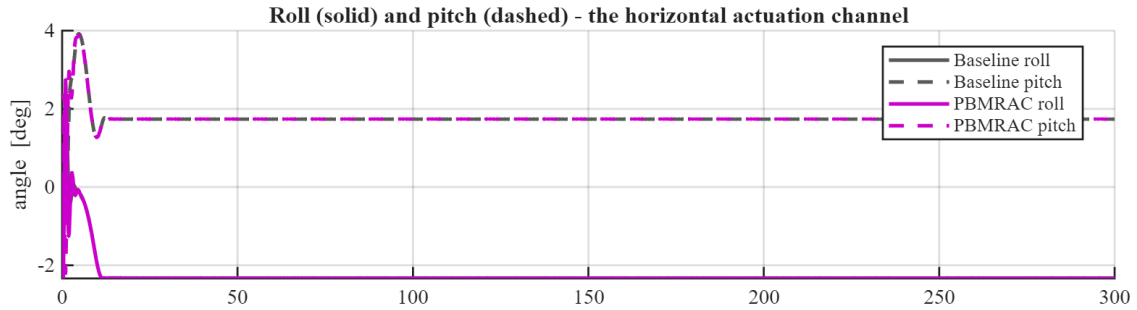
ROTOR SPEEDS, THRUST AND BODY MOMENTS

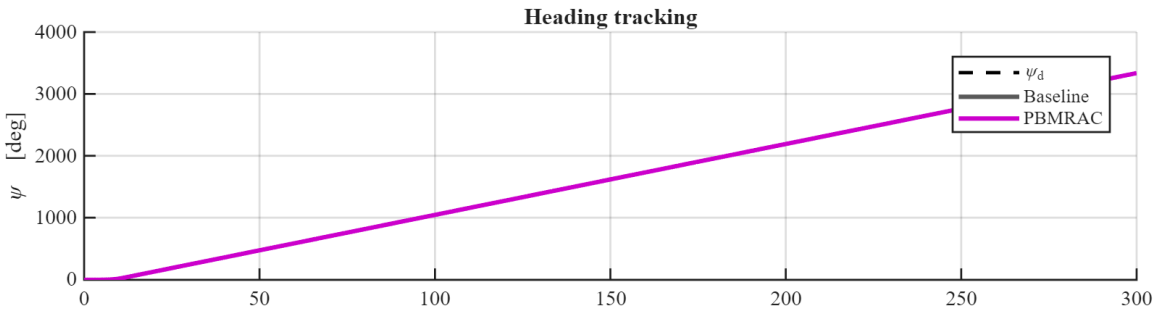


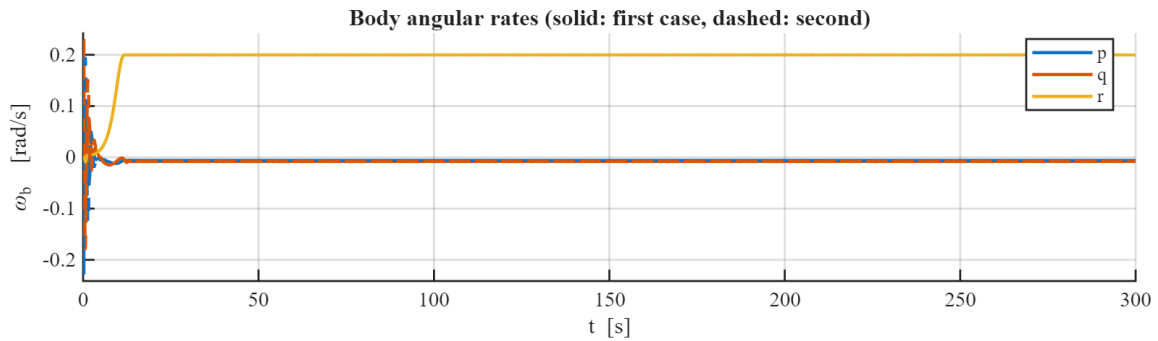




ROLL / PITCH, HEADING AND BODY ANGULAR RATES

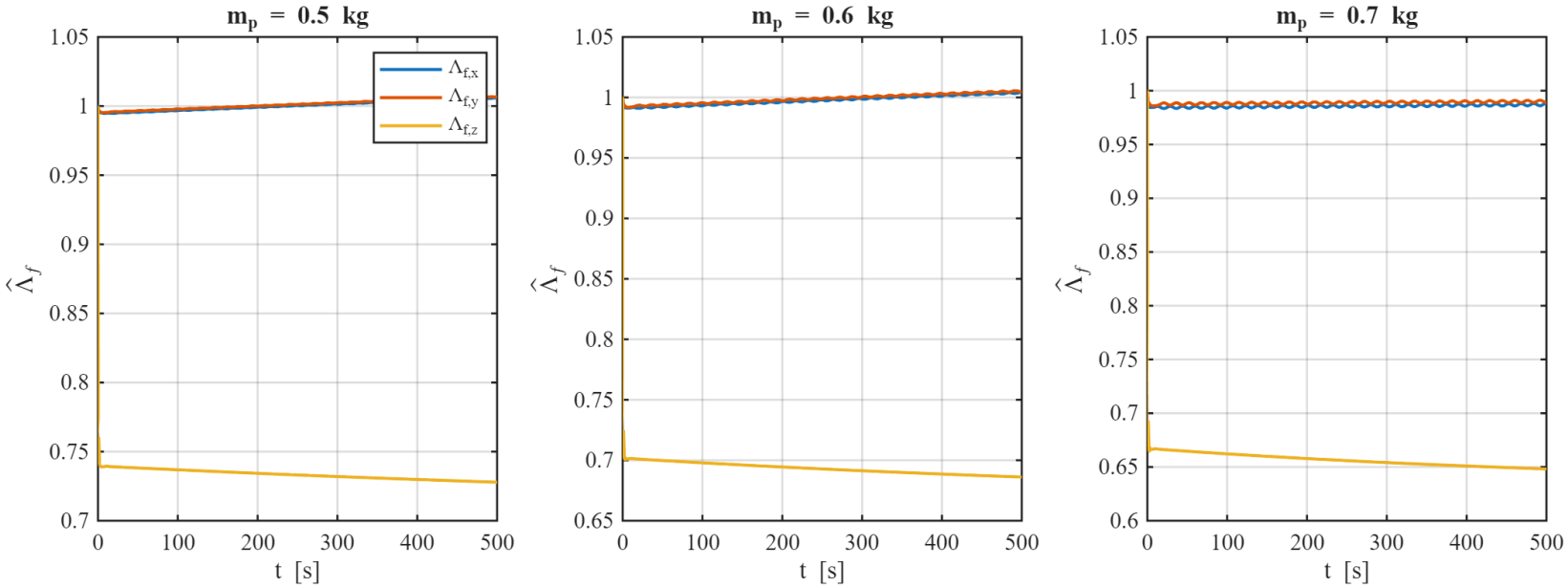






What added mass means for effectiveness

EFFECTIVENESS ESTIMATE vs PAYLOAD MASS (0.5, 0.6 AND 0.7 kg)



IDEAL MASS-ONLY REFERENCE

$$\begin{cases} \dot{\boldsymbol{v}} = -g\boldsymbol{e}_3 + \dfrac{\boldsymbol{u}}{m_0 + m_p} = -g\boldsymbol{e}_3 + \dfrac{\lambda_{mass}\,\boldsymbol{u}}{m_0} \\ \lambda_{mass} = \dfrac{m_0}{m_0 + m_p} \end{cases}$$

IDEAL vs ESTIMATED z-EFFECTIVENESS

Payload	Ideal λ_{mass}	$\hat{\lambda}_z$ at 500 s
0.50 kg	0.706	0.728
0.60 kg	0.667	0.686
0.70 kg	0.632	0.648

Tracking accuracy and identification accuracy differ

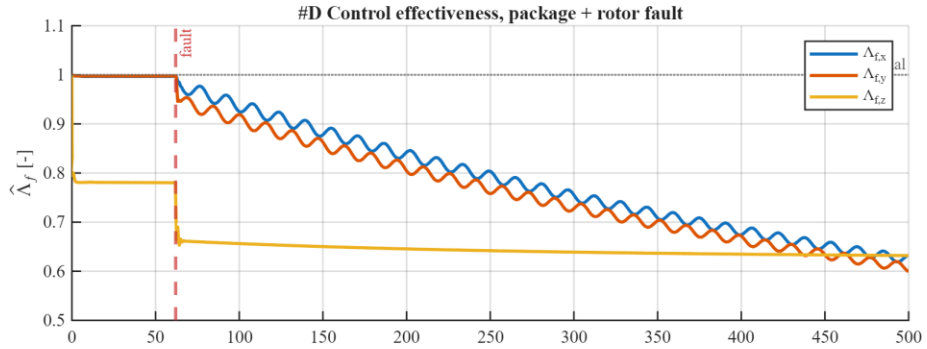
FORCE-CHANNEL AMBIGUITY

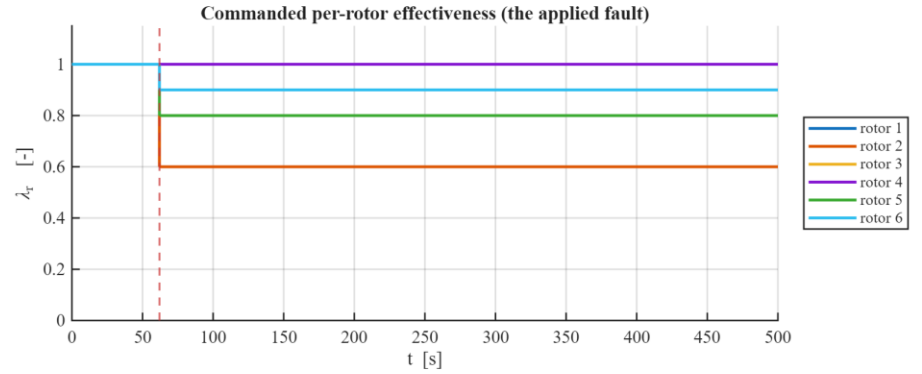
$$\begin{cases} \text{Predictor force} = \hat{\Lambda}_f \boldsymbol{u} + \hat{\boldsymbol{\theta}} \\ \hat{\boldsymbol{\theta}} \approx \boldsymbol{f}_e + (\Lambda_f - \hat{\Lambda}_f)\boldsymbol{u} \end{cases}$$

WHY TRACKING AND IDENTIFICATION DIFFER

- Different parameter combinations produce a similar predicted force.
- After the rotor fault $\hat{\lambda}_x$, $\hat{\lambda}_y$ keep drifting ($1 \rightarrow 0.63$ / 0.60 in 440 s) while the position error stays near 0.06 m.
- Good tracking and small prediction error do not establish physical identification.
- Persistent excitation and an adequate model are needed to separate the effects.

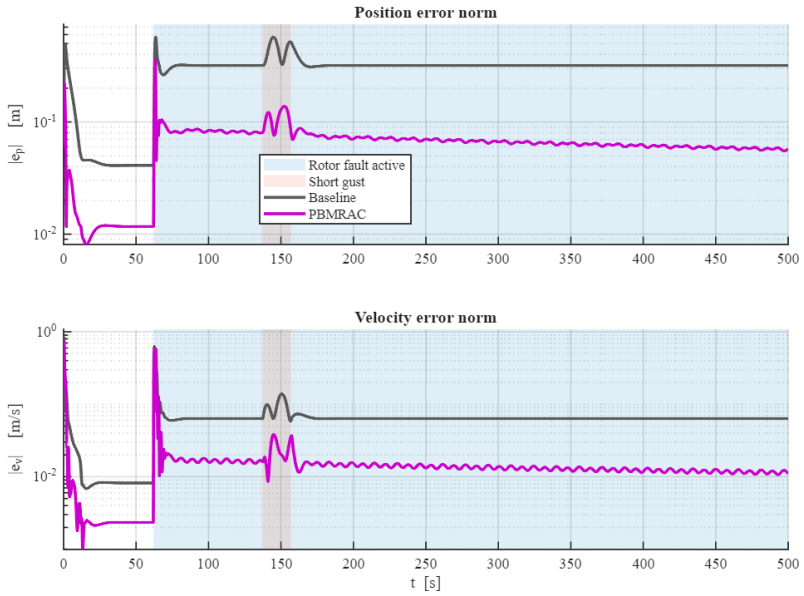
ESTIMATED EFFECTIVENESS — PACKAGE + ROTOR FAULT



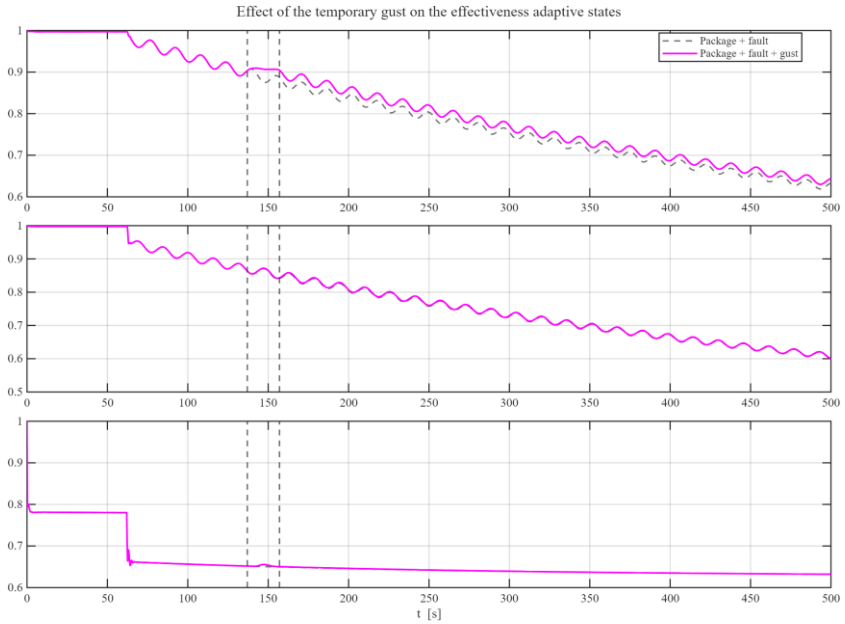


Robustness: rotor fault and a short gust

ERROR NORMS — PACKAGE + FAULT + GUST



EFFECTIVENESS ESTIMATE ACROSS THE GUST (DASHED: NO GUST)



RMS POSITION ERROR

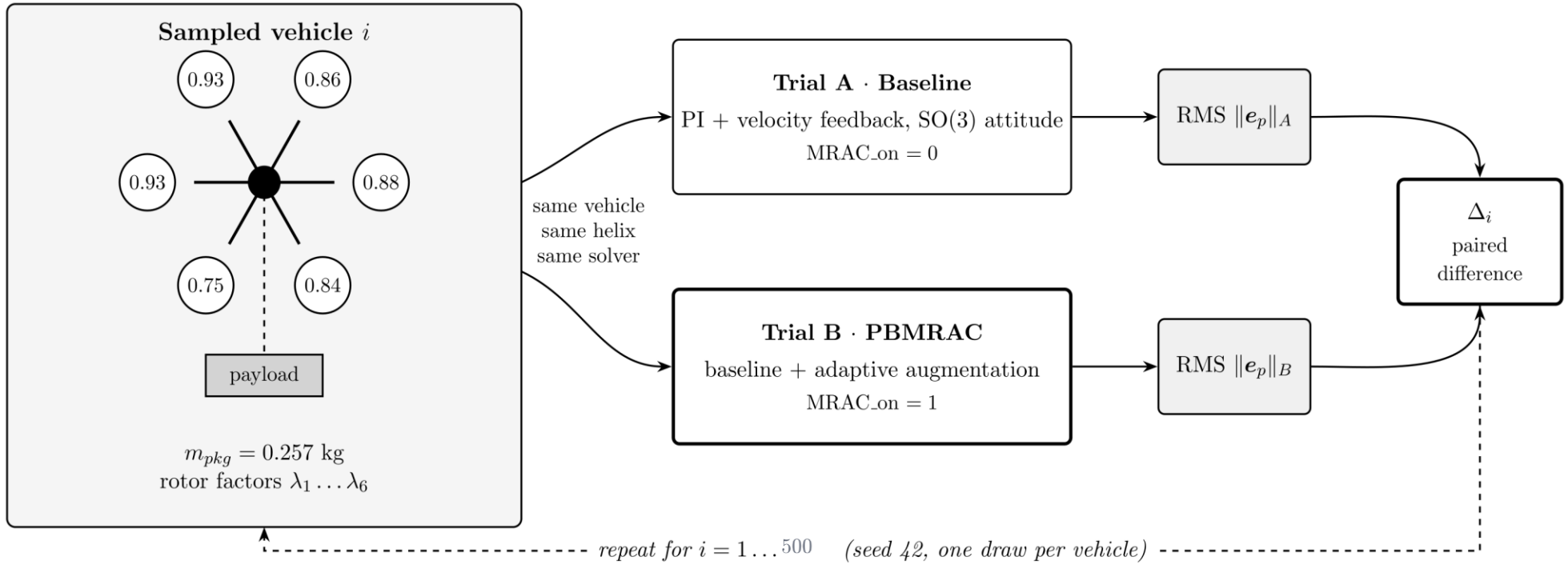
RMS position error	Baseline	PBMRAC
Package (B, 300 s)	0.062 m	0.017 m
+ rotor fault at 62 s (D, 500 s)	0.300 m	0.069 m
+ 10 m/s gust, 137 – 157 s (E)	0.296 m	0.068 m

NOTES

- Fault: $\lambda_r = [0.8,\ 0.6,\ 0.9,\ 1,\ 0.8,\ 0.9]$, persistent.
- The gust does not raise the RMS; peak error inside it: 0.33 m (baseline) vs 0.09 m (PBMRAC).
- The gust moves $\hat{\lambda}_x$, $\hat{\lambda}_y$ by about -0.06 and $\hat{\lambda}_z$ by only -0.007: a small lasting offset in the estimate.

Monte Carlo experiment design

EXPERIMENT FLOW — PAIRED BASELINE / PBMRAC TRIALS ON THE SAME SAMPLED VEHICLE



DESIGN

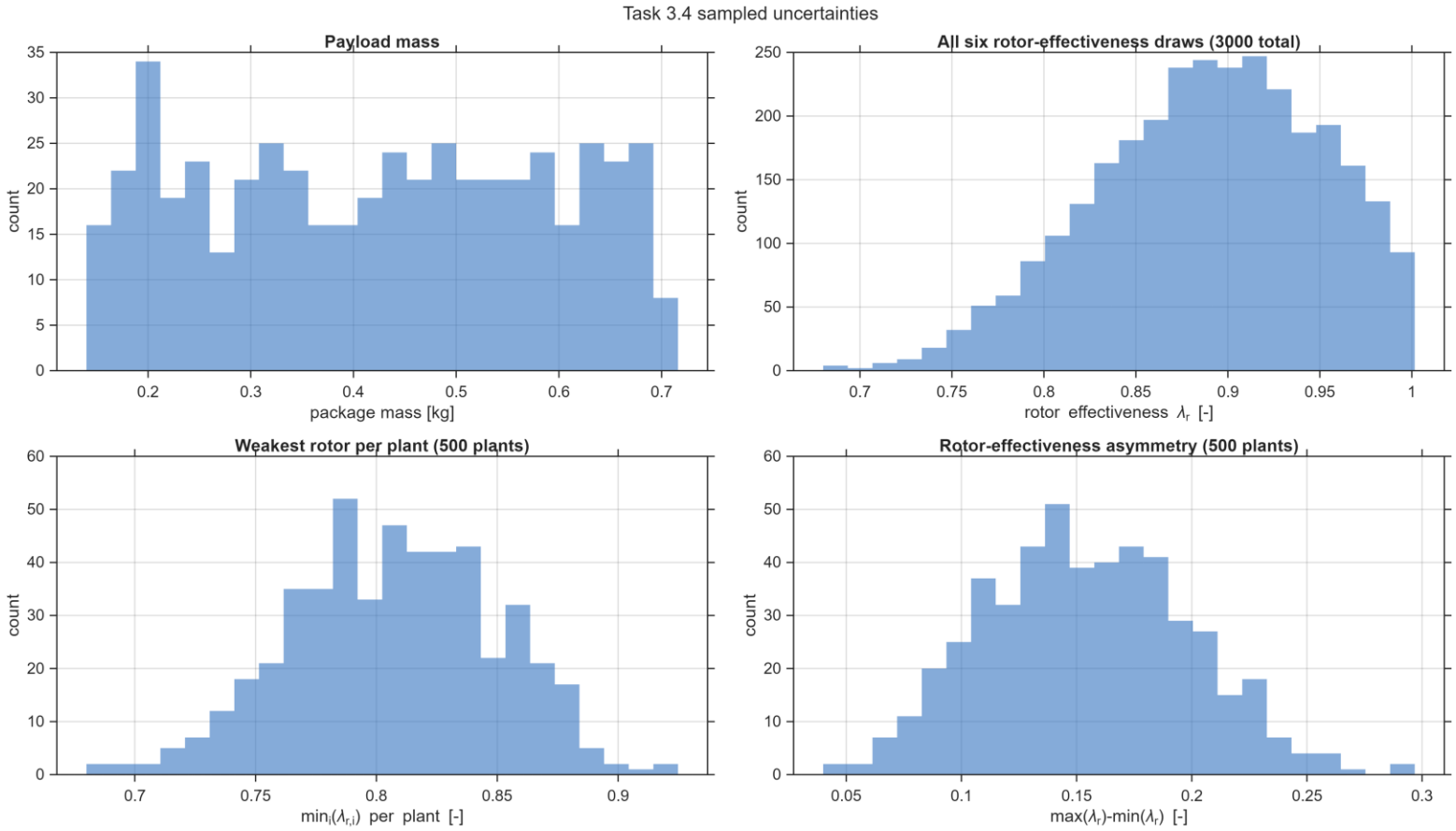
- 500 sampled vehicles; baseline and PBMRAC fly the same one.
- 1000 simulations: 100 s helix with 20 s approach, robust PBMRAC.
- Payload mass and all six rotor efficiencies vary jointly; the package CoM stays at its nominal position.

UNCERTAINTY MODEL

Uncertainty	Distribution	Interval
Payload mass	uniform	[0.15, 0.70] kg
Each rotor factor	$\mathcal{N}(0.90,\ 0.07^2)$, truncated	[0.65, 1.00]
Package CoM	fixed (not sampled)	[0.02, 0.02, −0.15] m

Monte Carlo experiment design

SAMPLED UNCERTAINTIES — 500 VEHICLES



Why 500 runs are adequate for the primary statistics

SAMPLE SIZE FOR A PAIRED PROPORTION

$$
\begin{cases}
X_i = \begin{cases} 1 & \mathrm{RMS}_{\mathrm{PBMRAC}}(i) < \mathrm{RMS}_{\mathrm{Base}}(i) \\ 0 & \text{otherwise} \end{cases} \\[2ex]
\hat{p} = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} X_i, \qquad \mathrm{SE}(\hat{p}) \approx \sqrt{\dfrac{p(1-p)}{N}} \le \dfrac{0.5}{\sqrt{N}} \\[2ex]
N \ge \dfrac{z^2 \cdot 0.25}{\varepsilon^2} = \dfrac{1.96^2 \cdot 0.25}{0.05^2} \approx 385
\end{cases}
$$

ACHIEVED PRECISION — WILSON INTERVAL

$$
\begin{cases}
\hat{p} = \dfrac{k}{N} = \dfrac{498}{500} = 99.6\,\% \\[2ex]
c = \dfrac{\hat{p} + z^2/2N}{1 + z^2/N}, \qquad h = \dfrac{z}{1 + z^2/N}\sqrt{\dfrac{\hat{p}(1-\hat{p})}{N} + \dfrac{z^2}{4N^2}} \\[2ex]
\mathrm{CI}_{95} = [\,c - h,\ c + h\,] \approx [\,98.6\,\%,\ 99.9\,\%\,]
\end{cases}
$$

WHY 500 AND NOT 1000

- The statistical error scales as $1/\sqrt{N}$: going from 500 to 1000 runs reduces it only by about 29 %.
- This costs twice the simulation time and does not change the qualitative conclusion.
- A larger campaign would help mainly to characterise the rare tail events (slides 24–25).

STABILITY AND CONCLUSION

- The requirement of 385 vehicles is conservative: it assumes the worst case $p = 0.5$.
- The running median, p95 and improvement fraction stay stable as samples are added (slide 26).
- The improvement probability is known to about ±1 percentage point, so $N = 500$ is sufficient for Task 3.4.

Under the chosen uncertainty distributions, PBMRAC improves RMS tracking in approximately 99.6% of sampled vehicles.

Define tracking metrics and precision

METRICS

$$
\begin{cases}
\boldsymbol{e}_p(t) = \boldsymbol{p}_d(t) - \boldsymbol{p}(t) \\
RMS_p = \sqrt{\dfrac{1}{N_t} \displaystyle\sum_k \|\boldsymbol{e}_p(t_k)\|^2} \\
Peak_p = \max_k \|\boldsymbol{e}_p(t_k)\| \\
Late_p = \text{mean}\, \|\boldsymbol{e}_p(t_k)\| \quad \text{(final 40\% of run)}
\end{cases}
$$

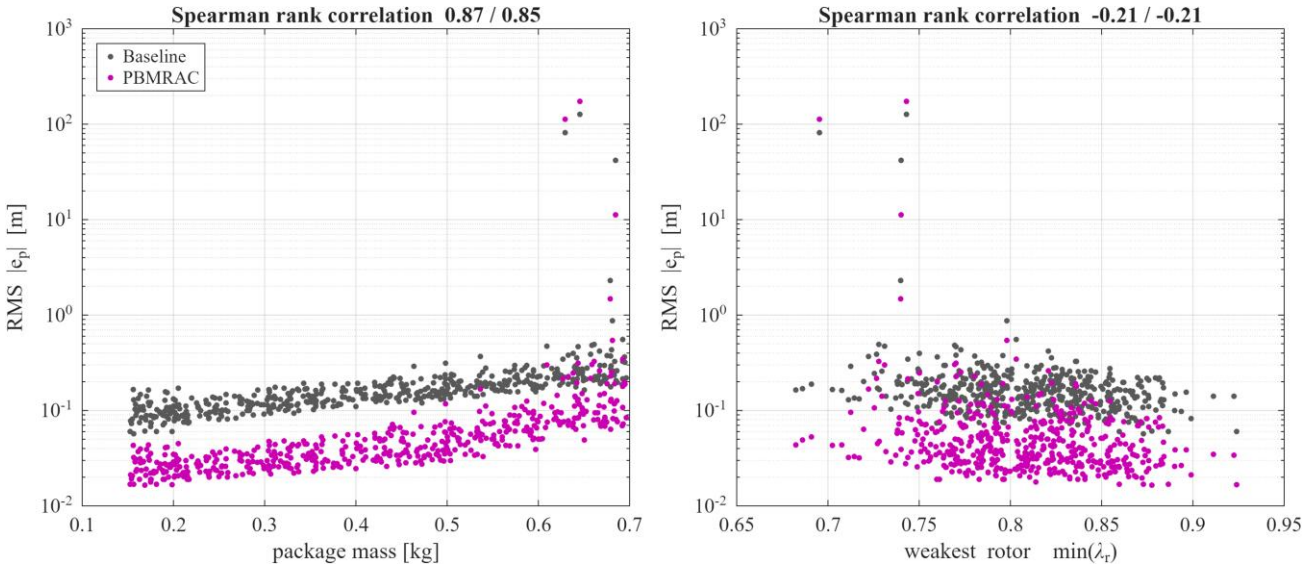
NOTES

- Metrics use 4000 uniformly spaced evaluation points.
- Across-run p95 is the 95th percentile of per-run RMS values.
- A few divergent runs (RMS up to 173 m) dominate the mean, so the median is the headline statistic.

PRECISION

Median: 95 % bootstrap CI, 5000 resamples of the 500 runs

RANK CORRELATION OF RMS WITH MASS AND WEAKEST ROTOR



Monte Carlo tracking results

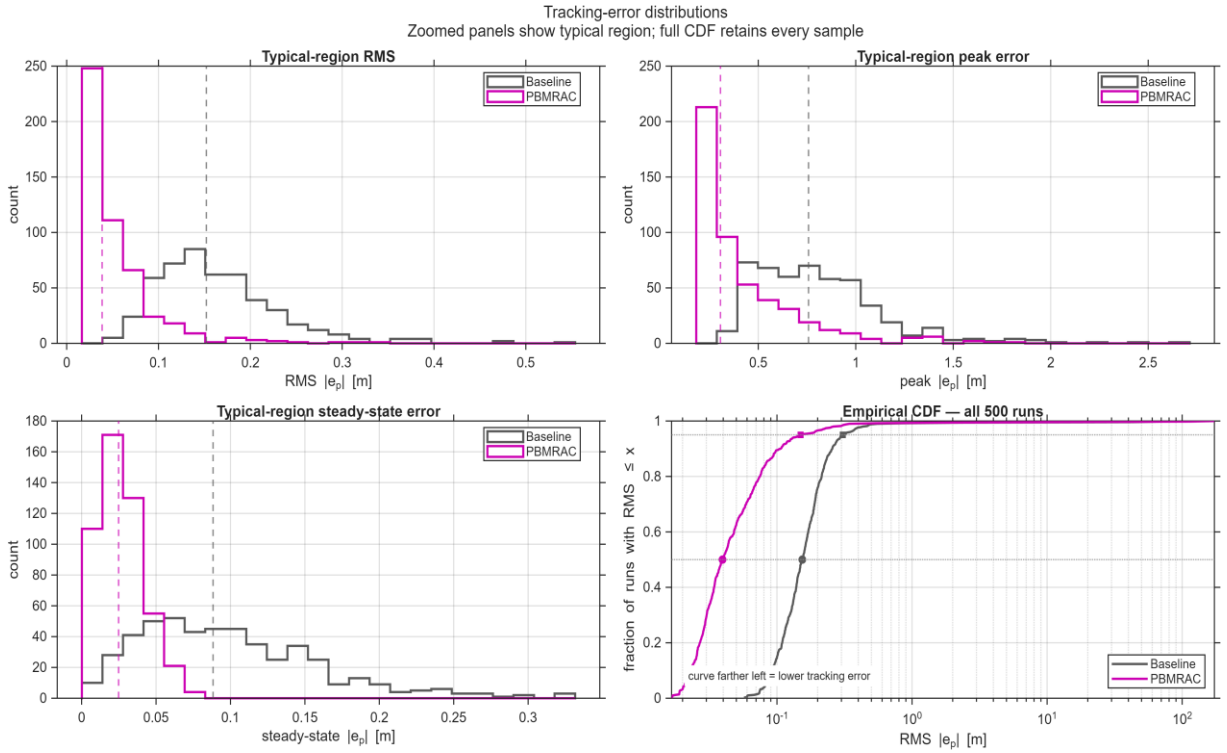
PER-RUN RMS POSITION ERROR

Per-run RMS position error	Baseline	PBMRAC
Mean	0.670 m	0.653 m
Median	0.154 m	0.039 m
p95	0.306 m	0.150 m
Maximum observed	126.2 m	173.2 m
Runs with RMS $>$ 1 m	4/500	4/500
Recorded failed runs	0/500	0/500

PAIRED COMPARISON

– PBMRAC has the lower RMS on 498 of 500 vehicles (99.6 %); median paired improvement 73.0 %.

DISTRIBUTIONS — RMS, PEAK, STEADY-STATE, CDF (dashed = median)

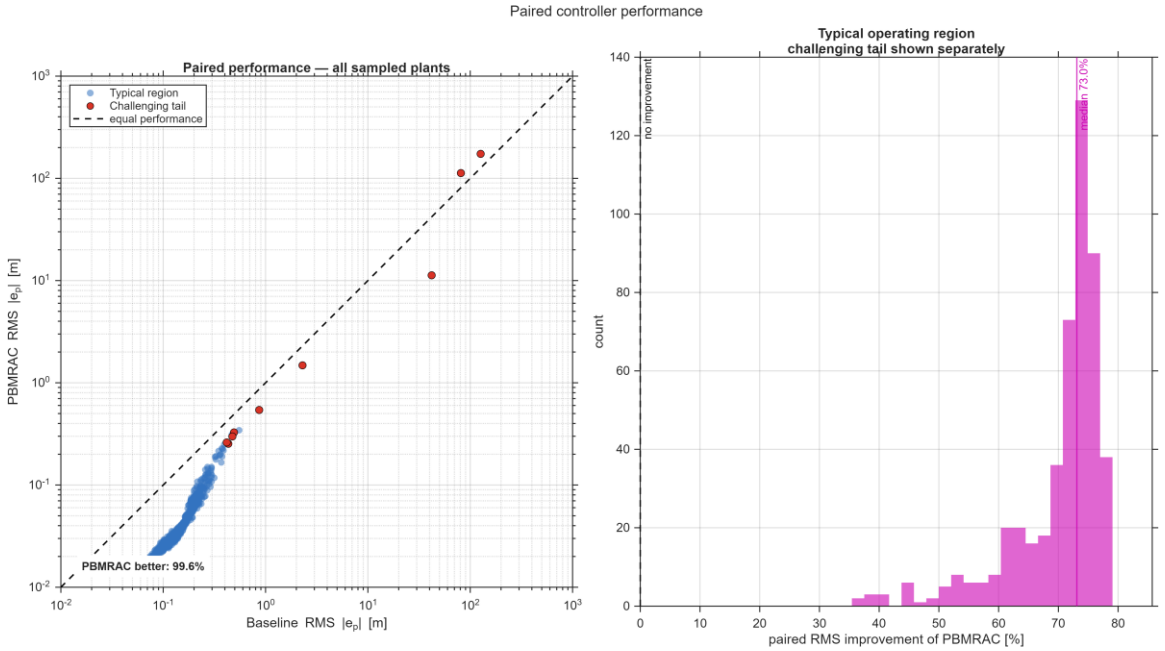


RELATIVE IMPROVEMENT

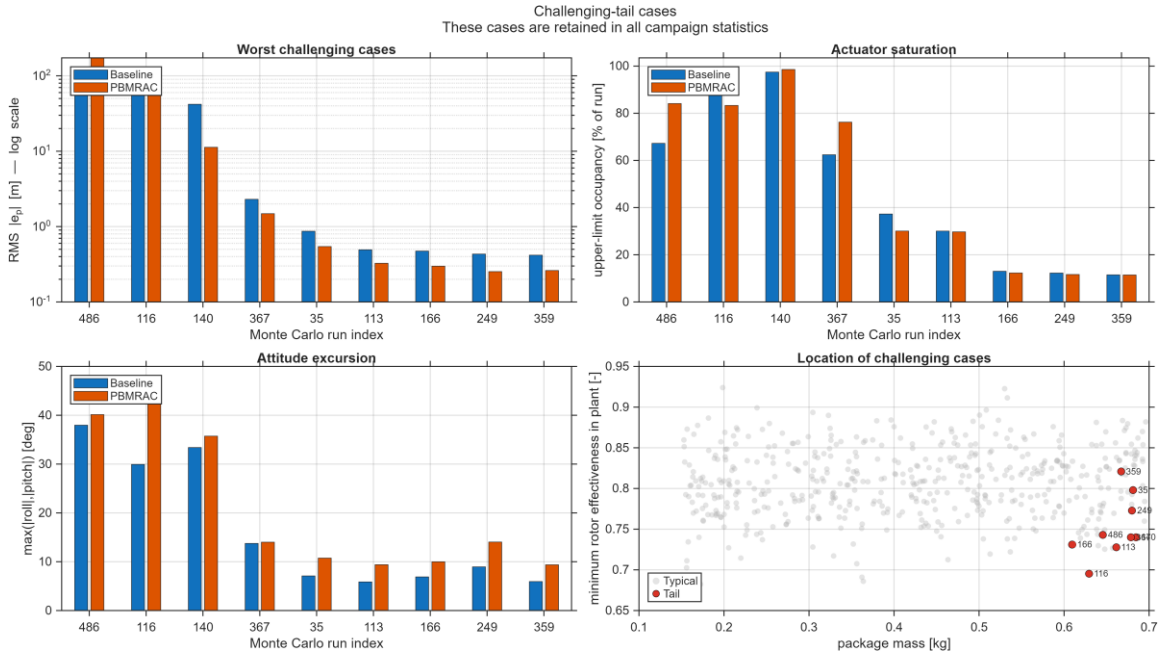
Median RMS −74.5 %
p95 RMS −51.1 %
Mean RMS −2.6 %

The challenging tail

PAIRED RMS — EVERY SAMPLED VEHICLE



THE TAIL VEHICLES (WITH HEDGING)



WHAT GOES WRONG

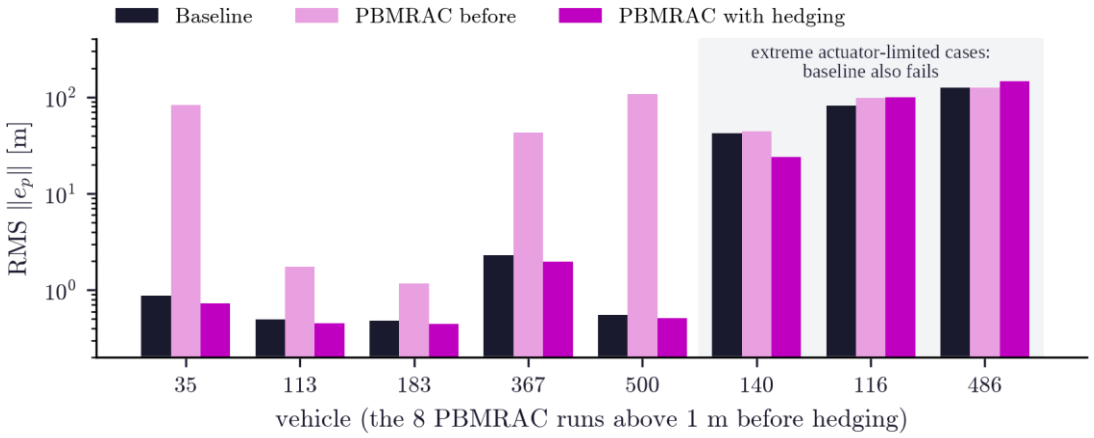
- Tail = RMS $> 1\,\mathrm{m}$, $> 5\,\%$ time at Ω_{max} or $> 30^\circ$ tilt for either controller: 9 of 500 vehicles (1.8 %).
- All nine carry $m_p > 0.6\,\mathrm{kg}$ and saturate the rotors for both controllers.
- PBMRAC loses tracking in a small number of severe actuator-limited cases, and only where the baseline also fails: the same 4 vehicles above 1 m RMS for both; $\hat{\lambda}_i$ reaches the 0.35 wall on 3 of them; largest PBMRAC tilt 46°.
- All runs stay in the statistics: PBMRAC is worse only on 2 extreme actuator-limited realizations (486: 126 vs 173 m; 116: 81 vs 112 m).

Fixing the tail: saturation hedging

WHAT WENT WRONG, AND THE FIX

- Saturated rotors deliver less force than commanded; the $\hat{\Lambda}$ law read the shortfall as lost effectiveness, drove $\hat{\lambda}$ to the 0.35 wall and asked for 2.9× more force.
- Fix: the predictor and the $\hat{\Lambda}$ law now see the force the vehicle can receive, $\mathbf{u}_h = \mathbf{u}\min(1,\ T_{max}/\|\mathbf{u}\|)$, $T_{max} = 22.93$ N (the attitude planner cap).
- The control law is unchanged; with no saturation $\mathbf{u}_h = \mathbf{u}$.

THE 8 VEHICLES THAT LOST TRACKING BEFORE



SAME 500 VEHICLES — BEFORE vs AFTER HEDGING

Same 500 vehicles, same gains	Before	After
Median RMS (PBMRAC)	0.069 m	0.069 m
p95 RMS (PBMRAC)	0.257 m	0.203 m
Mean RMS (PBMRAC)	1.091 m	0.660 m
PBMRAC better than baseline	488 / 500	498 / 500
PBMRAC runs > 1 m (baseline: 4)	8	4
Tail vehicles (tilt rule 30°)	13	9
$\hat{\lambda}$ at the 0.35 wall	28	3
Largest PBMRAC tilt	141°	49°

Precision and rare difficult cases

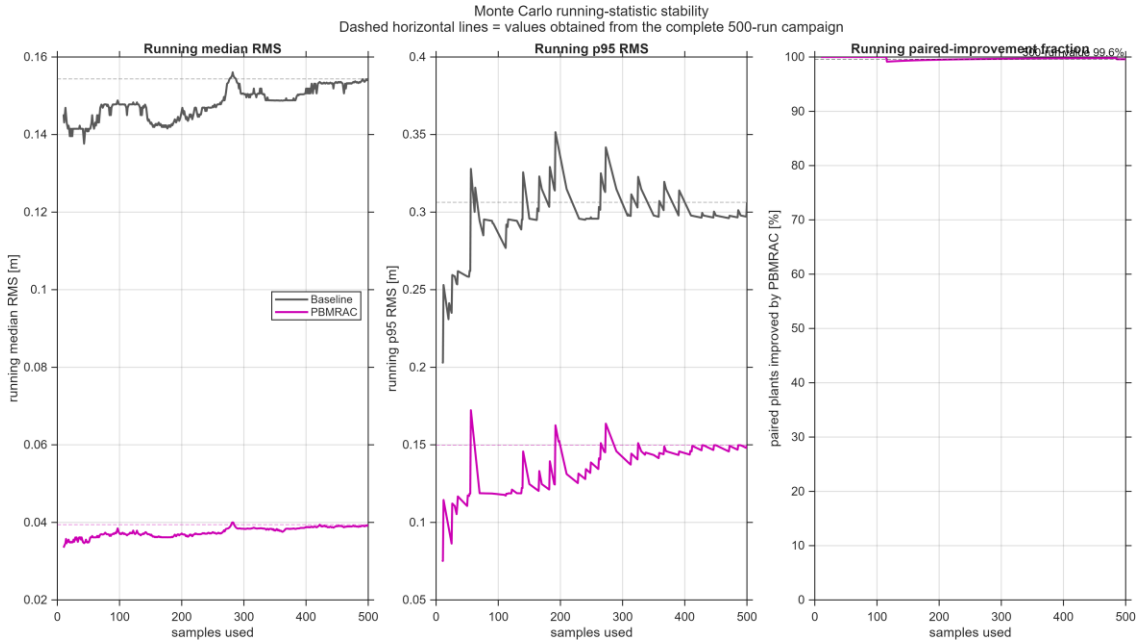
95% BOOTSTRAP CONFIDENCE ON THE MEDIAN RMS

Median RMS	Baseline	PBMRAC
Median	0.154 m	0.039 m
95 % bootstrap CI	[0.147, 0.162]	[0.037, 0.043]
Relative half-width	4.7 %	7.9 %

INTERPRETATION

- Bootstrap sampling uncertainty on the median is below $\sim 8\,\%$, conditional on the selected Monte Carlo distribution.
- The mean is not: its 95 % half-width is 92 % (baseline) and 117 % (PBMRAC), driven by one or two vehicles.

RUNNING MEDIAN, p95 AND IMPROVEMENT FRACTION vs SAMPLES



WHAT THE CONVERGENCE SHOWS

- The running median settles within about 200 samples.
- The p95 keeps jumping as tail vehicles arrive: 500 runs pin the median, not the extreme tail.

TASK 4

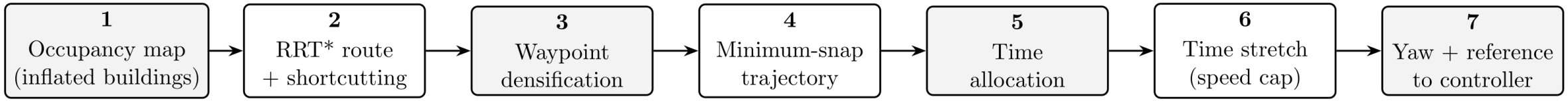
Delivery Mission

Route planning · Smooth trajectory · Disturbed flight test

POLITECNICO
MILANO 1863

The delivery mission

PLANNING PIPELINE



BUILDING MATRIX [x_min x_max y_min y_max z_min z_max] (m)

Bldg	x_{min}	x_{max}	y_{min}	y_{max}	z_{min}	z_{max}
B_1	22	38	28	42	0	28
B_2	52	68	48	62	0	26
B_3	52	68	13	27	0	32
B_4	76	92	32	46	0	32
B_5	102	118	13	27	0	24
B_6	102	118	48	62	0	30
B_7	148	164	27	43	0	30

MISSION PARAMETERS

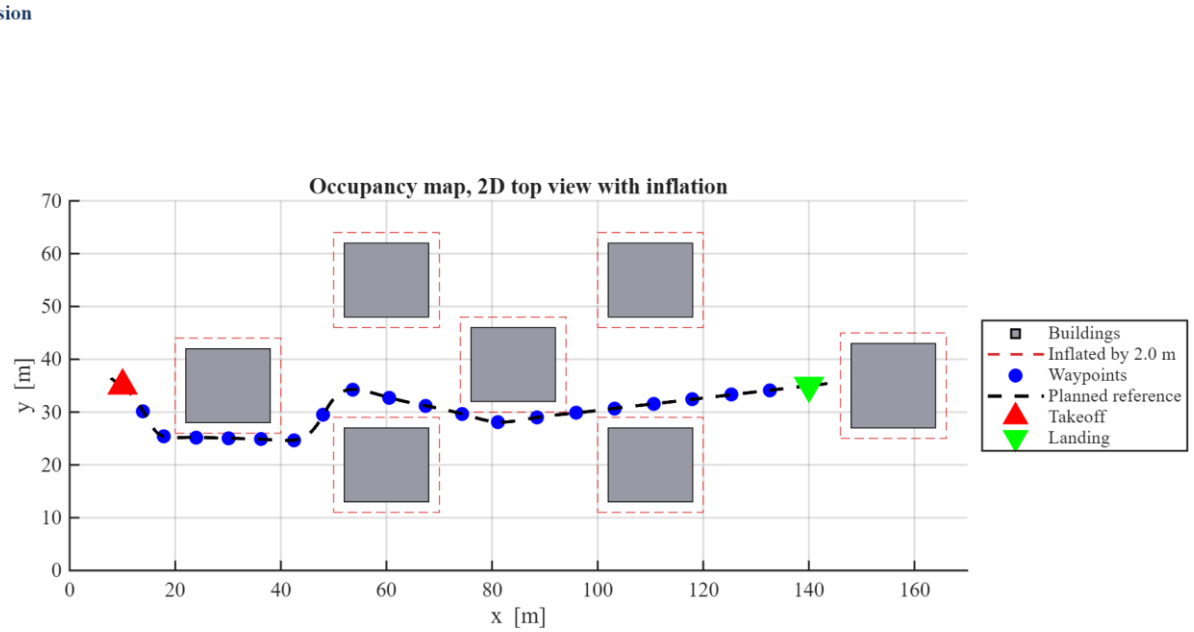
Cruise altitude (2-D)	20 m
Initial speed guess	3.0 m/s
Speed cap (2-D and 3-D)	4.5 m/s
Max. waypoint spacing	8 m
RRT* runs (restarts)	$\leq$ 200
Rotor start 2-D/3-D	hover/stopped
Takeoff hover	1 m, 5 s
End hold on the pad	10 s
Time step / seed	0.01 s / 1

MISSION

- Seven known buildings in a $170 \times 70 \times 40$ m map.
- Takeoff at [10, 35, 0] m; landing pad at [140, 35, 0] m, 8 m in front of building B_7.
- Wind gust and rotor degradation, 0.4 kg package.
- Two missions: 2-D (RRT* at 20 m) and 3-D (free z).

Represent buildings in an occupancy map

OCCUPANCY MAP — 2D TOP VIEW WITH INFLATION



BUILDING REPRESENTATION

- Each building is an axis-aligned box: $[x_{min}, x_{max}, y_{min}, y_{max}, z_{min}, z_{max}]$.

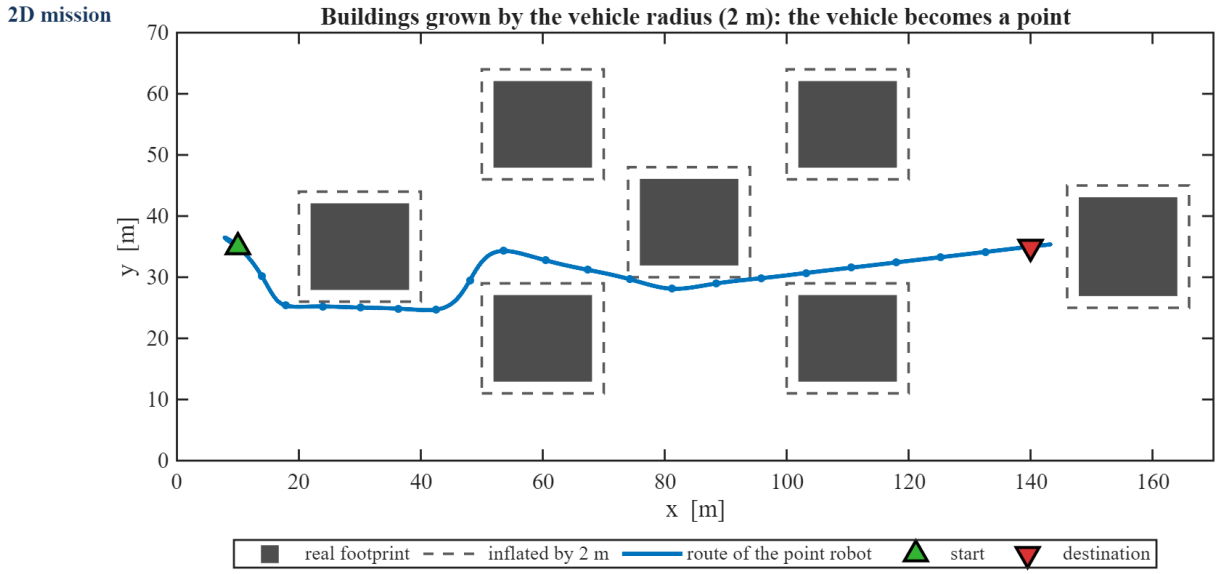
OCCUPANCY MAP

- `occupancyMap3D` stores occupied and free space.
- The default setting is one cell per metre.
- The code explicitly marks the sampled volume as occupied or free.

Enlarge obstacles to create clearance

OBSTACLE INFLATION (r = 2 m, PLANNED WITH 2.5 m)

$$\begin{cases} x\text{-interval} \rightarrow [x_{min} - r,\ x_{max} + r] \\ y\text{-interval} \rightarrow [y_{min} - r,\ y_{max} + r] \\ z\text{-interval} \rightarrow [z_{min},\ z_{max} + r] \end{cases}$$



RRT* selects a route through free space

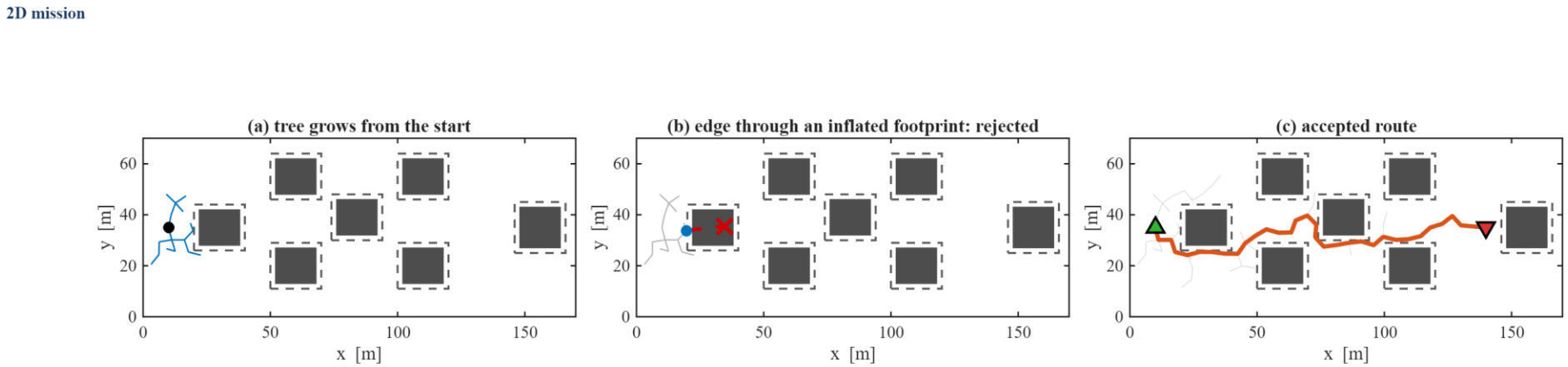
RRT* LOOP

1. Sample a candidate state, sometimes biased toward the goal.
2. Extend the existing tree toward it.
3. Reject connections that fail collision checking.
4. Choose favourable connections and rewire neighbours when their cost improves.
5. Return a route when the goal is reached or terminate when the search budget is exhausted.

CURRENT SETTINGS

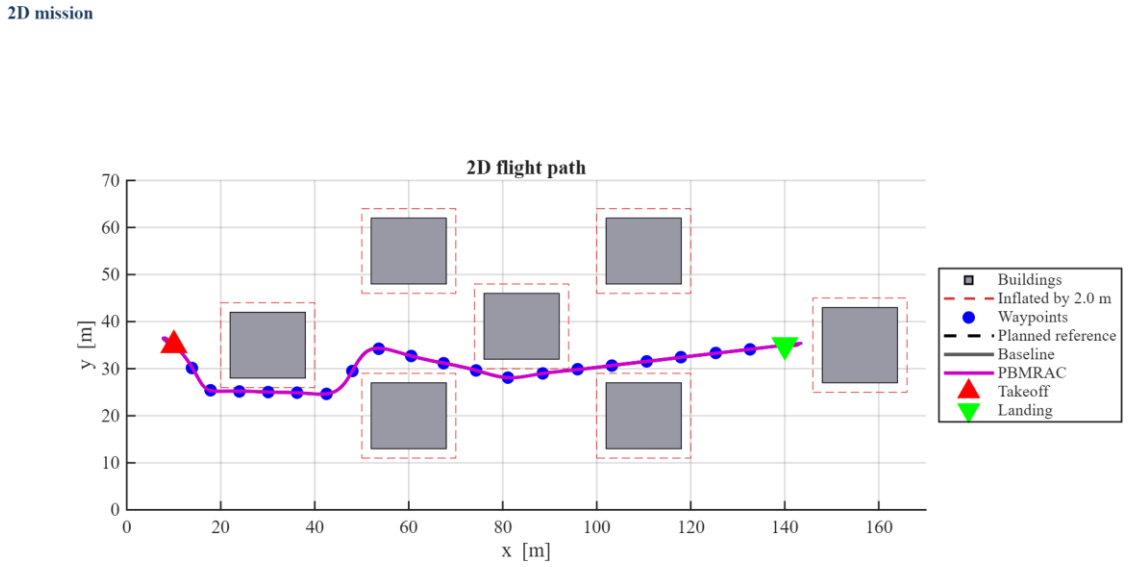
MaxConnectionDistance	5 m
MaxIterations	10000
GoalBias	0.12
Runs (new seed each)	$\leq$ 200
Stop after no gain for	60 runs
Collision-check spacing	0.25 m

RRT* IN THREE FRAMES — TREE GROWTH, REJECTED EDGE, ACCEPTED ROUTE



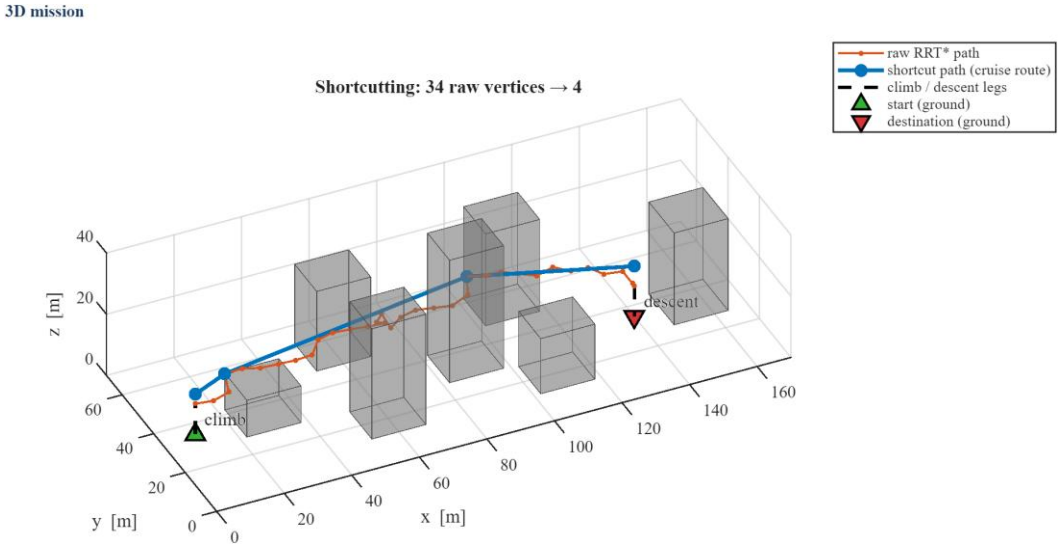
Two illustrative mission scenarios: RRT* restricted and free

2-D MISSION — RRT* RESTRICTED TO z = 20 m



- State space $z \in [20, 20]$ m: every sample and every collision check is made at the flight level.
- All roofs + inflation lie above 20 m: the route must go around the buildings.

3-D MISSION — RRT* FREE IN ALTITUDE



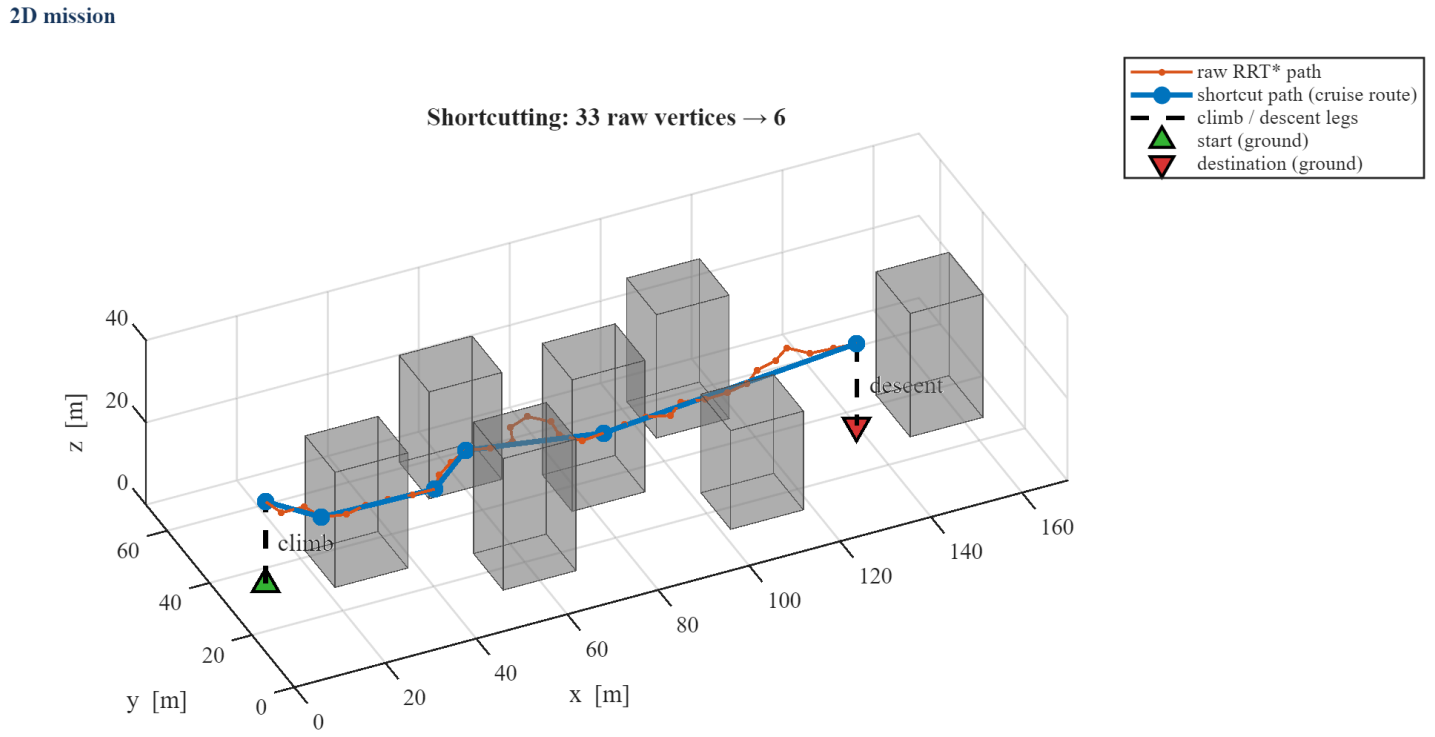
- $z \in [4, 37.5]$ m, chosen by the planner; buildings of varying height (B_1 12 m, B_5 18 m, B_4 40 m).
- The shortest route flies OVER B_1 (≈ 15.5 m) and passes north of the 40 m B_4 at ≈ 18 m: 134.6 m in total.

Define the route endpoints and simplify the path

MISSION ENDPOINTS

Takeoff	[10, 35, 0]	
Cruise start	[10, 35, 20]	← RRT* start
Cruise end	[140, 35, 20]	← RRT* goal
Landing	[140, 35, 0]	

RAW RRT* PATH vs SHORTCUT PATH (2-D: 33 RAW VERTICES → 6)

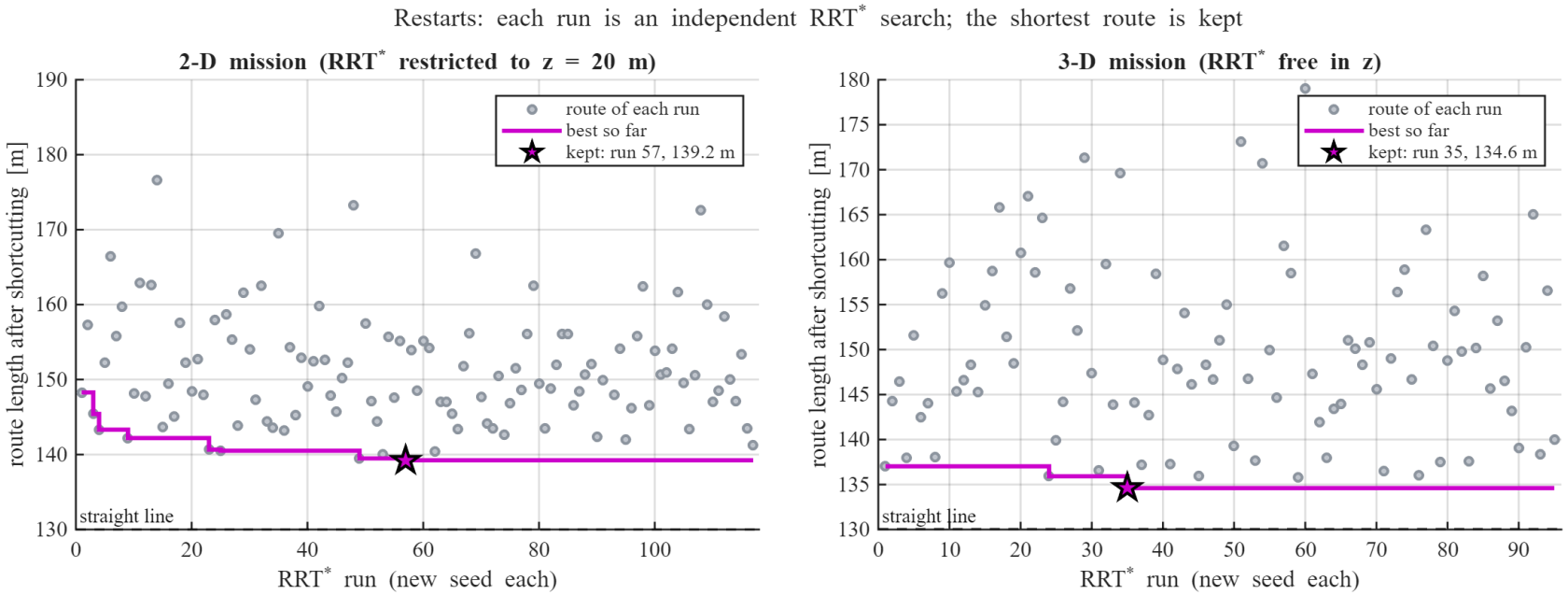


SHORTCUTTING

- Shortcutting connects each retained point to the furthest later point that passes the motion validator.
- Audit (2-D): 33 raw states became six shortcut points.

Iterating to improve the route: RRT* restarts

ROUTE LENGTH OF EVERY RRT* RUN — THE SHORTEST IS KEPT



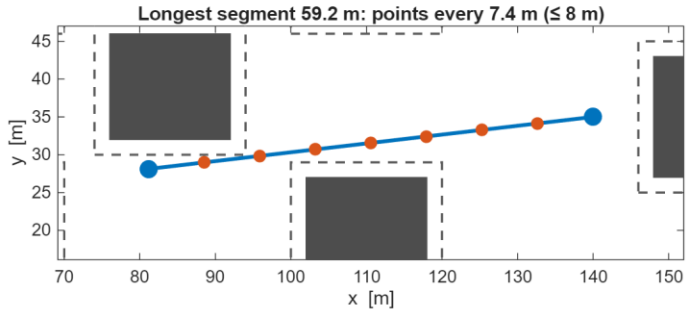
RESULT

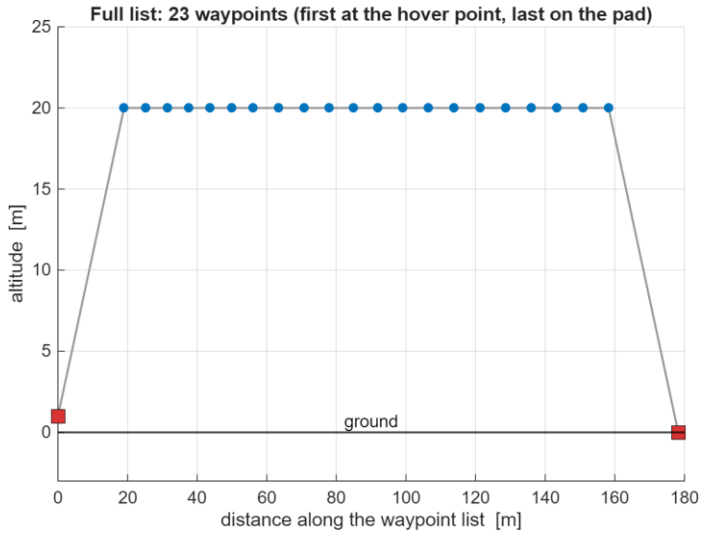
Route length	2-D	3-D
RRT* runs	117	95
First run	148.3 m	137.0 m
Kept (shortest)	139.2 m	134.6 m
Median of all runs	150.0 m	148.3 m
Straight line	130.0 m	130.0 m

Prepare the waypoints for smoothing

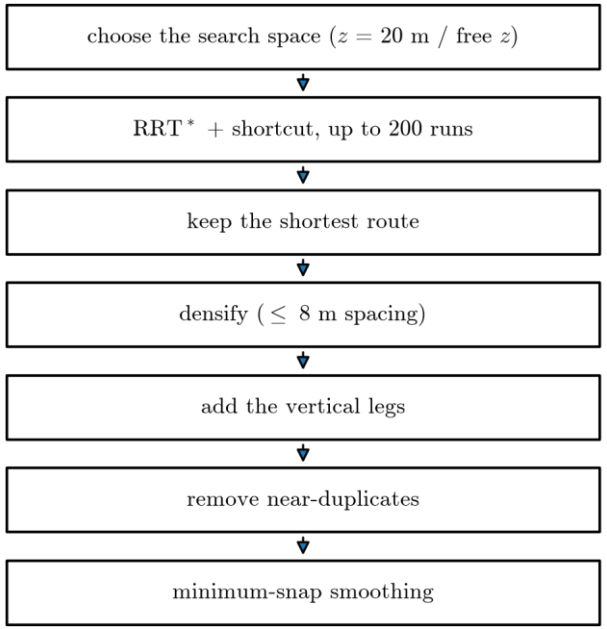
LONGEST SEGMENT (DENSIFICATION) AND FULL WAYPOINT LIST

2D mission





WAYPOINT PROCESSING CHAIN

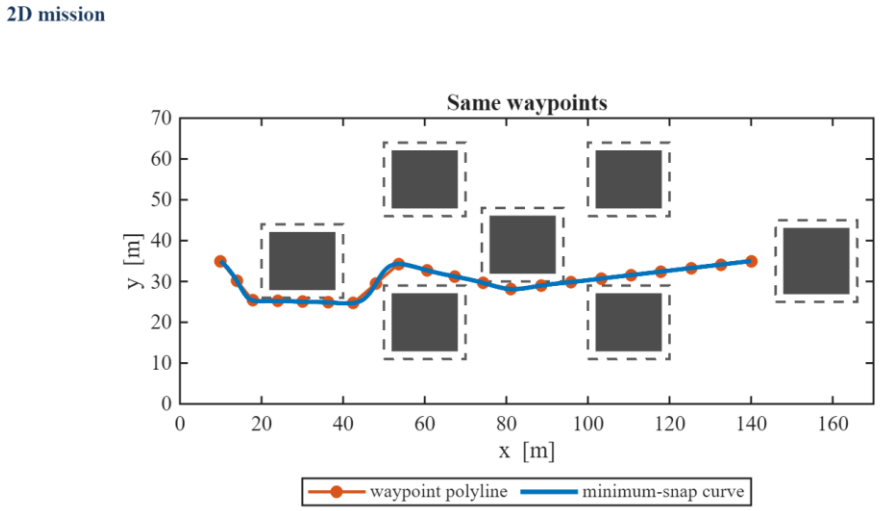


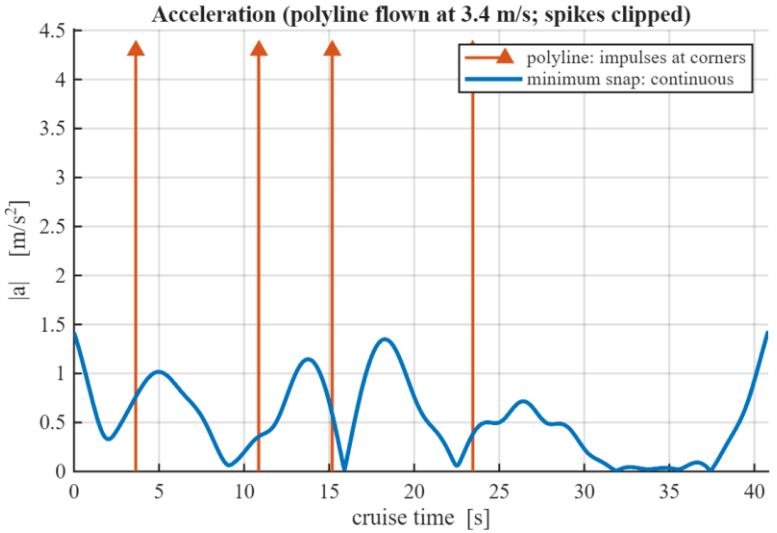
WAYPOINT LIST

– Points help constrain the later smooth curve.
– Final list: 23 waypoints (2-D), 21 (3-D).

Minimum snap turns waypoints into a smooth trajectory

SAME WAYPOINTS: POLYLINE vs MINIMUM-SNAP (MAP AND ACCELERATION)





DERIVATIVE CHAIN

$$\text{position} \rightarrow \text{velocity} \rightarrow \text{acceleration} \rightarrow \text{jerk} \rightarrow \text{snap}$$

$$p \qquad \dot{p} \qquad \ddot{p} \qquad p^{(3)} \qquad p^{(4)}$$

CONCEPTUAL OBJECTIVE

$$\begin{cases} \text{Minimize} \displaystyle\int \|p^{(4)}(t)\|^2 \, dt \\ \text{Subject to waypoint and derivative-continuity constraints} \end{cases}$$

Assign and optimize the segment times

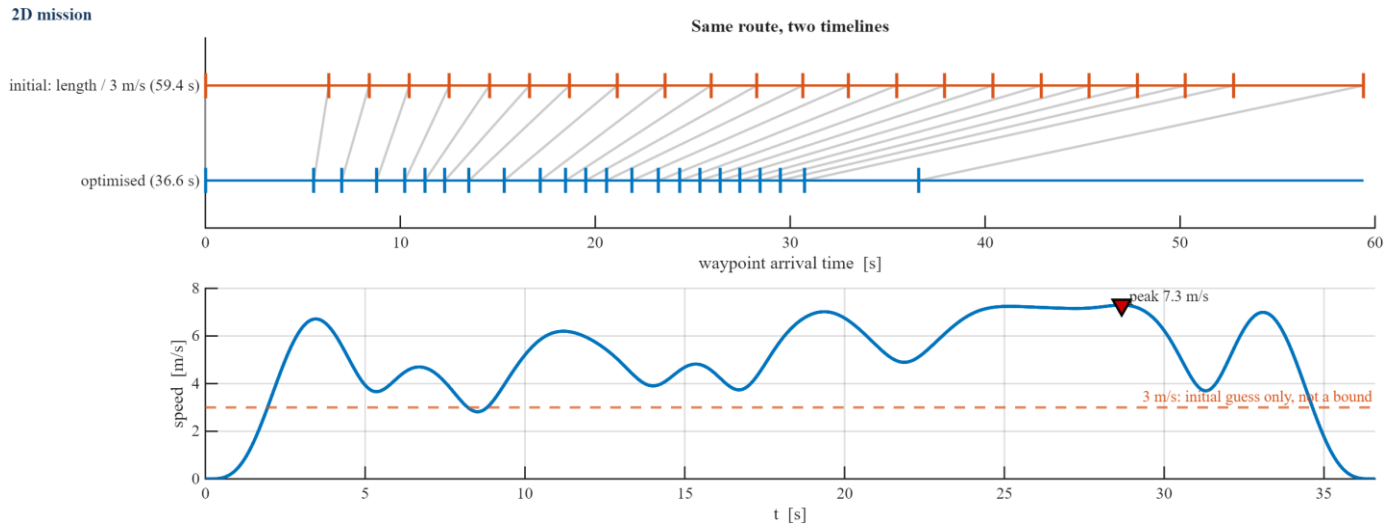
INITIAL TIMING GUESS

$$\begin{cases} \Delta t_j = \|w_{j+1} - w_j\|/3 \\ \text{Initial waypoint times} = \text{cumulative} \\ \qquad \text{sum of segment durations} \end{cases}$$

AUDIT

Audit: the optimised reference peaked at 7.30 m/s.

SAME ROUTE, TWO TIMELINES — INITIAL vs OPTIMISED, AND SPEED PROFILE



TIME ALLOCATION OPTIONS

- `TimeAllocation=true`: segment durations can change.
- `TimeWeight=80`: trades smoothness against total duration.

SEGMENT TIMES

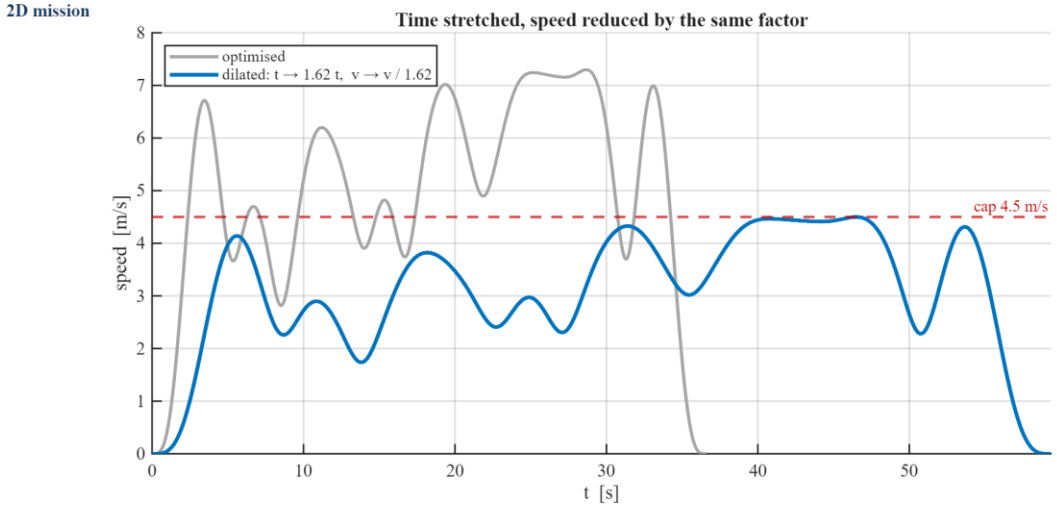
- Min./max. segment times are derived from the initial durations.

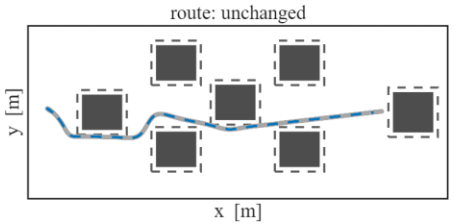
Enforce the reference speed by stretching time

UNIFORM TIME STRETCH

$$
\begin{cases}
k = \max(1,\ \text{sampled peak speed}/4.5) \\
t_{new} = k\, t_{old} \\
p_{new} = p_{old} \\
v_{new} = v_{old}/k \\
a_{new} = a_{old}/k^2
\end{cases}
$$

TIME STRETCHED, SPEED REDUCED BY THE SAME FACTOR (ROUTE UNCHANGED)



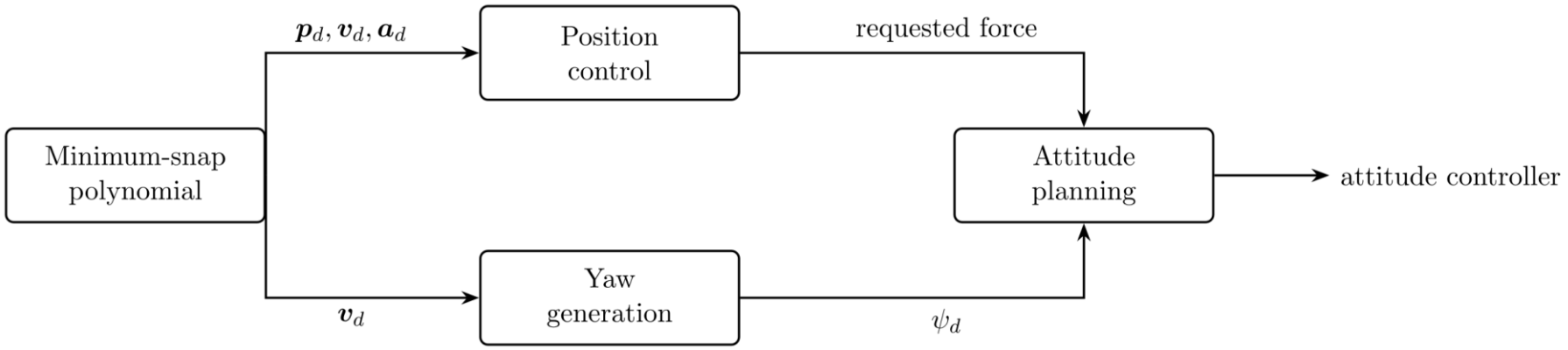


AUDIT VALUES

k (2-D)	7.30 / 4.5 = 1.62
Planned trajectory	36.7 s → 59.4 s
Of which descent	9.6 s
Sampled peak speed	4.500 m/s
Total simulation (+ rise, hover, hold)	76.4 s
3-D (same cap)	k = 1.96, 57.5 s (total 74.5 s)

Generate yaw and send the reference to the controller

REFERENCE FLOW TO THE CONTROLLER



YAW GENERATION

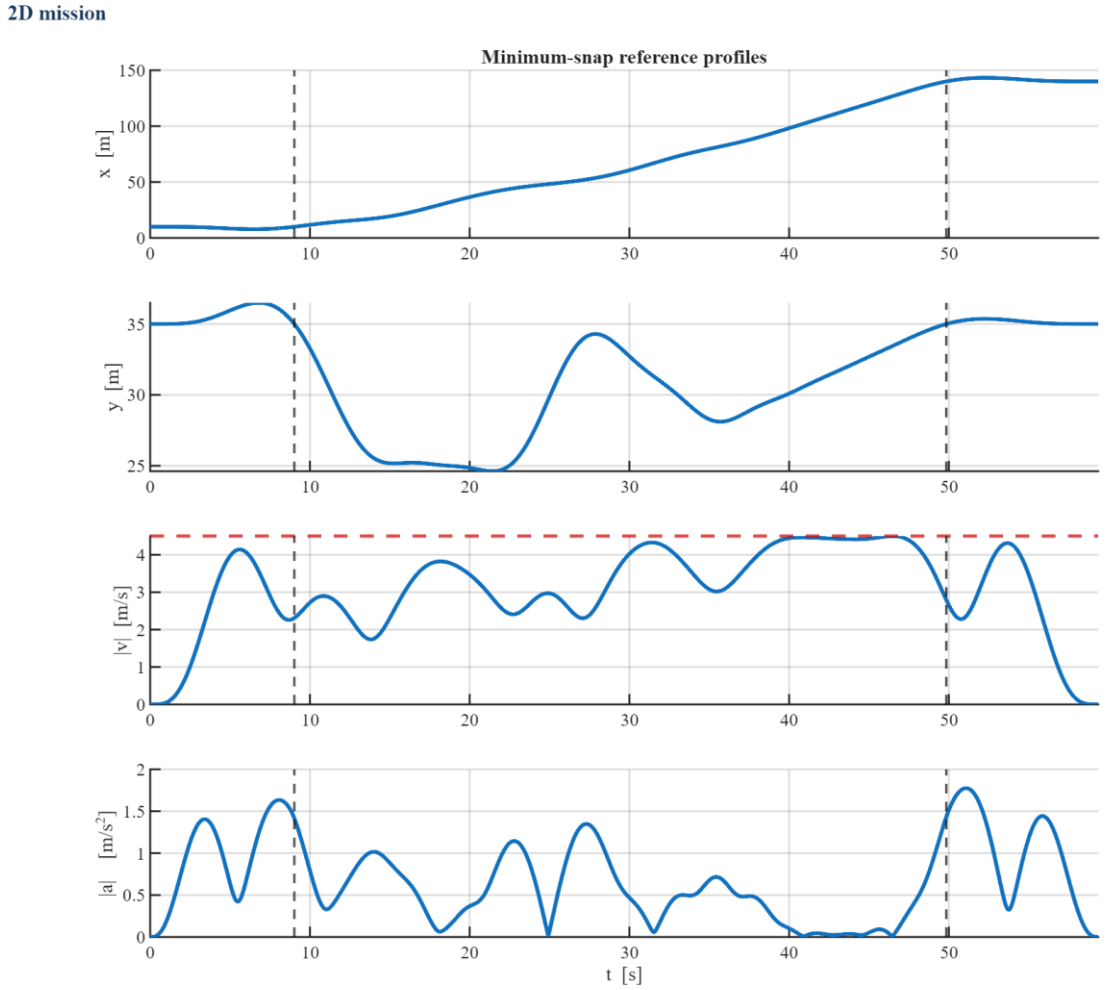
$$\begin{cases} \psi_d = \text{atan2}(v_{y,d},\, v_{x,d}) & \text{when horiz. speed} > 0.12 \text{ m/s} \\ \text{otherwise hold last valid heading (unwrapped)} \end{cases}$$

Validate the final reference

REFERENCE CHECK

Min. sampled reference clearance	2.55 m
Flown clearance, baseline / PBMRAC	2.59 / 2.58 m
Required (planned with 2.5 m)	2.0 m
Sampled peak speed	4.500 m/s
Sampled peak acceleration	1.78 m/s^2
Final position / velocity	[140, 35, 0] m / ≈ 0

MINIMUM-SNAP REFERENCE PROFILES: x, y, SPEED AND ACCELERATION



Mission phases and unforeseen events

PHASE / EVENT TIMING (2-D MISSION)

Rise to 1 m + hover	0–7.0 s
Ascent	7.0–16.0 s
Cruise	16.0–56.8 s
Descent	56.8–66.4 s
End hold on the pad	66.4–76.4 s
Wind gust	30.3–50.3 s
Rotor degradation	from 44.6 s, persists

EVENT PLACEMENT AND DISTURBANCES

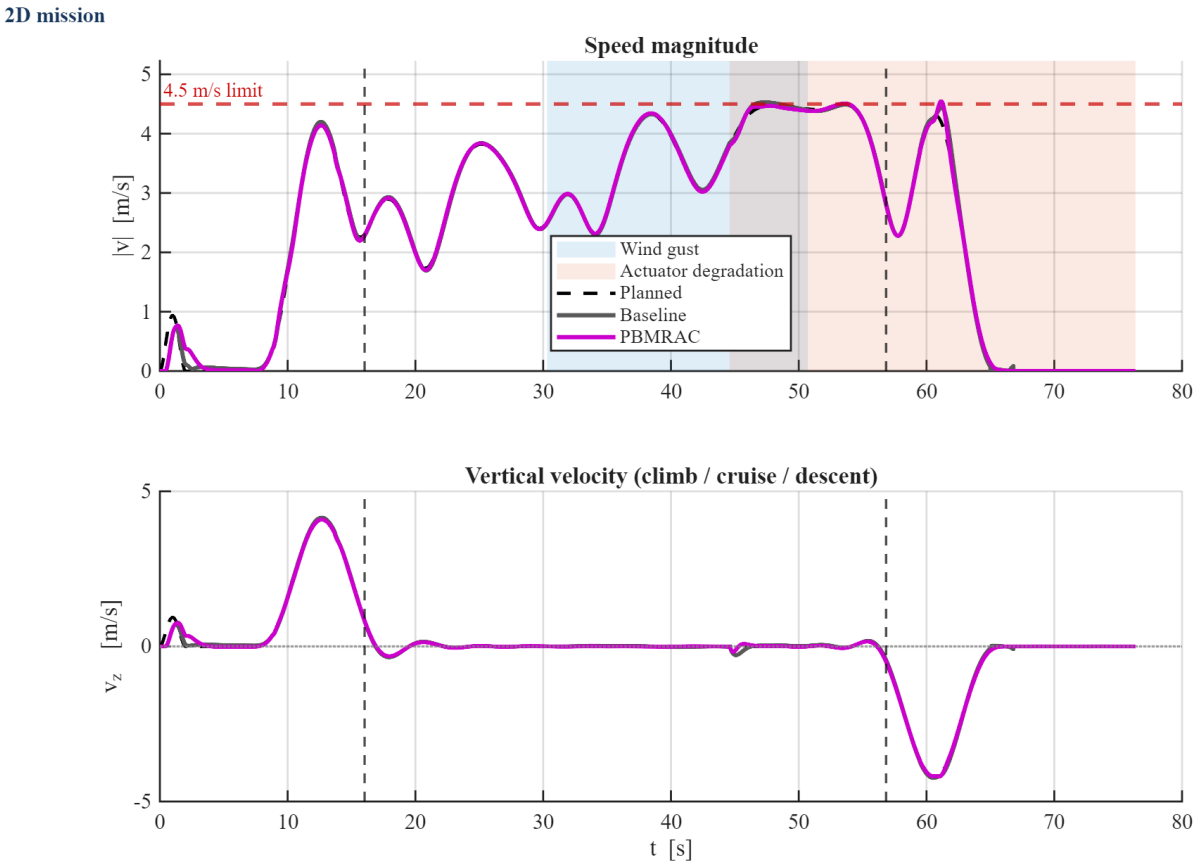
$$t_{gust} = t_{climb} + 0.35\,(t_{descend} - t_{climb})$$

$$t_{fail} = t_{climb} + 0.70\,(t_{descend} - t_{climb})$$

$$\mathbf{v}_{gust,\,peak} = [-12,\ 0,\ 0]^T \text{ m/s},\ \min(30 \text{ s},\ \tfrac{1}{2} \text{ span})$$

$$\lambda_r = [0.8,\ 0.9,\ 1,\ 0.8,\ 0.8,\ 1]^T$$

SPEED MAGNITUDE AND VERTICAL VELOCITY WITH PHASES AND EVENTS

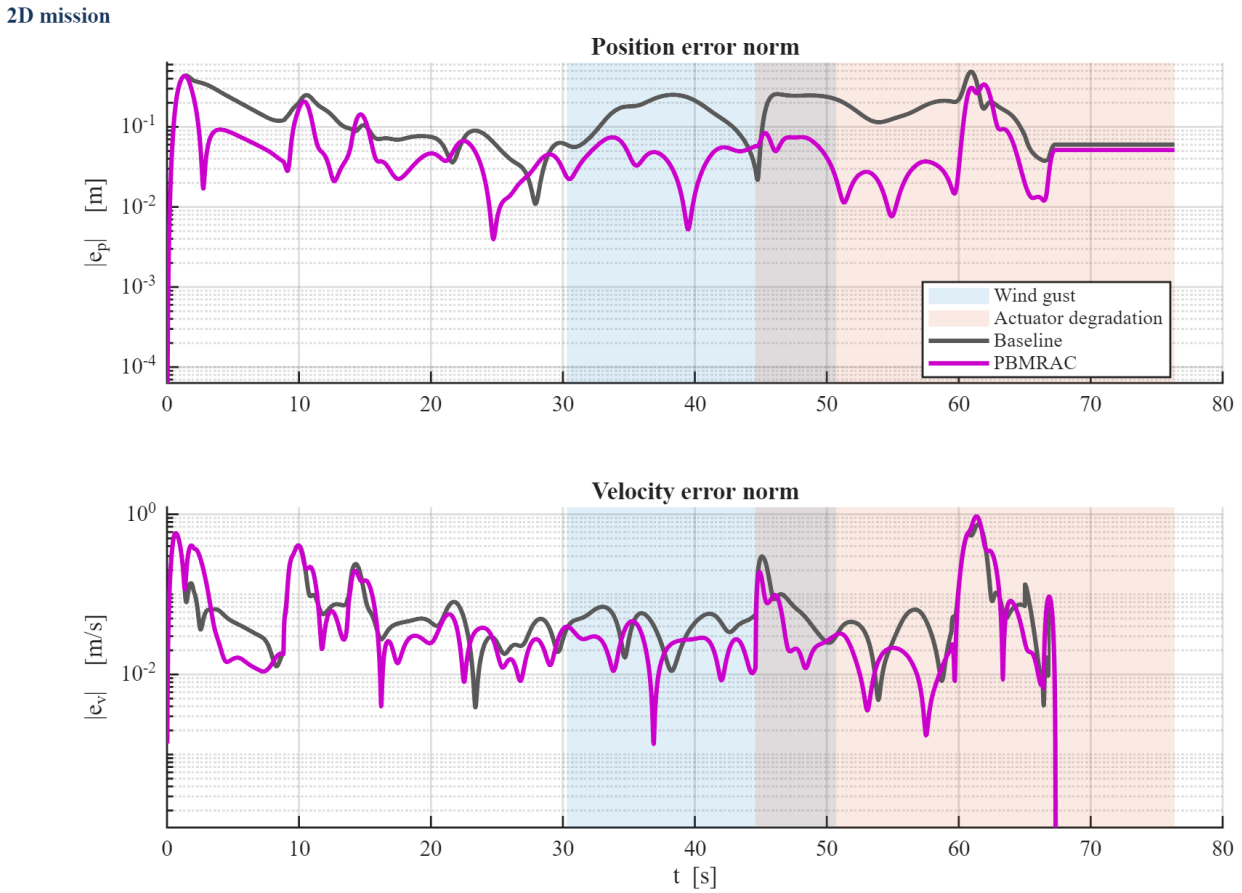


Baseline and PBMRAC fly the same mission

COMPARISON SETUP

Run	PBMRAC	Package	Gust	Rotor fault
Baseline	off	0.4 kg	yes	yes
PBMRAC	gated	0.4 kg	yes	yes

POSITION AND VELOCITY ERROR NORMS — BASELINE vs PBMRAC



Take-off, landing and when PBMRAC runs

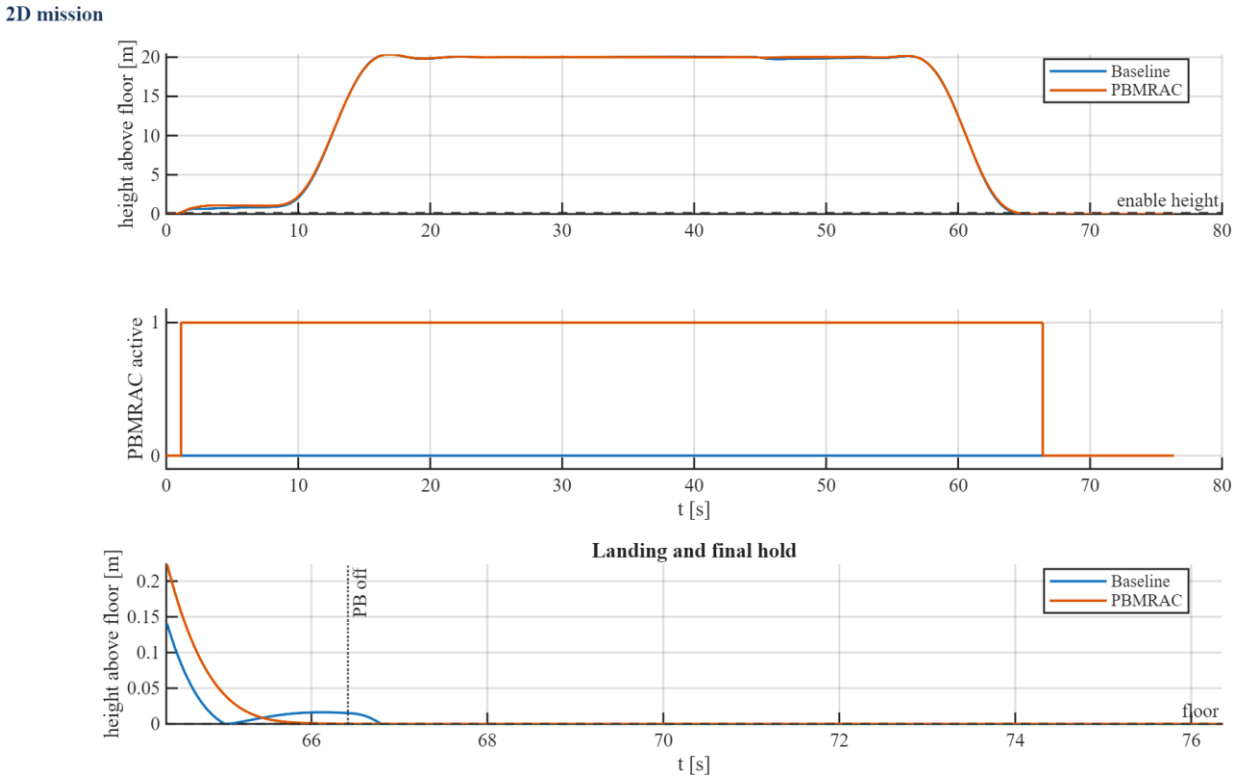
WHEN PBMRAC RUNS (FLIGHT GATE)

Takeoff	pad $\rightarrow$ 1 m in 2 s, hover 5 s, then the mission
PBMRAC on	height $\geq$ 0.15 m and climb $\geq$ 0.02 m/s, for 0.1 s
PBMRAC off	floor contact (latched to the end)
2-D / 3-D	on at 1.1 s; off at 66.4 s / 64.5 s

GROUND AND LANDING

- Ideal floor: height limited at the pad, impact stops the vehicle (no bounce or slide), support carries it at rest.
- PBMRAC is derived for free-flight translational dynamics; ground reaction is not part of the predictor model. Adaptation is therefore intentionally disabled during contact and initialized from the current measured state after confirmed lift-off, so the gate is a model-validity supervisor.
- PBMRAC switches on during the rise; the 5 s hover lets its initial adaptation transient settle before the main climb.
- At the end the reference is lowered smoothly to 5 cm below the pad over 1 s and kept there, so every controller touches down.
- The contact model arrests motion; speed just before contact: -0.07 m/s (baseline), -0.002 m/s (PBMRAC).

HEIGHT, PBMRAC ON/OFF AND LANDING — 2-D MISSION



2-D mission: each phase and control effort

RMS BY PHASE

$$\left\{ RMS_{phase} = \sqrt{\text{mean}\left(\|\boldsymbol{p}_d - \boldsymbol{p}\|^2\right)} \text{ over that phase} \right.$$

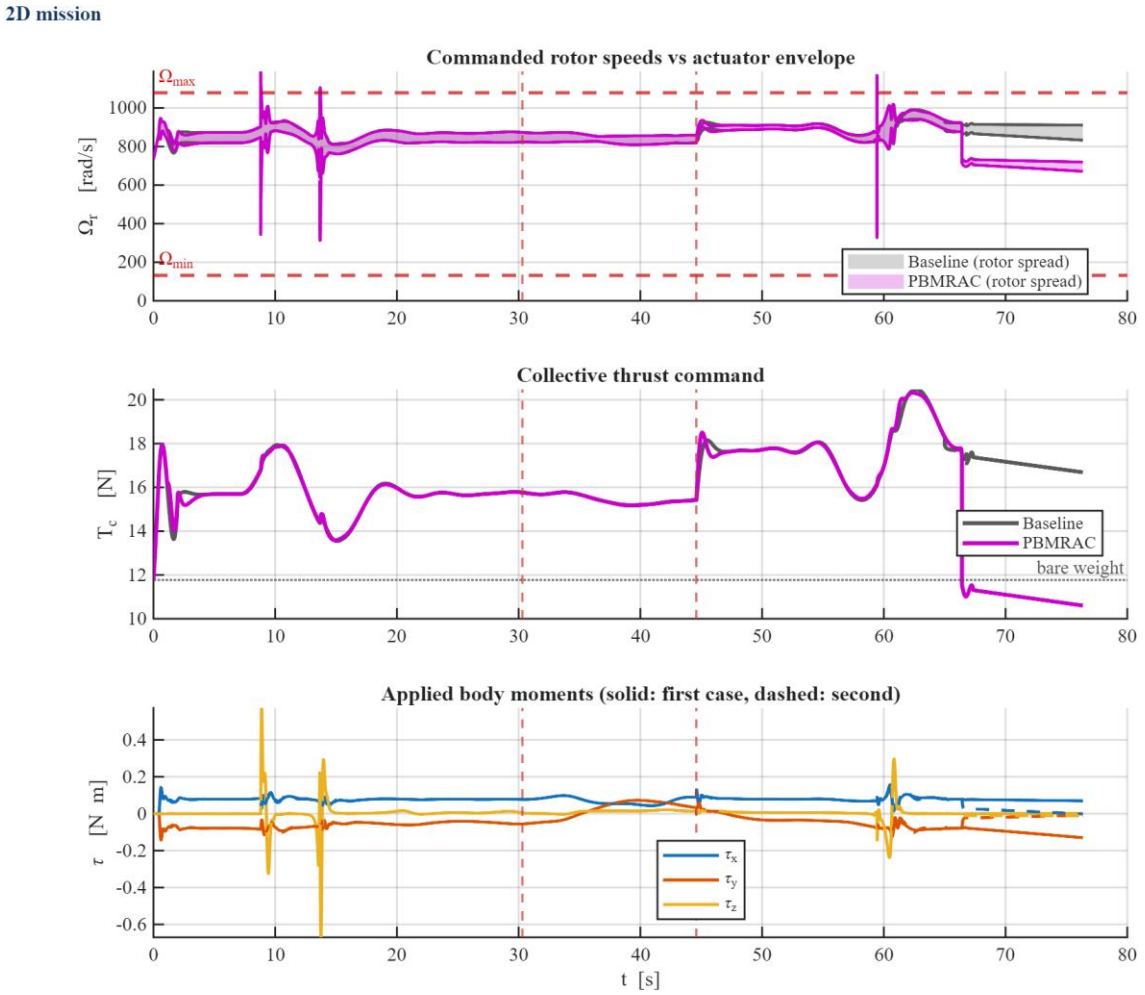
RMS POSITION ERROR PER PHASE [m] (2-D MISSION)

Controller	Ascent	Cruise	Descent	Whole	Peak	Pad err
Baseline	0.217	0.149	0.217	0.167	0.487	0.033
PBMRAC	0.134	0.044	0.142	0.087	0.433	0.012

NOTES

- Whole-run RMS: 0.167 → 0.087 m (−48 %); cruise 0.149 → 0.044 m.
- Ascent includes the rise and the 5 s hover; gust peak 0.26 vs 0.08 m.
- Rotor commands briefly exceed the allocation limits in ascent and descent: clipped 0.12 % (baseline) / 0.13 % (PBMRAC) of the time.
- Pad err = final distance to the pad centre; the error plots end near 5 cm because the reference sits 5 cm below the pad.

ROTOR SPEEDS, COLLECTIVE THRUST AND BODY MOMENTS



3-D mission: each phase

RMS BY PHASE

$$\left\{ RMS_{phase} = \sqrt{\text{mean}\left(\|\boldsymbol{p}_d - \boldsymbol{p}\|^2\right)} \text{ over that phase} \right.$$

RMS POSITION ERROR PER PHASE [m] (3-D MISSION)

Controller	Ascent	Transit	Descent	Whole	Peak	Pad err
Baseline	0.227	0.147	0.124	0.159	0.438	0.020
PBMRAC	0.157	0.035	0.061	0.085	0.433	0.009

NOTES

- Whole-run RMS: 0.159 → 0.085 m (−47 %); transit 0.147 → 0.035 m.
- Better in every phase. With the hover, the initial adaptation transient settles before the steep climb (same-session tests without the hover were worse in the climb; not a controlled proof).
- Commands clipped 0.10 % of the time for both; tilt up to 24° (20°).
- Flown clearance 2.40 m for both (reference 2.41 m).

POSITION AND VELOCITY ERROR NORMS — 3-D MISSION

